FACULTY OF ENGINEERING
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Development, simulation & implementation of precoding techniques for satellite links

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Abstract — Linear precoding has become a cornerstone of MIMO communications, as it enables spatial multiplexing gains that translate directly into higher spectral efficiency. These benefits are particularly valuable in multi-beam high-throughput satellite systems, where aggressive frequency reuse across the narrow spot beams turns the forward link into an interference-limited MU-MIMO channel. In this environment, precoding is crucial for mitigating inter-beam interference. In mobile satellite scenarios, however, the combination of long propagation delays and time-varying channels induced by user mobility introduce complications. More specific, the CSI available during the precoder computation is generally outdated with respect to the channel experienced at the time of transmission. This work investigates the performance degradation caused by this CSI mismatch and proposes a low-complexity compensation strategy.

Part I reviews the theoretical foundations underlying this work. After introducing the SU- and MU-MIMO system models, together with the most relevant information-theoretic concepts, a generalised WMMSE framework for joint optimisation of the precoder and combiner in a SU scenario is presented, and the capacity-achieving precoder is derived. For the MU-MIMO broadcast channel, we break down common linear precoding strategies and compare there performance. Although these results are well established, they are revisited here in a unified notation and validated through Monte-Carlo simulations, so that they can serve as a consistent baseline for the satellite-specific analysis that follows.

Part II applies this framework to the forward link of a multi-beam mobile satellite system. The impact of the CSI feedback delay on the system performance is quantified, revealing a substantial performance loss. To counter this degradation, a low-complexity prediction strategy based is proposed. Numerical results show that this scheme recovers a significant fraction of the performance loss under the considered channel model.

Keywords — MIMO, Precoding Techniques, Mobile Satellite Systems, Outdated CSI

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***Index Terms*—MIMO, Precoding Techniques, Mobile Satellite Systems, Outdated CSI**

I. Introduction

Satellite communications have recently regained considerable attention as a key enabler of ubiquitous broadband connectivity. They provide cost-efficient coverage of large geographical areas, including regions where deploying terrestrial infrastructure is impractical, and are envisioned as an integral component of future 5G-and-beyond networks [1], [2]. To meet the growing demand for throughput, modern satellite systems have evolved from single-beam architectures towards multi-beam systems, in which the coverage area is partitioned into many narrow spot beams. Combined with aggressive frequency reuse across these beams, this multi-beam architecture turns the forward link into an interference-limited multi-user (MU)-multiple-input multiple-output (MIMO) channel, where linear precoding proves itself a powerful tool for managing the resulting interference [3].

The vast majority of the satellite-precoding literature assumes perfect, instantaneous channel state information (CSI) at the gateway (GW). This assumption becomes unrealistic in mobile satellite scenarios, where user terminal (UT) mobility introduces a Doppler spread that makes the channel time-varying. In addition, propagation delays remain large compared to terrestrial systems: even for a LEO constellation, the round-trip time is on the order of a few milliseconds. As a result, the CSI available at the GW when the precoder is computed is systematically outdated with respect to the channel actually experienced at the time of transmission.

Only a limited number of works have explicitly addressed linear precoding for mobile multi-beam satellite systems under outdated CSI [3]. Vázquez et al. [4] studied delayed CSI in a GEO mobile interactive system. Joroughi et al. [5] proposed robust precoding strategies for a maritime multi-beam scenario. In contrast, the present work starts from a generic multi-user MIMO downlink formulation and studies the effect of outdated CSI on standard linear precoders. Rather than modifying the gateway-side precoder structure, we seek to compensate the mismatch at its source.

The contributions of this paper are twofold. First, the impact of outdated CSI on the achievable sum-rate of three linear precoders, namely zero-forcing (ZF), block diagonalization (BD), and weighted minimum mean square error (WMMSE), is quantified as a function of the feedback delay. Second, a low-complexity compensation strategy based on element-wise autoregressive channel prediction is integrated into the feedback loop and evaluated by means of numerical simulations.

The remainder of the paper is organized as follows. Section II presents the system and channel model. Section III summarizes the considered linear precoders. Section IV introduces the proposed channel-prediction strategy for outdated CSI compensation. Section V reports the simulation results. Section VI outlines directions for future work, and Section VII concludes this work.

II. System Description

We consider the forward link of a multi-beam high-throughput satellite (HTS) system. The ground GW centralizes all baseband processing and communicates with K UTs through the satellite, which is equipped with N_T antenna feeds. Each UT has N_R receive antennas, and the system is modeled as a MU-MIMO downlink channel.

System Model. The GW assigns a dedicated precoding matrix $\mathbf{F}_k(t) \in \mathbb{C}^{N_T \times N_S}$ to the N_S-dimensional data symbol vector $\mathbf{a}_k(t)$ intended for UT k. Defining the compound precoder $\mathbf{F}(t) = [\mathbf{F}_1(t) \ \cdots \ \mathbf{F}_K(t)] \in \mathbb{C}^{N_T \times KN_S}$ and the stacked symbol vector $\mathbf{a}(t) = [\mathbf{a}_1^T(t) \ \cdots \ \mathbf{a}_K^T(t)]^T$, the received signal at UT k is

$$\mathbf{y}_k(t) = \mathbf{H}_k(t)\,\mathbf{F}(t)\,\mathbf{a}(t) + \mathbf{n}_k(t), \tag{1}$$

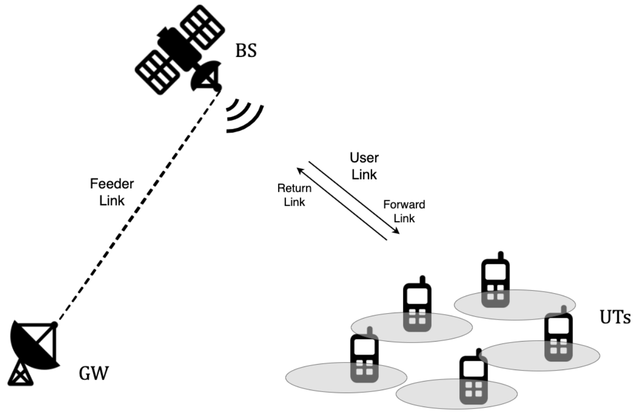


Fig. 1. Illustration of the considered system architecture for mobile satellite communications.

where $\mathbf{H}_k(t) \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix and $\mathbf{n}_k(t) \sim \mathcal{CN}(\mathbf{0}, N_0\mathbf{I}_{N_R})$ is additive white Gaussian noise. The data symbols are assumed zero-mean and spatially uncorrelated, and the total transmit power satisfies $\mathrm{tr}(\mathbf{F}(t)\mathbf{F}^H(t)) \leq P_T$ at any moment in time.

Channel Model. The user link of UT k follows a frequency-flat Rician fading model [6], expressed as:

$$\mathbf{H}_k(t) = \sqrt{\frac{\kappa}{\kappa+1}}\,\mathbf{H}_k^{\mathrm{LoS}} + \sqrt{\frac{1}{\kappa+1}}\,\mathbf{H}_k^{\mathrm{NLoS}}(t), \qquad (2)$$

where κ is the Rician factor. The deterministic line-of-sight (LoS) component is treated as time-invariant and modeled as a rank-one matrix, in which all entries share a common random phase offset. The non-line-of-sight (NLoS) component $\mathbf{H}_k^{\mathrm{NLoS}}(t)$ captures the Doppler-induced fading due to UT mobility. Its entries are modeled as i.i.d. circularly symmetric (CS) zero-mean unit-variance complex Gaussian random processes whose temporal correlation follows Jakes' isotropic scattering model. Specifically, the power spectral density of each NLoS channel coefficient is given by

$$S_{h^{\mathrm{NLoS}}}(f) = \frac{1}{\pi f_D\sqrt{1-(f/f_D)^2}}, \quad |f| < f_D, \qquad (3)$$

where $f_D = v f_c/c$ is the maximum Doppler frequency, set by the UT velocity v, the carrier frequency f_c, and the speed of light c. The corresponding autocorrelation function is equal to the zero-order Bessel function of the first kind: $R_{h^{\mathrm{NLoS}}}(\tau) = J_0(2\pi f_D \tau)$.

CSI Feedback Delay. Before the GW can compute the precoder, channel estimates are acquired at the UTs and fed back over the return link. Denoting the total elapsed time as the CSI feedback delay τ_{CSI}, the precoder at time t is designed from the outdated channel state $\mathbf{H}(t-\tau_{\mathrm{CSI}})$. The degree of mismatch is governed by the NLoS coherence time, defined as the lag at which the autocorrelation first falls to one half:

$$T_c^{\mathrm{NLoS}} \triangleq \max\left\{\tau > 0 : R_{h^{\mathrm{NLoS}}}(\tau) \geq \tfrac{1}{2}\right\}. \qquad (4)$$

Throughout this work, results are reported as a function of the normalized delay $\tau_{\mathrm{CSI}}/T_c^{\mathrm{NLoS}}$. This normalization renders the conclusions independent of any specific combination of carrier frequency, UT velocity, or orbital altitude.

III. Precoder Design

Linear precoding designs the transmitted signal in order to limit the interference and efficiently deliver the desired data streams at each UT, by exploiting the available CSI and the spatial degrees of freedom offered by the multi-antenna system. For the system model introduced in the previous section, the design objective is the maximization of the achievable weighted sum-rate (WSR):

$$\max_{\{\mathbf{F}_k\}} \sum_{k=1}^{K} w_k R_k \qquad \text{s.t.} \qquad \sum_{k=1}^{K} \mathrm{tr}\big(\mathbf{F}_k\mathbf{F}_k^H\big) \leq P_T, \qquad (5)$$

where $w_k \geq 0$ is the weight assigned to user k and R_k denotes its achievable rate. In this work, equal weights are used for all users, so that the problem reduces to sum-rate (SR) maximization. Three linear precoders are considered: ZF, BD, and WMMSE.

Zero-Forcing Precoding [7]. ZF precoding cancels all inter-user and inter-stream interference by choosing the compound precoder as the right pseudo-inverse of the compound channel matrix. As a result, each transmitted stream is received over an independent virtual scalar Gaussian subchannel, which allows simple symbol-by-symbol detection. Transmit power is then allocated across these effective subchannels according to the well-known waterfilling solution [8]. The main advantage of ZF is its low complexity and transparent structure. However, when the channel directions are strongly correlated, the required channel inversion becomes inefficient, as lots of power is needed to maintain interference cancellation.

Block Diagonalization Precoding [9]. BD relaxes the ZF constraint by suppressing only the inter-user interference. For each user, the precoder is restricted to the null space of the channel matrices of all other users, such that the multi-user channel is decomposed into parallel K single-user MIMO subchannels. Each effective single-user (SU) channel is then diagonalized by means of its singular value decomposition (SVD), after which waterfilling is applied across the resulting eigenmodes. Compared with ZF, BD exploits the receive-side spatial processing more efficiently and achieves a higher sum rate, especially when each user is equipped with multiple receive antennas.

WMMSE Precoding [10]. Unlike ZF and BD, WMMSE does not enforce hard interference-nulling constraints. Instead, it targets the WSR objective directly through an equivalent weighted minimum mean square error formulation, hence the name. The resulting problem is solved iteratively by alternating between three closed-form updates: the receive combiners, the MSE weights, and the transmit precoder. Among the considered linear precoders, WMMSE provides the best SR performance and serves as the main performance benchmark in the remainder of the paper.

The next sections investigate how these designs degrade when the available CSI is outdated and to what extent this degradation can be mitigated through channel prediction.

IV. Outdated CSI Compensation

The precoders described in the previous section are designed under the implicit assumption that the CSI available at the GW matches the channel experienced at the moment of transmission. In a mobile satellite system, however, this assumption is violated by the combination of a time-varying channel due to UT mobility and the non-negligible feedback delay. As a result, the precoder is computed from an outdated channel realization and is therefore no longer matched to the instantaneous channel. To mitigate this mismatch, a low-complexity channel-prediction strategy is adopted at the UTs.

The key idea is to exploit the temporal correlation of the fading process. Instead of feeding back the most recent channel estimate, each UT predicts the channel that will be experienced after the feedback delay and reports this predicted channel to the GW. The precoder computation itself is left unchanged: only the CSI fed back to the GW is replaced by a prediction. Under the simplified Rician model of Section II, the time variation is entirely contained in the NLoS component, whose entries are spatially uncorrelated and share the same temporal correlation function. For this reason, prediction is performed element-wise using the same scalar predictor for each channel coefficient.

A pth-order autoregressive predictor is used. Let $\hat{h}_{k,(m,n)}[j]$ denote the estimate of the (m,n)-th coefficient of the channel matrix of user k at the j-th pilot instant. The predicted coefficient Δ pilot intervals ahead is then written as

$$\hat{h}_{k,(m,n)}[j+\Delta] = \sum_{\ell=0}^{p-1} \alpha_\ell \, \hat{h}_{k,(m,n)}[j-\ell], \tag{6}$$

where the prediction horizon is chosen such that $\Delta T_p = \tau_{\mathrm{CSI}}$, with T_p the pilot period. The predictor coefficients $\{\alpha_\ell\}$ are obtained from the Yule-Walker equations associated with the known channel autocorrelation function of the entire channel:

$$\mathbf{R}_h \, \boldsymbol{\alpha} = \mathbf{r}_h, \tag{7}$$

where the Toeplitz matrix $\mathbf{R}_h$ and vector $\mathbf{r}_h$ are constructed from $R_h(\tau) = \frac{\kappa}{\kappa+1} + \frac{1}{\kappa+1} J_0(2\pi f_D \tau)$. Since this autocorrelation is fully determined by the maximum Doppler frequency, the predictor coefficients can be computed offline once the model parameters have been selected.

The predictor is governed by two design parameters: the pilot period T_p, which sets the temporal resolution of the channel observations, and the context window $L \triangleq pT_p$, which controls how far back in time the predictor reaches. The model order $p = L/T_p$ follows directly. A tuning study, omitted here for brevity, selected a pilot rate of 6 messages per coherence period and a context window equal to one coherence period. These values are used in all subsequent simulations.

Three idealizations underlie the proposed scheme. First, channel-estimation errors at the UTs are neglected. Second, each UT is assumed to know the second-order statistics of its channel, which here reduces to knowledge of f_D. Third, the pilot period T_p is treated as a free design variable rather than being constrained by a pilot-overhead budget.

V. Simulation Results

This section evaluates the proposed framework through numerical Monte Carlo simulations. We first recall the relative performance of the considered linear precoders under ideal CSI, and then focus on the impact of CSI aging and the gain achieved by the proposed prediction strategy. Unless stated otherwise, the results correspond to an HTS system serving $K = 4$ UTs, with $N_T = 8$ feeds at the satellite and $N_R = 2$ receive antennas per UT. The Rician factor is set to $\kappa = 5$ dB, a common value for urban environments.

As a baseline, Fig. 2 compares the achievable sum rate of ZF, BD, and WMMSE precoding under a purely Rayleigh fading channel and perfect instantaneous CSI. BD consistently outperforms ZF by exploiting the receive-side spatial degrees of freedom of each UT, while WMMSE achieves the highest sum rate across the entire signal-to-noise ratio (SNR) range by trading off interference suppression against desired-signal power. The gap between BD and WMMSE disappears at high SNR, where the regularization in the WMMSE update vanishes and the design effectively reduces to BD. In the remainder of this section, the analysis focuses on WMMSE.

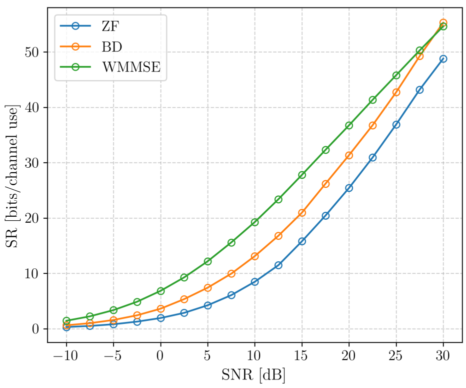


Fig. 2. Comparison of the achievable sum-rate for different precoding schemes under Rayleigh fading and perfect CSI in a $8 \times \{2,2,2,2\}$ system.

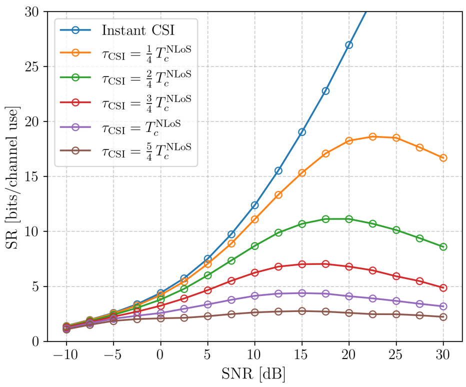
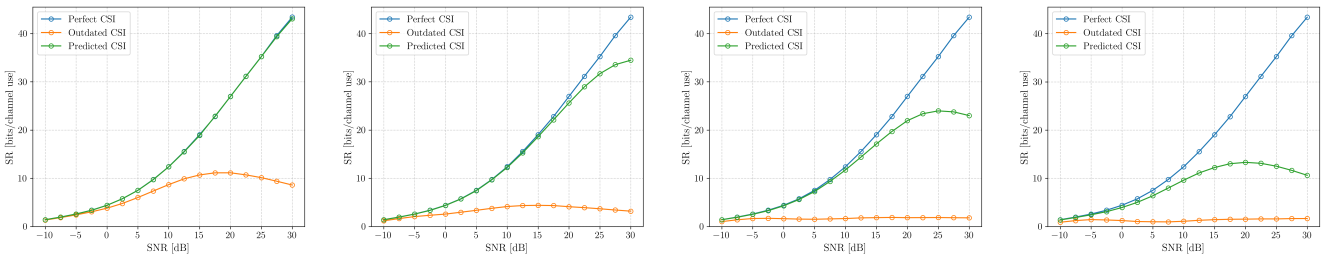


Fig. 3. Comparison of the achievable sum-rate using a WMMSE precoder for different values of the normalized CSI feedback delay.



(a) $\tau_{\text{CSI}} = \frac{1}{2}\, T_c^{\text{NLoS}}$ (b) $\tau_{\text{CSI}} = T_c^{\text{NLoS}}$ (c) $\tau_{\text{CSI}} = \frac{3}{2}\, T_c^{\text{NLoS}}$ (d) $\tau_{\text{CSI}} = 2\, T_c^{\text{NLoS}}$

Fig. 4. Comparison of the achievable sum-rate using a WMMSE precoder with and without the AR-based channel prediction, for different values of the normalized CSI feedback delay.

The conclusions for ZF and BD are in general qualitatively similar but quantitatively more pessimistic.

Fig. 3 reports the sum rate of WMMSE precoding under the Rician channel of Section II for several values of the normalized feedback delay, ranging from instantaneous CSI ($\tau_{\text{CSI}} = 0$) to $\tau_{\text{CSI}} = 5/4\, T_c^{\text{NLoS}}$. Three observations stand out. First, even when the CSI is perfectly instantaneous, the achievable rate lies noticeably below the Rayleigh reference of Fig. 2. This is a direct consequence of the rank-one LoS component of the Rician channel, which concentrates a substantial fraction of the channel energy along a single spatial direction and thereby limits the spatial multiplexing gain. Second, the precoder is highly sensitive to outdated CSI: even a delay of a quarter of the coherence time causes a significant performance loss at moderate-to-high SNR, and the rate degrades monotonically with an increasing feedback delay. Beyond the coherence time, the precoder fails to exploit the spatial dimension of the channel altogether, causing the sum-rate to collapse. The third observation is a characteristic feature of WMMSE precoding with outdated CSI. The sum-rate becomes non-monotonic in function of the SNR, as the precoder increasingly concentrates power along spatial directions that no longer correspond to the desired channel directions and thus actively generates inter-user interference at high SNR.

Fig. 4 evaluates the proposed channel-prediction strategy for four operating points $\tau_{\text{CSI}}/T_c^{\text{NLoS}} \in \{0.5, 1.0, 1.5, 2.0\}$, spanning moderate to severe CSI aging. In each subplot, the sum-rate curve obtained with the autoregressive (AR) predictor at the UTs is compared against the uncompensated case and the perfect-CSI reference. For moderate delays (Figs. 4a and 4b), the predictor recovers most of the performance loss and brings the achievable rate close to the perfect-CSI upper bound across the entire SNR range. The non-monotonic high-SNR behaviour observed without compensation disappears, as the predicted CSI is accurate enough to keep the precoder aligned with the true channel directions. For larger delays (Figs. 4c and 4d), the predictor still provides a substantial improvement, but a residual gap with respect to the perfect-CSI reference becomes visible. Overall, these results show that a low-complexity AR predictor, requiring only knowledge of the channel autocorrelation and no modification of the GW-side precoder, is sufficient to restore a large fraction of the performance lost to outdated CSI in the operating regime of practical interest, under the considered channel model.

VI. Future Work

The simplified Rician model adopted in this paper neglects spatial correlation between channel entries and assumes perfect pre-compensation of the LoS Doppler shift. A natural next step is therefore to extend the analysis to more realistic ray-based satellite channel [11] model that can explicitly capture both effects. First results obtained in the dissertation suggest that the proposed AR-based prediction strategy remains effective in this setting. At the same time, the realistic model reveals an important structural limitation: the channel of each user is effectively rank one, so that each UT can support at most one spatial stream. Future work should investigate the impact of this rank deficiency on different precoding strategies and develop prediction methods that exploit the joint spatial-temporal correlation structure of the channel.

VII. Conclusion

This paper studied the impact of outdated CSI on linear precoding for mobile satellite systems and proposed a low-complexity compensation strategy based on channel prediction at the UTs. Starting from a multi-user MIMO downlink model with Rician fading, three linear precoders were considered: ZF, BD, and WMMSE. The simulations showed that all of them are highly sensitive to CSI aging.

To mitigate this degradation, an element-wise AR predictor was introduced at the UTs. The numerical results showed that this simple strategy restores a large fraction of the performance lost to outdated CSI. For feedback delays up to about one coherence period of the NLoS component, the predicted CSI performance approaches the instantaneous CSI reference very closely. Even for more severe delay values, the prediction remains clearly beneficial and suppresses much of the high-SNR degradation observed without compensation.

A practical example illustrates the relevance of these results. Consider an mobile satellite system (MSS) forward link in the L-band at $f_c = 1.5$ GHz, with a LEO satellite at an altitude of 550 km and a UT velocity of 20 km/h. Since the satellite's trajectory is known with high accuracy, its Doppler shift is deterministic and can be pre-compensated [12]. Under the model adopted in this work, the resulting total CSI feedback delay is approximately $\tau_{\mathrm{CSI}} = 7.34$ ms, while the coherence time of the NLoS component is about $T_c^{\mathrm{NLoS}} = 8.71$ ms. This yields a ratio of $\tau_{\mathrm{CSI}}/T_c^{\mathrm{NLoS}} \approx 0.84$, which lies in the regime where the proposed predictor is most effective.

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Symbols and Notations

Abbreviations

acf	autocorrelation function
AR	autoregressive
AWGN	additive white Gaussian noise
BS	base station
BD	block diagonalization
BER	bit error rate
CS	circularly symmetric
CSI	channel state information
DPC	dirty paper coding
EVD	eigenvalue decomposition
GEO	geostationary earth orbit
GW	gateway
HTS	high-throughput satellite
IBR	information bit rate
i.i.d.	independent and identically distributed
KKT	Karush-Kuhn-Tucker
LEO	low earth orbit
LoS	line-of-sight
LSV	left singular vector
MI	mutual information
MIMO	multiple-input multiple-output
MMSE	minimum mean squared error

MSE	mean squared error
MSS	mobile satellite system
MU	multi-user
NLoS	non-line-of-sight
pdf	probability density function
PSD	power spectral density
QoS	Quality–of–Service
RTT	round-trip time
RX	receiver
SINR	signal-to-interference-plus-noise ratio
SISO	single-input single-output
SNR	signal-to-noise ratio
SR	sum rate
SU	single-user
SVD	singular value decomposition
THP	Tomlinson–Harashima precoding
TX	transmitter
UT	user terminal
WMSE	weighted mean squared error
WMMSE	weighted minimum mean squared error
WSR	weighted sum rate
ZF	zero–forcing

System Parameters

K	Number of UTs
N_T	Number of transmit antennas at the BS
N_R	Number of receive antennas at each UT
N_S	Number of data streams per UT
P_t	Total transmit power
N_0	Noise power on each receive antenna

Symbols

We limit ourselves to the notation for a MU-MIMO downlink system, as this is the most general scenario discussed in this dissertation. The notation for a SU-MIMO system can be obtained by setting $K = 1$ and omitting the user index k.

System Components

$h_{k,(m,n)} \in \mathbb{C}$ — channel coefficient from the n-th transmit antenna of the BS to the m-th receive antenna of user k

$\mathbf{H}_k \in \mathbb{C}^{N_R \times N_T}$ — channel matrix for user k

$\mathbf{H} = \left[\mathbf{H}_1^T \;\ldots\; \mathbf{H}_K^T\right]^T \in \mathbb{C}^{KN_R \times N_T}$ — compound channel matrix

$n_{k,m} \in \mathbb{C}$ — additive noise sample on the m-th receive antenna of user k

$\mathbf{n}_k = [n_{k,1} \;\ldots\; n_{k,N_R}]^T \in \mathbb{C}^{N_R}$ — noise vector for user k

$\mathbf{n} = [\mathbf{n}_1^T \;\ldots\; \mathbf{n}_K^T]^T \in \mathbb{C}^{KN_R}$ — compound noise vector

$\mathbf{f}_{k,i} \in \mathbb{C}^{N_T}$ — precoder vector for the i-th data stream intended for user k

$\mathbf{F}_k = \left[\mathbf{f}_{k,1} \;\ldots\; \mathbf{f}_{k,N_S}\right] \in \mathbb{C}^{N_T \times N_S}$ — precoder matrix for user k

$\mathbf{F} = \left[\mathbf{F}_1 \;\ldots\; \mathbf{F}_K\right] \in \mathbb{C}^{N_T \times KN_S}$ — compound precoder matrix

$\mathbf{g}_{k,i} \in \mathbb{C}^{N_R}$ — combiner vector for the i-th data stream intended for user k

$\mathbf{G}_k = \left[\mathbf{g}_{k,1}^T \;\ldots\; \mathbf{g}_{k,N_S}^T\right]^T \in \mathbb{C}^{N_S \times N_R}$ — combiner matrix for user k

$\mathbf{G} = \mathrm{diag}\big(\mathbf{G}_1, \;\ldots,\; \mathbf{G}_K\big) \in \mathbb{C}^{KN_S \times KN_R}$ — compound combiner matrix

System Signals

$a_{k,i} \in \mathbb{C}$ — data symbol in the i-th data stream for user k

$\mathbf{a}_k = [a_{k,1} \;\ldots\; a_{k,N_S}]^T \in \mathbb{C}^{N_S}$ — data symbol vector for user k

$\mathbf{a} = [\mathbf{a}_1^T \;\ldots\; \mathbf{a}_K^T]^T \in \mathbb{C}^{KN_S}$ — compound data symbol vector

$x_n \in \mathbb{C}$ — transmit signal on the n-th transmit antenna at the BS

$\mathbf{x} = [x_1 \;\ldots\; x_{N_T}]^T \in \mathbb{C}^{N_T}$ — transmit signal vector $\left(\mathbf{x} = \mathbf{F}\,\mathbf{a} = \sum_{k=1}^{K}\sum_{i=1}^{N_S} \mathbf{f}_{k,i}\, a_{k,i}\right)$

$y_{k,m} \in \mathbb{C}$ — receive signal on the m-th receive antenna of user k

$\mathbf{y}_k = [y_{k,1} \;\ldots\; y_{k,N_R}]^T \in \mathbb{C}^{N_R}$ — receive signal vector at user k $\left(\mathbf{y}_k = \mathbf{H}_k\,\mathbf{x} + \mathbf{n}_k\right)$

$\mathbf{y} = [\mathbf{y}_1^T \;\ldots\; \mathbf{y}_K^T]^T \in \mathbb{C}^{KN_R}$ — compound receive signal vector $\left(\mathbf{y} = \mathbf{H}\,\mathbf{x} + \mathbf{n}\right)$

$z_{k,i} \in \mathbb{C}$ — decision variable in the i-th data stream at user k

$\mathbf{z}_k = [z_{k,1} \ \dots \ z_{k,N_S}]^T \in \mathbb{C}^{N_S}$ — decision variable vector at user k $(\mathbf{z}_k = \mathbf{G}_k \, \mathbf{y}_k)$

$\mathbf{z} = [\mathbf{z}_1^T \ \dots \ \mathbf{z}_K^T]^T \in \mathbb{C}^{KN_S}$ — compound decision variable vector $(\mathbf{z} = \mathbf{G} \, \mathbf{y})$

Part I

Background Material

Chapter 1

Communication Systems

This chapter establishes the system models and information-theoretic framework that underlie the analysis presented throughout this work. We begin by describing the general architecture of a digital communication system and then specialize to the system model adopted in this work. Finally, we review the fundamental information-theoretic concepts required to analyze the performance of these systems.

1.1 Digital Communication Systems

A general digital communication system transfers digital information from a source to a destination over a physical transmission medium, referred to as the channel. The corresponding block diagram is shown in Fig. 1.1.

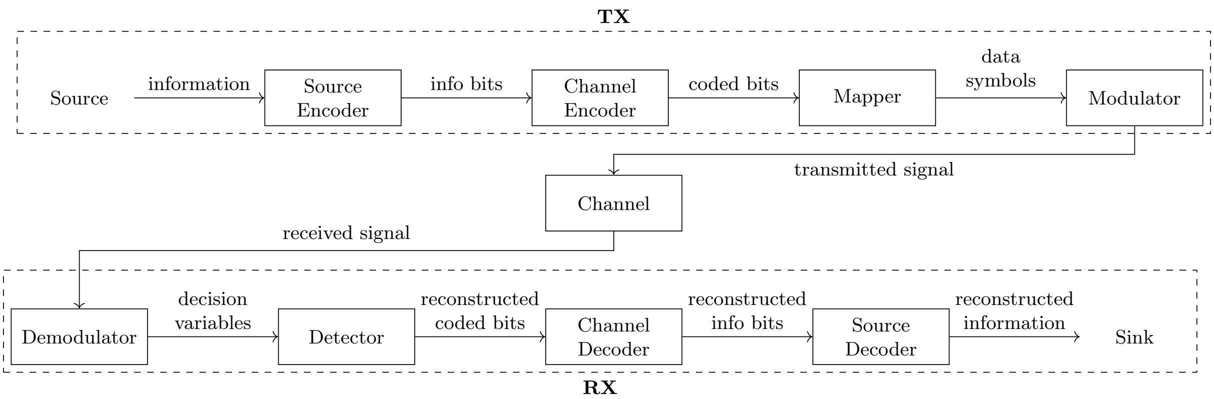


Figure 1.1: Block diagram of a general digital communication system.

At the transmitter (TX), the information sequence is first processed by a source encoder to remove redundancy, followed by a channel encoder that introduces controlled redundancy to enable error detection. The resulting coded bit sequence is partitioned into blocks of m_c bits, each of which is mapped onto a complex-valued symbol drawn from a constellation $\mathcal{X}$ of size 2^{m_c}. The modulator then converts the symbol sequence into a baseband signal suitable for transmission.

During transmission, the channel distorts the signal due to attenuation, multipath propagation, and additive noise, such that the received signal differs from the transmitted signal.

At the receiver (RX), the demodulator converts the received signal into a sequence of decision variables. These are processed by the detector to obtain estimates of the originally transmitted symbols, which are then mapped back to coded bits. Finally, the channel and source decoders reconstruct the original information sequence.

In this work, we consider a simplified version of this general architecture in order to focus on the aspects most relevant to our study. Source coding and decoding are omitted, and the information is assumed to be available as a sequence of independent and identically distributed (i.i.d.) bits without prior compression or digitization. Channel coding and decoding are also not considered, such that the analysis is restricted to uncoded transmission.

The transmission medium is assumed to be a wireless channel affected by fading and additive white Gaussian noise. Furthermore, the channel is considered to be frequency-flat, meaning that its response remains constant over the entire signal bandwidth. The specific channel models used in this work are introduced in the relevant chapters.

1.2 MIMO Communication Systems

When both the TX and the RX are equipped with multiple antennas, the system is referred to as a MIMO system.

Compared to a single-antenna link, the multiple antennas provide additional spatial degrees of freedom that can be exploited to increase data throughput, improve reliability, or achieve a trade-off between both. Spatial multiplexing techniques enable significant capacity gains by transmitting multiple independent data streams simultaneously over different antennas. A well-known example is the BLAST architecture [1]. Alternatively, spatial diversity can be achieved through space-time coding schemes, which distribute symbols across both space and time to improve robustness against fading. A classical example is the Alamouti scheme [2]. Throughout this work, the focus is on spatial multiplexing and, in particular, on how the simultaneous transmission of multiple data streams over a shared channel can be made most efficient through precoding.

1.2.1 SU-MIMO System Model

In the point-to-point case, where a single TX with N_T antennas communicates with a single RX with N_R antennas, the system is called a single-user (SU) MIMO system. A block diagram is shown in Fig. 1.2.

The discrete-time complex baseband input-output relation of a SU-MIMO Gaussian channel is characterized by

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \tag{1.1}$$

where $\mathbf{x} \in \mathbb{C}^{N_T}$ is the transmitted signal vector, $\mathbf{y} \in \mathbb{C}^{N_R}$ is the received signal vector, $\mathbf{H} \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix, and $\mathbf{n} \in \mathbb{C}^{N_R}$ is the additive noise vector. The (n_r, n_t)-th entry of $\mathbf{H}$ represents the complex channel coefficient between transmit antenna n_t and receive

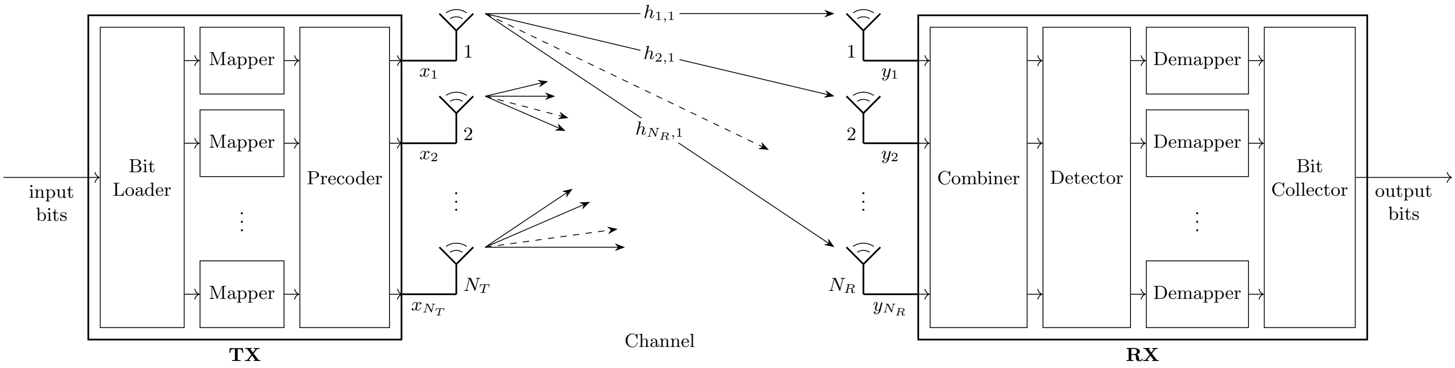


Figure 1.2: Block diagram of a SU-MIMO system.

antenna n_r. The noise vector is circularly symmetric (CS) complex Gaussian distributed, i.e. $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, N_0\,\mathbf{I}_{N_R})$, such that its entries are spatially independent, each with an expected power of N_0. The time index is omitted for brevity.

At the TX, the information bits are divided into N_S streams and mapped onto data symbols, which are collected in the vector $\mathbf{a} \in \mathbb{C}^{N_S \times 1}$. Then, the transmitted signal is obtained by mapping the N_S data symbol streams onto the N_T transmit antennas according to a linear precoding scheme, represented by a matrix $\mathbf{F} \in \mathbb{C}^{N_T \times N_S}$. The transmitted signal can therefore be expressed as

$$\mathbf{x} = \mathbf{F}\,\mathbf{a}. \tag{1.2}$$

The data symbols are assumed to be zero-mean, mutually uncorrelated, and normalized, such that $\mathbb{E}\big[\mathbf{a}\mathbf{a}^{\mathsf{H}}\big] = \mathbf{I}_{N_S}$. Throughout this work, the power allocation is assumed to be incorporated into the precoding matrix, unless explicitly stated otherwise. Under this convention, the total transmit power is constrained by

$$\mathbb{E}\big[\|\mathbf{x}\|^2\big] = \mathrm{tr}\Big(\mathbf{F}^{\mathsf{H}}\,\mathbf{F}\Big) \le P_t. \tag{1.3}$$

At the RX, the N_R received signals are linearly combined to obtain the decision-variable vector $\mathbf{z} \in \mathbb{C}^{N_S \times 1}$. The combiner is represented by a matrix $\mathbf{G} \in \mathbb{C}^{N_S \times N_R}$. The decision variables can therefore be expressed as

$$\mathbf{z} = \mathbf{G}\,\mathbf{y}. \tag{1.4}$$

Based on $\mathbf{z}$, the detector forms estimates of the transmitted data symbols, which are then mapped back to bits.

Substituting (1.2) into (1.1) and applying (1.4) yields the relation between the transmitted data symbols and the decision variables:

$$\mathbf{z} = (\mathbf{G}\,\mathbf{H}\,\mathbf{F})\ \mathbf{a} + \mathbf{G}\,\mathbf{n}, \tag{1.5a}$$

$$= \mathbf{T}\,\mathbf{a} + \mathbf{n}_{\mathbf{G}}, \tag{1.5b}$$

where $\mathbf{T} \in \mathbb{C}^{N_S \times N_S}$ is the MIMO transfer matrix and $\mathbf{n}_{\mathbf{G}}$ represents spatially correlated Gaussian noise. It is evident that, whenever the transfer matrix is not diagonal, the data symbols of different streams interfere with each other. The design of the precoding and combining matrices is therefore essential to suppress this inter-stream interference and improve overall system performance.

1.2.2 MU-MIMO System Model

When multiple receivers are served simultaneously by a single multi-antenna transmitter, the system is referred to as a downlink MU MIMO system. In this context, the transmitter is called the base station (BS) and the individual receivers are referred to as user terminals (UTs). A block diagram is shown in Fig. 1.3.

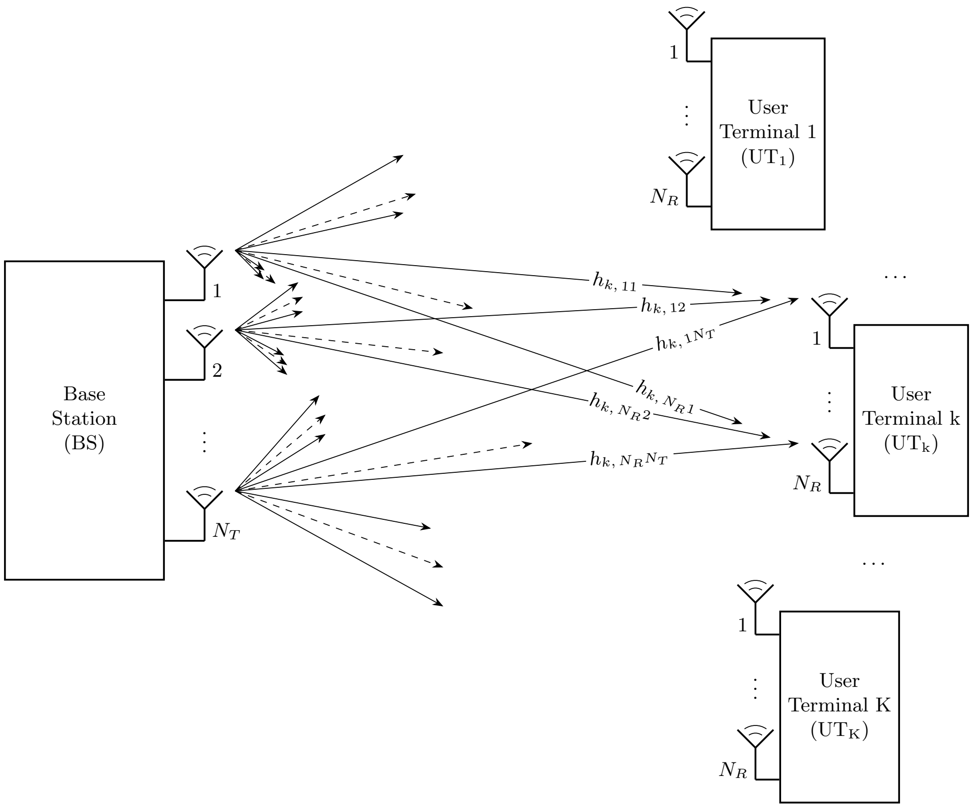


Figure 1.3: Block diagram of a MU-MIMO system.

Throughout this work, the BS is assumed to be equipped with more antennas than all UTs combined, and each UT is assumed to receive at most as many data streams as it has receive antennas. For notational convenience, all UTs are assumed to receive the same number of data streams, denoted by N_S. In practical scenarios where this does not hold, N_S is defined as the maximum number of data streams intended for any single UT. The data symbol vectors intended for a UT receiving fewer streams are padded with zeros.

Analogous to the SU-MIMO case, the communication between the BS and a certain $\mathrm{UT_k}$ at a particular time instant is characterized by

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{x} + \mathbf{n}_k, \qquad k = 1, \ldots, K, \tag{1.6}$$

where $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix that describes the link between the BS and $\mathrm{UT_k}$.

The transmitted signal vector $\mathbf{x} \in \mathbb{C}^{N_T}$ is now a function of the data symbol vectors intended for different UTs. The BS assigns a dedicated precoding scheme $\mathbf{F}_k \in \mathbb{C}^{N_T \times N_S}$ to the data symbol vector $\mathbf{a}_k$ intended for $\mathrm{UT_k}$, and transmits the superposition of all precoded streams:

$$\mathbf{x} = \sum_{k=1}^{K} \mathbf{F}_k \, \mathbf{a}_k. \tag{1.7}$$

Since the data symbols intended for different UTs are assumed to be uncorrelated, the total transmit power constraint can now be formulated as

$$\mathbb{E}\left[\|\mathbf{x}\|^2\right] = \sum_{k=1}^{K} \mathrm{tr}\left(\mathbf{F}_k^{\mathsf{H}} \, \mathbf{F}_k\right) \leq P_t. \tag{1.8}$$

Applying the combining scheme $\mathbf{G}_k \in \mathbb{C}^{N_S \times N_R}$ at $\mathrm{UT_k}$ to the received signal vector and substituting (1.7) into (1.6) yields

$$\mathbf{z}_k = \underbrace{(\mathbf{G}_k \, \mathbf{H}_k \, \mathbf{F}_k) \;\, \mathbf{a}_k}_{\text{desired signal}} + \underbrace{\sum_{k'=1,\, k' \neq k}^{K} (\mathbf{G}_k \mathbf{H}_k \mathbf{F}_{k'}) \;\, \mathbf{a}_{k'}}_{\text{inter-user interference}} + \underbrace{\mathbf{G}_k \, \mathbf{n}_k}_{\text{noise}}. \tag{1.9}$$

The key difference from the SU-MIMO scenario is the presence of inter-user interference. Each UT receives not only its desired signal but also the signals intended for all other UTs. Suppressing this interference through precoder design at the BS is an important challenge in MU-MIMO systems.

The overall input-output relation of the MU-MIMO Gaussian channel can be written compactly as

$$\mathbf{z} = (\mathbf{G}\,\mathbf{H}\,\mathbf{F}) \;\, \mathbf{a} + \mathbf{G}\,\mathbf{n}, \quad \text{with} \tag{1.10}$$

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_1 \\ \vdots \\ \mathbf{z}_K \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \mathbf{G}_1 & & \\ & \ddots & \\ & & \mathbf{G}_K \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_1 \\ \vdots \\ \mathbf{H}_K \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_1 & \dots & \mathbf{F}_K \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_K \end{bmatrix}.$$

where $\mathbf{G} \in \mathbb{C}^{KN_S \times KN_R}$ denotes the compound combining matrix, $\mathbf{H} \in \mathbb{C}^{KN_R \times N_T}$ the compound channel matrix, and $\mathbf{F} \in \mathbb{C}^{N_T \times KN_S}$ the compound precoding matrix. The block-diagonal structure of $\mathbf{G}$ indicates that each UT processes only its own received signal vector, i.e., no cooperation is assumed between different UTs.

1.3 Information Theory Concepts

In this section, we introduce the information-theoretic concepts needed for the analysis of MIMO communication systems. We first discuss mutual information, then define the achievable rate and capacity for SU-MIMO channels, and finally extend these concepts to the MU scenario.

1.3.1 Mutual Information

Consider a single use of a SU-MIMO Gaussian channel, as described by (1.1). The transmitter generates a random vector $\mathbf{x} \in \mathbb{C}^{N_T}$ according to the input distribution with probability density function (pdf) $p(\mathbf{x})$, while the receiver observes the random vector $\mathbf{y} \in \mathbb{C}^{N_R}$. The relation between the channel input $\mathbf{x}$ and the channel output $\mathbf{y}$ is probabilistic and is fully characterized by the channel law, described by the conditional pdf $p(\mathbf{y} \mid \mathbf{x})$.

The number of information bits transmitted per channel use is referred to as the information bit rate (IBR). The mutual information (MI) between $\mathbf{x}$ and $\mathbf{y}$ quantifies how much information the received vector $\mathbf{y}$ provides about the transmitted vector $\mathbf{x}$. In other words, it represents the average reduction in the uncertainty about $\mathbf{x}$ obtained from observing $\mathbf{y}$. It is given by

$$I(\mathbf{x};\mathbf{y}) = \mathbb{E}\left[\log_2\left(\frac{p(\mathbf{y} \mid \mathbf{x})}{p(\mathbf{y})}\right)\right], \tag{1.11}$$

where the expectation is taken with respect to the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ and $p(\mathbf{y})$ is the marginal distribution of $\mathbf{y}$.

According to Shannon's channel coding theorem [3], reliable communication is possible if the transmission IBR is strictly smaller than the MI between $\mathbf{x}$ and $\mathbf{y}$. Here, reliable communication means that the decoding error probability can be made arbitrarily small by using sufficiently long channel codes.

1.3.2 Achievable Rate and Capacity

For a given input distribution $p(\mathbf{x})$ and channel law $p(\mathbf{y} \mid \mathbf{x})$, any IBR r smaller than the MI is said to be achievable. The MI is thus a supremum (least upper bound) on the achievable rate.

The capacity C of a SU-MIMO channel is the largest achievable rate. Equivalently, it is the highest spectral efficiency for which reliable communication is possible. It is defined as the maximum MI between $\mathbf{x}$ and $\mathbf{y}$ over all possible input distributions $p(\mathbf{x})$:

$$C = \max_{p(\mathbf{x})} I(\mathbf{x};\mathbf{y}). \tag{1.12}$$

Typically, the maximization is subject to the transmit power constraint, as formulated in (1.3).

In [4], the capacity for a SU-MIMO Gaussian communication system is derived. First, it is shown that under the total transmit power constraint, the MI is maximized in case of Gaussian signaling, i.e. the channel input $\mathbf{x}$ is CS complex Gaussian distributed. Then, the mutual information between $\mathbf{x}$ and $\mathbf{y}$ for a given covariance matrix $\mathbf{C}_{\mathrm{x}}$ is obtained as

$$I(\mathbf{x};\mathbf{y}) = \log_2\left(\det\left(\mathbf{I}_{N_R} + \mathbf{C}_{\mathrm{n}}^{-1}\,\mathbf{H}\,\mathbf{C}_{\mathrm{x}}\,\mathbf{H}^{\mathsf{H}}\right)\right). \tag{1.13}$$

The capacity then results from maximizing (1.13) over all positive semidefinite covariance matrices $\mathbf{C}_\mathrm{x}$ that satisfy the transmit power constraint:

$$\begin{aligned} C = \max_{\mathbf{C}_\mathrm{x}} \ & \log_2 \left(\det\left(\mathbf{I}_{N_R} + \mathbf{C}_\mathrm{n}^{-1}\,\mathbf{H}\,\mathbf{C}_\mathrm{x}\,\mathbf{H}^\mathsf{H}\right)\right) \\ \text{s.t.} \quad & \mathbf{C}_\mathrm{x} \succeq 0, \quad \mathrm{tr}(\mathbf{C}_\mathrm{x}) \le P_t. \end{aligned} \tag{1.14}$$

1.3.3 Rate Region and Sum Capacity

In contrast to the SU scenario, where the IBR is a scalar r, a MU-MIMO channel is characterized by a rate tuple $(r_1, \ldots, r_K)$, where each element r_k denotes the IBR of a single UT. The rate region $\mathcal{R}$ is the set of all rate tuples whose individual IBRs are simultaneously achievable under the given system constraints. The rate region is a convex set, since any convex combination of achievable rate tuples is itself achievable through time-sharing between the corresponding transmission strategies. An example rate region for $K = 2$ users is shown in Fig. 1.4.

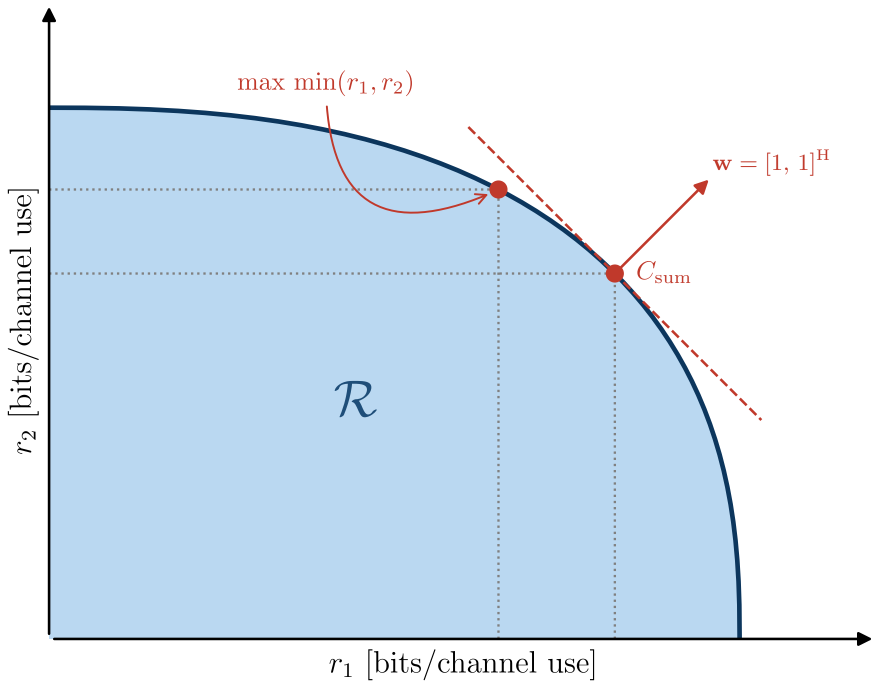


Figure 1.4: Example of a rate region $\mathcal{R}$ for a MU-MIMO system with $K = 2$ users. The sum capacity point is the boundary point whose outward normal is the all-ones vector. The max–min point is given by the intersection of the boundary with the line $r_1 = r_2$. It corresponds to a fair operating point where both users achieve the same IBR.

The rate region captures the trade-offs inherent to multi-user transmission: increasing the IBR of one user may reduce the rates achievable by others due to interference and shared resources. It is therefore desirable to operate on the boundary of the rate region, where no user's rate can be increased without reducing that of another.

The sum capacity C_{sum} is the maximum achievable sum rate (SR), i.e., the sum of all individual IBRs:

$$C_{\text{sum}} = \max_{(r_1,\dots,r_K)\in\mathcal{R}} \sum_{k=1}^{K} r_k. \tag{1.15}$$

More generally, we define the weighted sum rate (WSR) as

$$\text{WSR} = \sum_{k=1}^{K} w_k\, r_k, \tag{1.16}$$

where $w_k \geq 0$ is the weight assigned to UT_k. The full boundary of the rate region can then be traced by maximizing the WSR over all achievable rate tuples for varying weight vectors $\mathbf{w} = (w_1, \dots, w_K)$. The sum capacity corresponds to the special case where all weights are equal to one. The maximization in both cases is typically subject to the total transmit power constraint (1.8).

In [5, 6, 7], the sum capacity of the MU-MIMO Gaussian channel is derived by exploiting the uplink–downlink duality: the downlink broadcast channel and the uplink multiple-access channel yield the same rate region under the same total transmit power constraint. It therefore suffices to solve the uplink problem, which is more tractable.

In the uplink channel, all K UTs transmit simultaneously, with UT_k sending a vector $\mathbf{x}_k^{\text{up}} \in \mathbb{C}^{N_R}$ to the BS. The MI between the transmitted vectors $\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}$ and the received vector $\mathbf{y}$, given input covariance matrices $\mathbf{C}_{\text{x},k}$, is defined as

$$I(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}; \mathbf{y}) = \mathbb{E}\left[\log_2\left(\frac{p(\mathbf{y} \mid \mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}})}{p(\mathbf{y})}\right)\right], \tag{1.17}$$

where the expectation is taken with respect to the joint pdf of $(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}})$ and $\mathbf{y}$, and $p(\mathbf{y})$ is the marginal pdf of $\mathbf{y}$. It is further obtained as

$$I(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}; \mathbf{y}) = \log_2\left(\det\left(\mathbf{I}_{N_T} + \mathbf{C}_\text{n}^{-1} \sum_{k=1}^{K} \mathbf{H}_k\, \mathbf{C}_{\text{x},\,k}^{\text{up}}\, \mathbf{H}_k^{\mathsf{H}}\right)\right). \tag{1.18}$$

The uplink SR is achievable if and only if the sum of the IBRs of all UTs is strictly smaller than the MI as given in (1.18). Therefore, the sum capacity of the downlink channel is obtained by maximizing (1.18) over all positive semidefinite covariance matrices satisfying the total transmit power constraint and applying the duality result:

$$\begin{aligned} C_{\text{sum}} = \max_{\mathbf{C}_{\text{x},\,1}^{\text{up}},\dots,\mathbf{C}_{\text{x},\,K}^{\text{up}}} \;& \log_2\left(\det\left(\mathbf{I}_{N_T} + \mathbf{C}_\text{n}^{-1} \sum_{k=1}^{K} \mathbf{H}_k\, \mathbf{C}_{\text{x},\,k}^{\text{up}}\, \mathbf{H}_k^{\mathsf{H}}\right)\right) \\ \text{s.t.} \quad & \sum_{k=1}^{K} \operatorname{tr}\left(\mathbf{C}_{\text{x},\,k}^{\text{up}}\right) \leq P_t \quad \text{and} \quad \mathbf{C}_{\text{x},\,k}^{\text{up}} \succeq 0 \quad \forall k = 1, \dots, K. \end{aligned} \tag{1.19}$$

Chapter 2

Precoding for SU-MIMO systems

This chapter develops linear precoding strategies for SU-MIMO systems under a total transmit–power constraint. First, we analyze the capacity in greater detail. Next, a generalized WMMSE design framework for the joint optimization of the precoder and combiner is derived. Finally, the performance is evaluated through Monte-Carlo simulations for different system configurations.

2.1 Capacity Achieving Precoding

Consider the SU-MIMO system model introduced in Section 1.2.1 together with the associated capacity optimization problem in (1.14). We first derive a more explicit expression for the channel capacity and then show that this capacity can be achieved by means of linear precoding. Parts of this section follow a structure similar to that in Chapter 3 of [8].

2.1.1 SU-MIMO Capacity

The singular value decomposition (SVD) [9] of the channel matrix yields

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{H}}, \tag{2.1}$$

where $\mathbf{U} \in \mathbb{C}^{N_R \times N_R}$ and $\mathbf{V} \in \mathbb{C}^{N_T \times N_T}$ are unitary matrices, and $\mathbf{\Sigma} \in \mathbb{R}^{N_R \times N_T}$ is a rectangular diagonal matrix which contains the singular values of $\mathbf{H}$ on its main diagonal in decreasing order. The columns of $\mathbf{U}$ and $\mathbf{V}$ contain the left and right singular vectors of $\mathbf{H}$, respectively.

Now define the transformed variables $\tilde{\mathbf{y}} = \mathbf{U}^{\mathsf{H}}\mathbf{y}$, $\tilde{\mathbf{x}} = \mathbf{V}^{\mathsf{H}}\mathbf{x}$, and $\tilde{\mathbf{n}} = \mathbf{U}^{\mathsf{H}}\mathbf{n}$. Since multiplication by a unitary matrix preserves the distribution of CS complex Gaussian noise, $\tilde{\mathbf{n}}$ has the same distribution as $\mathbf{n}$. Moreover, because $\mathbf{U}$ and $\mathbf{V}$ are unitary, the transformations between $(\mathbf{x}, \mathbf{y})$ and $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ are bijective. The original channel from (1.1) can therefore be represented equivalently as

$$\tilde{\mathbf{y}} = \mathbf{\Sigma}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}, \tag{2.2}$$

which is obtained by applying the SVD of the channel matrix and performing a change of variables. The channel in (2.2) is equivalent to the original channel in the sense that the transformation preserves the mutual information between input and output. Consequently, both channels have the same capacity.

The channel matrix $\mathbf{H}$ has exactly as many non-zero singular values $\{\sigma_i\}$ as its rank, denoted as R_H. Written component-wise, the equivalent channel becomes

$$\tilde{y}_i = \sigma_i\,\tilde{x}_i + \tilde{n}_i \qquad \text{for} \quad i = 1, \ldots, R_H, \tag{2.3}$$

with $\tilde{y}_i = \tilde{n}_i$ for $R_H < i \leq N_R$. The equivalent channel decomposes into R_H independent parallel scalar Gaussian subchannels, each with its own channel gain σ_i. These subchannels and their gains are referred to as eigen subchannels and eigenmodes, respectively. Since the components $\tilde{x}_i$ for $R_H < i \leq N_T$ have no effect on the output, they carry no information and can be set to zero.

For each eigen subchannel, the mutual information (MI) is maximised by a zero-mean, CS complex Gaussian input [4]. Under this choice, the optimization problem (1.14) simplifies to

$$\begin{aligned} C = &\max_{\{\mathbb{E}[|\tilde{x}_i|^2]\}} \sum_{i=1}^{R_H} \log_2\left(1 + \mathbb{E}\left[|\tilde{x}_i|^2\right]\,\frac{\sigma_i^2}{N_0}\right) \\ \text{s.t.} \quad &\sum_{i=1}^{R_H} \mathbb{E}\left[|\tilde{x}_i|^2\right] \leq P_t. \end{aligned} \tag{2.4}$$

This maximization problem is solved using the Lagrange multiplier method (see Appendix A). The variances of the channel input need to be chosen according to the waterfilling solution [10]:

$$\mathbb{E}\left[\|\tilde{x}_i\|^2\right] = \left(\mu_{\mathrm{w}} - \frac{N_0}{\sigma_i^2}\right)_+, \tag{2.5}$$

where the water level μ_{w} is found iteratively to satisfy the transmit power constraint with equality.

The resulting capacity is given by

$$C = \sum_{i=1}^{R_H} \log_2\left(\max(1,\, \mu_{\mathrm{w}}\frac{\sigma_i^2}{N_0})\right). \tag{2.6}$$

2.1.2 Optimal Precoder Design

In this section, we design a linear precoder and combiner that together achieve the capacity of the SU-MIMO channel. We assume that the data symbol vector is CS complex Gaussian distributed and spatially uncorrelated, and restrict ourselves to the case where the number of data streams is not fixed in advance, but can be chosen freely as part of the design.

For an arbitrary precoding matrix $\mathbf{F}$ and combining matrix $\mathbf{G}$, the data processing inequality [11] yields $I(\mathbf{x};\mathbf{y}) \geq I(\mathbf{a};\mathbf{z})$, where the MI between the transmitted data symbols $\mathbf{a}$ and the decision variables $\mathbf{z}$ is given by:

$$I(\mathbf{a};\mathbf{z}) = \log_2\Big(\det\Big(\mathbf{I}_{N_s} + (\mathbf{G}\,\mathbf{C}_\mathrm{n}\,\mathbf{G}^\mathsf{H})^{-1}\,(\mathbf{GHF})\,\mathbf{C}_\mathrm{a}\,(\mathbf{GHF})^\mathsf{H}\Big)\Big) \tag{2.7a}$$

$$= \log_2\Big(\det\Big(\mathbf{I}_{N_s} + \tfrac{1}{N_0}(\mathbf{GG}^\mathsf{H})^{-1}\,(\mathbf{GHF})\,(\mathbf{GHF})^\mathsf{H}\Big)\Big). \tag{2.7b}$$

Expression (2.7a) follows directly from (1.13) and the structural similarity between (1.1) and (1.5). Expression (2.7b) then follows from (2.7a) upon substituting $\mathbf{C}_\mathrm{a} = \mathbf{I}_{N_s}$ and $\mathbf{C}_\mathrm{n} = N_0\mathbf{I}_{N_R}$.

Motivated by the capacity derivation in the previous subsection, we choose the columns of $\mathbf{F}$ and the rows of $\mathbf{G}$ as the R_H most dominant right and left singular vectors of $\mathbf{H}$, respectively. These are precisely the singular vectors associated with the non-zero singular values of $\mathbf{H}$. The number of data streams is therefore chosen equal to the rank of the channel matrix. In addition, the precoder incorporates the power allocation through a diagonal matrix $\mathbf{P}$. More specifically, we set

$$\mathbf{F} = \mathbf{V}_{N_s}\sqrt{\mathbf{P}} \qquad \text{and} \qquad \mathbf{G} = \mathbf{U}_{N_s}^\mathsf{H}. \tag{2.8}$$

It follows from (2.8) that $I(\mathbf{a};\mathbf{z}) = I(\mathbf{x};\mathbf{y})$ for an appropriate choice of the power allocation matrix $\mathbf{P}$. This result can be verified in a brute-force way by substituting the particular precoder and combiner matrices from (2.8) into (2.7) and replacing the channel matrix by its SVD.

An alternative and more elegant approach consists in substituting the particular precoder and combiner into (1.5) and comparing the resulting expression with the equivalent channel given in (2.2). This yields

$$\mathbf{z} = \mathbf{U}_{N_s}^\mathsf{H}\,\mathbf{U\Sigma V}^\mathsf{H}\,\mathbf{V}_{N_s}\sqrt{\mathbf{P}}\,\mathbf{a} + \mathbf{U}_{N_s}^\mathsf{H}\mathbf{n} \tag{2.9a}$$

$$= \mathbf{\Sigma}_{N_s}\sqrt{\mathbf{P}}\,\mathbf{a} + \tilde{\mathbf{n}}, \tag{2.9b}$$

where $\mathbf{\Sigma}_{N_s} \in \mathbb{R}^{N_s\times N_s}$ contains the R_H non-zero singular values of $\mathbf{H}$ on its diagonal. Comparing (2.9) with (2.2), one can readily verify that $I(\mathbf{a};\mathbf{z}) = I(\mathbf{x};\mathbf{y})$, provided that the power allocated to each stream is set according to the waterfilling solution. Hence, transmission with the precoder and combiner from (2.8) achieves the capacity of the SU-MIMO channel.

Denoting the sth element of the decision variable vector $\mathbf{z}$ by z_s, one can write (2.9) componentwise as

$$z_s = \sigma_s\sqrt{p_s}\,a_s + \tilde{n}_s \qquad \text{for} \quad s = 1,\ldots,N_s. \tag{2.10}$$

This expression shows that the precoder and combiner in (2.8) diagonalize the SU-MIMO transfer matrix from (1.5). The original SU-MIMO channel is decoupled into $N_s = R_H$ independent parallel scalar single-input single-output (SISO) Gaussian subchannels with gains σ_s, which coincides exactly with the decomposition of the equivalent channel in (2.3). A visualization of this decomposition is shown in Fig. 2.1.

As a result, the individual data streams become statistically independent. Detection therefore reduces to symbol-by-symbol processing rather than joint detection of the full decision-variable vector. This leads to a significant reduction in receiver complexity, especially for systems with a large number of antennas and data streams.

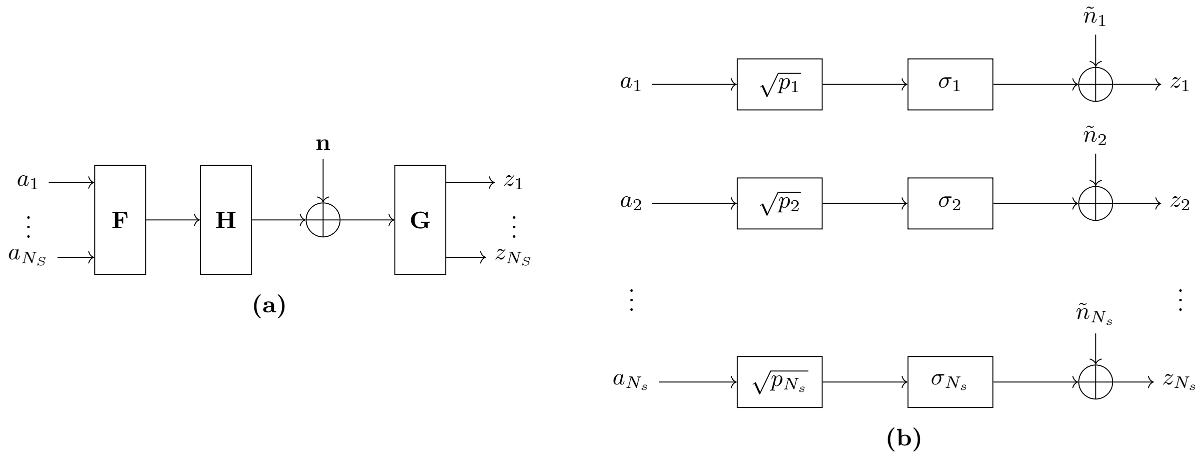


Figure 2.1: A block diagram of the original SU-MIMO channel (a) and its decomposition into parallel SISO Gaussian eigen subchannels (b) by means of the optimal precoder and combiner.

No general closed-form expression exists for the optimal water level μ_{w}, and hence for the resulting power allocation. Nevertheless, the optimal power allocation is fully characterized by

$$\sum_{i=1}^{N_s} p_i = P_t \qquad \text{and} \qquad p_i = \left(\mu_{\mathrm{w}} - \frac{N_0}{\sigma_i^2}\right)_+ . \tag{2.11}$$

Since the total allocated power is a monotonically increasing function of the water level, the latter can be determined numerically by means of a bisection method. A more efficient approach exploits the geometric interpretation of the waterfilling solution. In the present SU-MIMO setting, each eigen subchannel can be represented as a container with unit width and floor level N_0/σ_i^2, while the total transmit power is interpreted as water that is poured into these containers until a common water level is reached. The complete procedure is summarized in Appendix A.2. The

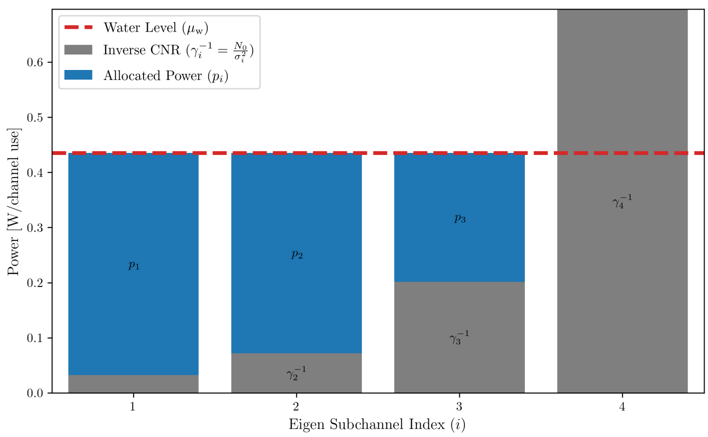
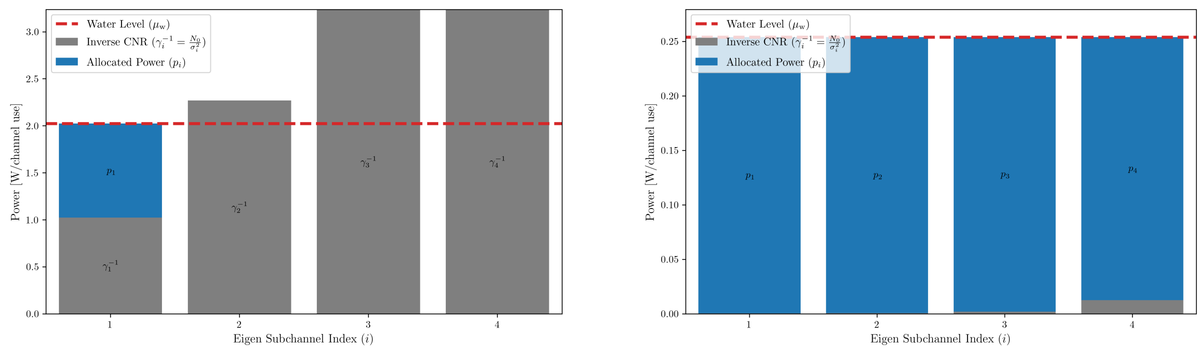


Figure 2.2: Optimal power allocation for a SU-MIMO system with 4 transmit and 4 receive antennas and a total transmit power P_t of 1 W per channel use. The system operates at an signal-to-noise ratio (SNR) of 5 dB.

structure of the optimal power allocation is illustrated in Fig. 2.2 for a SU-MIMO system with four transmit and four receive antennas operating at an SNR of 5 dB: eigen subchannels with stronger channel conditions, i.e. larger gains σ_i, receive more power than weaker ones.

At extremely low SNR, the waterfilling solution corresponds to the eigenbeamforming power allocation strategy, which concentrates all available power on the strongest eigen subchannel. As a result, only a single data stream is transmitted in this SNR regime. This behavior is illustrated in Fig. 2.3a. Conversely, at an extremely high SNR, the optimal power allocation strategy reduces to the uniform power allocation, which distributes power equally across all eigen subchannels, as illustrated in Fig. 2.3b. It is worth noting that uniform power allocation does not require knowledge of the current channel matrix, in contrast to the general waterfilling solution.



(a) SNR $= -10$ dB **(b)** SNR $= 25$ dB

Figure 2.3: Waterfilling power allocation for a SU-MIMO system with 4 transmit and 4 receive antennas at different SNR extremes. The maximum transmit power P_t is 1 W per channel use.

2.2 Joint Design for Linear Precoder and Combiner

In the previous section, a linear precoder and combiner that jointly decompose the SU-MIMO channel into independent parallel eigen subchannels were shown to achieve the channel capacity. In this section, we generalize this result by adopting the minimization of a weighted sum of mean–square estimation errors as the design criterion. It will be shown that the optimum precoder and combiner under this criterion likewise decouple the channel into parallel eigen subchannels, regardless of the specific choice of error weights. By an appropriate selection of these weights, the framework specializes to the maximum information rate design, the Quality–of–Service (QoS) design, and the unweighted minimum mean squared error (MMSE) design.

2.2.1 Generalized WMMSE Design

The objective is to jointly design the precoder $\mathbf{F}$ and the combiner $\mathbf{G}$ so as to minimize a wweighted combination of mean–square estimation errors. To this end, the estimation error vector $\mathbf{e} \in \mathbb{C}^{N_s}$ is defined as the difference between the transmitted data symbol vector $\mathbf{a}$ and the decision variable vector $\mathbf{z}$:

$$\mathbf{e} = \mathbf{a} - \mathbf{z} = \mathbf{a} - (\mathbf{GHF}\,\mathbf{a} + \mathbf{Gn}). \tag{2.12}$$

The mean squared error (MSE) matrix, also referred to as the error covariance matrix, is defined as

$$\mathrm{MSE} = \mathbb{E}\Big[\mathbf{e}\,\mathbf{e}^{\mathsf{H}}\Big] = (\mathbf{GHF} - \mathbf{I}_{N_s})(\mathbf{GHF} - \mathbf{I}_{N_s})^{\mathsf{H}} + \mathbf{G}\mathbf{C}_{\mathrm{n}}\mathbf{G}^{\mathsf{H}} \tag{2.13}$$

where the last equality follows from the independence of the data symbols $\mathbf{a}$ and the noise vector $\mathbf{n}$, and the assumption that the data symbols are spatially uncorrelated and normalized.

The weighted mean squared error (WMSE) is defined as

$$\mathrm{WMSE} = \mathbb{E}\left[\left\|\mathbf{W}_{\mathrm{e}}^{1/2}\,\mathbf{e}\right\|^2\right] = \mathbb{E}\Big[\mathrm{tr}(\mathbf{W}_{\mathrm{e}}^{1/2}\mathbf{e}\,\mathbf{e}^{\mathsf{H}}\mathbf{W}_{\mathrm{e}}^{1/2\,H})\Big] = \mathrm{tr}(\mathbf{W}_{\mathrm{e}}\,\mathbb{E}\Big[\mathbf{e}\mathbf{e}^{\mathsf{H}}\Big]), \tag{2.14}$$

where $\mathbf{W}_{\mathrm{e}}$ is a diagonal positive definite weight matrix that assigns a relative importance to the error of each data stream.

The weighted minimum mean squared error (WMMSE) optimization problem is then formulated as

$$\begin{aligned} &\min_{\mathbf{F},\mathbf{G}}\ \mathrm{tr}(\mathbf{W}_{\mathrm{e}}\,\mathbb{E}\Big[\mathbf{e}\mathbf{e}^{\mathsf{H}}\Big]) \\ &\text{s. t.}\quad \mathrm{tr}(\mathbf{F}\mathbf{F}^{\mathsf{H}}) \le P_t. \end{aligned} \tag{2.15}$$

The optimization problem in(2.15) can be solved using the Lagrange duality together with the Karush-Kuhn-Tucker (KKT) conditions. This leads to the following necessary optimality conditions [12]:

$$\mathbf{HF} = \mathbf{HF}\,(\mathbf{GHF})^{\mathsf{H}} + \mathbf{C}_{\mathrm{n}}\mathbf{G}^{\mathsf{H}} \tag{2.16a}$$

$$\mathbf{W}_{\mathrm{e}}\mathbf{GH} = (\mathbf{GHF})^{\mathsf{H}}\,\mathbf{W}_{\mathrm{e}}\mathbf{GH} + \mu\mathbf{F}^{\mathsf{H}} \tag{2.16b}$$

where $\mu > 0$ is the Lagrange multiplier associated with the total transmit power constraint.

Solving these coupled matrix equations yields the optimum linear precoder and combiner. As shown in [12], they admit the following closed-form expressions:

$$\mathbf{F} = \mathbf{V}\boldsymbol{\Phi}_{\mathrm{f}} \tag{2.17a}$$

$$\mathbf{G} = \boldsymbol{\Phi}_{\mathrm{g}}\mathbf{V}^{\mathsf{H}}\mathbf{H}^{\mathsf{H}}\mathbf{C}_{\mathrm{n}}^{-1} \tag{2.17b}$$

The matrices $\boldsymbol{\Phi}_{\mathrm{f}}, \boldsymbol{\Phi}_{\mathrm{g}} \in \mathbb{R}^{R_H \times R_H}$ are diagonal matrices with non-negative diagonal entries, and are given by

$$\boldsymbol{\Phi}_{\mathrm{f}} = (\mu^{-1/2}\boldsymbol{\Lambda}^{-1/2}\mathbf{W}_{\mathrm{e}}^{1/2} - \boldsymbol{\Lambda}^{-1})_{+}^{1/2} \tag{2.18a}$$

$$\boldsymbol{\Phi}_{\mathrm{g}} = (\mu^{1/2}\boldsymbol{\Lambda}^{-1/2}\mathbf{W}_{\mathrm{e}}^{-1/2} - \mu\boldsymbol{\Lambda}^{-1}\mathbf{W}_{\mathrm{e}}^{-1})_{+}^{1/2}\boldsymbol{\Lambda}^{1/2} \tag{2.18b}$$

The matrices $\mathbf{V} \in \mathbb{C}^{N_T \times R_H}$ and $\boldsymbol{\Lambda} \in \mathbb{R}^{R_H \times R_H}$ are obtained from the eigenvalue decomposition (EVD) of the noise-whitened Gram matrix of $\mathbf{H}$:

$$\mathbf{H}^{\mathsf{H}}\mathbf{C}_{\mathrm{n}}^{-1}\mathbf{H} = \begin{bmatrix}\mathbf{V} & \bar{\mathbf{V}}\end{bmatrix}\begin{bmatrix}\boldsymbol{\Lambda} & \mathbf{0} \\ \mathbf{0} & \bar{\boldsymbol{\Lambda}}\end{bmatrix}\begin{bmatrix}\mathbf{V} & \bar{\mathbf{V}}\end{bmatrix}^{\mathsf{H}}, \tag{2.19}$$

where the columns of $\mathbf{V}$ form an orthonormal basis for the range space of $(\mathbf{H}^{\mathsf{H}}\mathbf{C}_{\mathrm{n}}^{-1}\mathbf{H})$, and the columns of $\bar{\mathbf{V}}$ span its null space. The diagonal matrix $\mathbf{\Lambda}$ contains the R_H non-zero eigenvalues arranged in descending order, whereas $\bar{\mathbf{\Lambda}}$ contains the remaining zero eigenvalues. Note that, as in Section 2.1.2, the number of data streams is set equal to the rank of the channel matrix, i.e. $N_s = R_H$. Moreover, the rank of the channel matrix is equal to the rank of $(\mathbf{H}^{\mathsf{H}}\mathbf{C}_{\mathrm{n}}^{-1}\mathbf{H})$ since $\mathbf{C}_{\mathrm{n}}^{-1}$ is positive definite and therefore does not alter the rank of the product.

Now, it only remains to show how to obtain the Lagrange multiplier that satisfies the power constraint and also is such that the bracketed matrices in both (2.18a) and (2.18b) are positive-definite. Let $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_{R_H})$ and $\mathbf{W}_{\mathrm{e}} = \mathrm{diag}(w_1, \ldots, w_{R_H})$. Suppose $k \leq R_H$ eigen subchannels are effectively used for transmission, i.e. they are allocated non-zero power. From (2.18a) and the total power constraint in (2.15), we have

$$\mathrm{tr}(\mathbf{F}\mathbf{F}^{\mathsf{H}}) = \mathrm{tr}(\mathbf{\Phi}_{\mathrm{f}}^2) = \mu^{-1/2}\sum_{i=1}^{k}\lambda_i^{-1/2}w_i^{1/2} - \sum_{i=1}^{k}\lambda_i^{-1} = P_t. \tag{2.20}$$

Therefore, we can obtain the following expression for the Lagrange multiplier μ:

$$\mu = \left(\frac{\sum_{i=1}^{k}\lambda_i^{-1/2}w_i^{1/2}}{P_t + \sum_{i=1}^{k}\lambda_i^{-1}}\right)^2. \tag{2.21}$$

Let $\rho_k = \lambda_k w_k$ and let them be ordered in decreasing manner. To compute $\mathbf{\Phi}_{\mathrm{f}}$ optimally and ensure that it is positive-definite, we can compute μ according to (2.21) for each $k = 1, \ldots, R_H$ iteratively, and select the largest k such that $\mu \leq \rho_k$ [12].

From (2.17a) and (2.17b), we have:

$$\mathbf{T} = \mathbf{G}\mathbf{H}\mathbf{F} = \mathbf{\Phi}_{\mathrm{g}}\mathbf{\Lambda}\mathbf{\Phi}_{\mathrm{f}}. \tag{2.22}$$

In other words, the optimum precoder and combiner diagonalize the SU-MIMO transfer matrix $\mathbf{T}$, and hence decouple the channel into parallel eigen subchannels, regardless of the specific choice of error weights. This result is similar to the capacity-achieving precoder and combiner derived in Section 2.1.2, which likewise decompose the channel matrix into parallel eigen subchannels.

By choosing the error weight matrix appropriately, the generalized WMMSE framework can be specialized to different design criteria, such as maximum information rate, unweighted MMSE, or prescribed quality-of-service requirements. The following subsections discuss these special cases in more detail.

2.2.2 Maximum Information Rate Design

It was shown in Section 2.1.2 that the capacity of the SU-MIMO channel can be achieved by a linear precoder and combiner that diagonalize the transfer matrix and allocate power according to the waterfilling solution. Rewriting (2.11) in matrix format and switching to the parameter notation of the generalized design yields [12]

$$\mathbf{\Phi}_{\mathrm{f}} = (\mu^{-1/2}\mathbf{I} - \mathbf{\Lambda}^{-1})_{+}^{1/2}. \tag{2.23}$$

The above equation is obtained by choosing the error weight matrix as $\mathbf{W}_\mathrm{e} = \mathbf{\Lambda}$ in (2.18a). It is shown in [12] that, for this particular choice of error weights, the closed-form solutions in (2.17a) and (2.17b) yield the linear precoder and combiner that indeed maximize the information rate. In other words, the maximum information rate design is just a special case of the generalized WMMSE design. The choice of the error weights $\mathbf{W}_\mathrm{e} = \mathbf{\Lambda}$ indicates that stronger eigen subchannels support higher rates compared to the weaker ones, and hence need to be more heavily weighted in the cost function. This is consistent with the waterfilling solution, which allocates more power to stronger subchannels.

The maximum information rate design is particularly relevant in adaptive modulation systems, where more power and higher order modulation are used on eigen subchannels with higher gains to improve the total data rate. The performance of these systems will be evaluated in Section 2.3, based on simulation results for different system configurations.

2.2.3 Quality of Service Design

Consider an application in which different data streams carry information of heterogeneous types with different MSE requirements for successful transmission. A typical example is a multimedia system in which audio and video streams are transmitted simultaneously but require different levels of reliability. For such Quality–of–Service (QoS)-sensitive applications, the system must be designed to achieve prescribed MSE levels on the individual eigen subchannels.

First, we note that the expression for the MSE matrix simplifies upon substituting the optimum precoder and combiner from (2.17a) and (2.17b):

$$\mathrm{MSE} = \mathbb{E}\left[\mathbf{e}\,\mathbf{e}^\mathsf{H}\right] = \mathbf{W}_\mathrm{e}^{-1/2}\mathbf{\Lambda}^{-1/2}\mu^{1/2}. \tag{2.24}$$

Next, define the output SNR matrix as

$$\begin{aligned} \mathrm{SNR} &= (\mathbf{GHF})^\mathsf{H}\,(\mathbf{G}\mathbf{C}_\mathrm{n}\mathbf{G}^\mathsf{H})^{-1}\,(\mathbf{GHF}) && (2.25\mathrm{a})\\ &= \mathbf{\Phi}_\mathrm{f}^2\,\mathbf{\Lambda} = (\mu^{-1/2}\mathbf{\Lambda}^{1/2}\mathbf{W}_\mathrm{e}^{1/2} - \mathbf{I})_+, && (2.25\mathrm{b}) \end{aligned}$$

where we again used the expressions for the optimum precoder and combiner to simplify the first equality. The relation $\mathrm{SNR} = (\mathrm{MSE}^{-1} - \mathbf{I})_+$ follows directly from these expressions, showing that the SNR and MSE on each subchannel are in a one-to-one inverse correspondence. Hence, prescribing relative MSEs is equivalent to prescribing relative output SNRs. In particular, equal SNRs across all eigen subchannels imply equal MSEs, regardless of the specific choice of error weights.

We now determine the error weight matrix $\mathbf{W}_\mathrm{e}$ required to realize a desired set of relative output SNRs across the eigen subchannels. To this end, let $\mathbf{D}$ be a diagonal matrix containing the desired relative SNRs, and let $d > 0$ denote an arbitrary scaling factor. The design requirement reduces to the following equation

$$d\,\mathbf{D} = (\mu^{-1/2}\mathbf{\Lambda}^{1/2}\mathbf{W}_\mathrm{e}^{1/2} - \mathbf{I})_+, \tag{2.26}$$

which must be solved for the weight matrix. Assuming for simplicity that all subchannels are active, i.e. $\mathbf{\Phi}_\mathrm{f} > 0$, the corresponding solution is given by [12]

$$\mathbf{W}_\mathrm{e}^{1/2} = \mu^{1/2}(d\,\mathbf{D} + \mathbf{I})\mathbf{\Lambda}^{-1/2} \quad \text{with} \quad \mu^{1/2} = \frac{\mathrm{tr}(\mathbf{\Lambda}^{-1/2}\mathbf{W}_\mathrm{e}^{1/2})}{P_t + \mathrm{tr}(\mathbf{\Lambda}^{-1})} \quad \text{and} \quad d = \frac{P_t}{\mathrm{tr}(\mathbf{D}\mathbf{\Lambda}^{-1})}. \tag{2.27}$$

As a simple example, consider an application that requires all data streams to achieve the same reliability, using a fixed rate and an identical modulation and coding scheme for all streams. Within the present framework, this corresponds to imposing equal error rates and thus equal output SNRs on all eigen subchannels. The latter requirement can be satisfied trivially by selecting $\mathbf{D} = \mathbf{I}$. The resulting optimum precoder and combiner are then given by [12]

$$\mathbf{F} = \mathbf{V}\mathbf{\Phi}_\mathrm{f} \quad \text{with} \quad \mathbf{\Phi}_\mathrm{f} = \sqrt{\frac{P_t}{\mathrm{tr}(\mathbf{\Lambda}^{-1})}}\,\mathbf{\Lambda}^{-1/2} \tag{2.28a}$$

$$\mathbf{G} = \mathbf{\Phi}_\mathrm{g}\mathbf{V}^\mathsf{H}\mathbf{H}^\mathsf{H}\mathbf{C}_\mathrm{n}^{-1} \quad \text{with} \quad \mathbf{\Phi}_\mathrm{g} = \frac{\sqrt{P_t\,\mathrm{tr}(\mathbf{\Lambda}^{-1})}}{P_t + \mathrm{tr}(\mathbf{\Lambda}^{-1})}\,\mathbf{\Lambda}^{-1/2} \tag{2.28b}$$

This QoS design for equal error rates allocates more power to eigen subchannels with smaller channel gains and less power to stronger eigen subchannels, such that all active subchannels achieve the same output SNRs and MSEs. In contrast to the maximum-information-rate design, no eigen subchannels are dropped, regardless of the channel realization. Furthermore, it can be shown that the optimum precoder and combiner transform the transfer matrix of the system into a scaled identity matrix [12].

2.2.4 Unweighted MMSE Design

If the error weight matrix is chosen as $\mathbf{W}_\mathrm{e} = \mathbf{I}$, the WMMSE problem reduces to the unweighted minimum mean squared error (MMSE) design. The closed-form solutions in (2.17a) and (2.17b) for the optimum precoder and combiner then minimize the sum of the expected squared symbol estimation errors across all eigen subchannels. Unlike the QoS design, this criterion does not enforce balanced performance across the individual subchannels, but improves the overall performance of the system. In particular, substituting $\mathbf{W}_\mathrm{e} = \mathbf{I}$ into (2.24) shows that weaker subchannels generally exhibit larger MSEs, and therefore lower output SNRs, than stronger ones.

The corresponding power allocation follows from (2.18a) by setting $\mathbf{W}_\mathrm{e} = \mathbf{I}$, which yields

$$\phi_{\mathrm{f}\,i}^2 = \frac{1}{\sqrt{\lambda_i}}\left(\mu_\mathrm{w} - \frac{1}{\sqrt{\lambda_i}}\right)_+ \qquad \text{for} \quad i = 1, \ldots, R_H \tag{2.29}$$

where $\phi_{\mathrm{f}\,i}^2$ is the ith element in the main diagonal of $\mathbf{\Phi}_\mathrm{f}$ and $\mu_\mathrm{w} = \mu^{-1/2}$ is a water level chosen to satisfy the total power constraint with equality. This expression has exactly the same form as the weighted waterfilling solution derived in Appendix A. In particular, it is obtained from (A.6) by identifying the allocated power with $p_i = \phi_{\mathrm{f},i}^2$, the channel weight with $\alpha_i = \lambda_i^{-1/2}$, and the channel-gain-to-noise ratio with $\gamma_i = \lambda_i$. Hence, the power allocation can be computed efficiently using Algorithm 5 with these parameter choices.

As in the maximum-information-rate design, eigen subchannels whose gains fall below a threshold receive no power. Among the remaining active subchannels, however, the weighting factor $\alpha_i = \lambda_i^{-1/2}$ gives relatively more emphasis to weaker subchannels than in the standard sum-rate case with $\alpha_i = 1$. This reflects the fact that the present design minimizes the total sum MSE, rather than maximizing the information rate.

2.3 Performance Evaluation

The preceding section derived optimal linear precoders and combiners for the SU-MIMO channel under several design criteria. This section evaluates the performance of the maximum information rate design through Monte Carlo simulations. As established in Section 2.2.2, this design is equivalent to the capacity-achieving SVD-based precoder and combiner with waterfilling power allocation.

All simulations are conducted under the following assumptions. The channel is modeled as an i.i.d. Rayleigh block fading channel, whose statistical model is derived in Appendix C. Each entry of the channel matrix $\mathbf{H}$ is drawn independently from $\mathcal{CN}(0, 1)$, and the channel is held constant within a transmission block and drawn independently across blocks. The reported performance metrics are averaged over at least 100 independent channel realizations. Perfect channel state information (CSI) is assumed at both the TX and the RX, so that the precoder and combiner are computed from the exact channel matrix.

The following subsections examine the effect of different system configurations and design parameters on the system performance. At the end of this section, analytical results are included to serve as a cross-check for the simulation curves. All results are presented as a function of the SNR, in terms of the total bit error rate (BER) and the total information bit rate (IBR) of the system.

2.3.1 Bit Allocation Strategies

As a first experiment, the BER is evaluated as a function of the SNR for a 2×2 SU-MIMO system using three modulation schemes: quadrature amplitude modulation (QAM), phase shift keying (PSK), and pulse amplitude modulation (PAM), each carrying the same number of bits per symbol. Figure 2.4 shows the results under a fixed bit allocation, where the same modulation order is applied on all eigen subchannels across the entire SNR range. As expected, the BER decreases monotonically with increasing SNR for all three constellations. The BER curve falls most steeply for QAM, followed by PSK and then PAM, confirming that QAM reaches any given target BER at a strictly lower SNR than the other two schemes at equal spectral efficiency. QAM is therefore adopted as the modulation scheme throughout this dissertation, unless otherwise stated. The number of bits per symbol is always chosen to be even, so that the QAM constellation forms a regular square grid.

As established in Section 2.2.2, the maximum information rate design is particularly relevant in adaptive modulation systems, where the number of bits per symbol is not fixed but adapted to the instantaneous capacity of each eigen subchannel. Specifically, the bit allocation on subchannel i

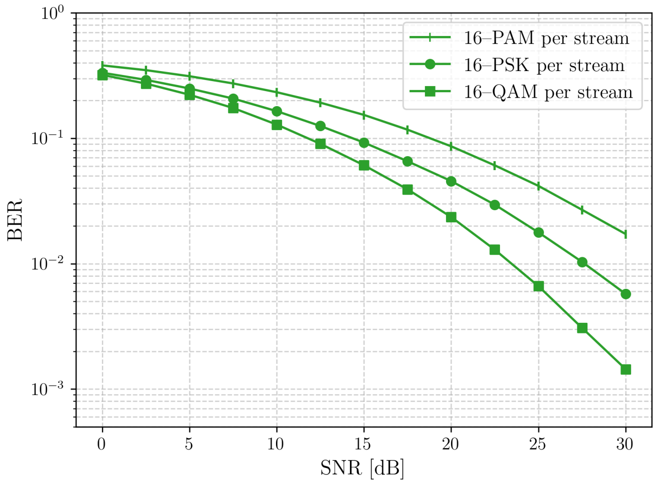
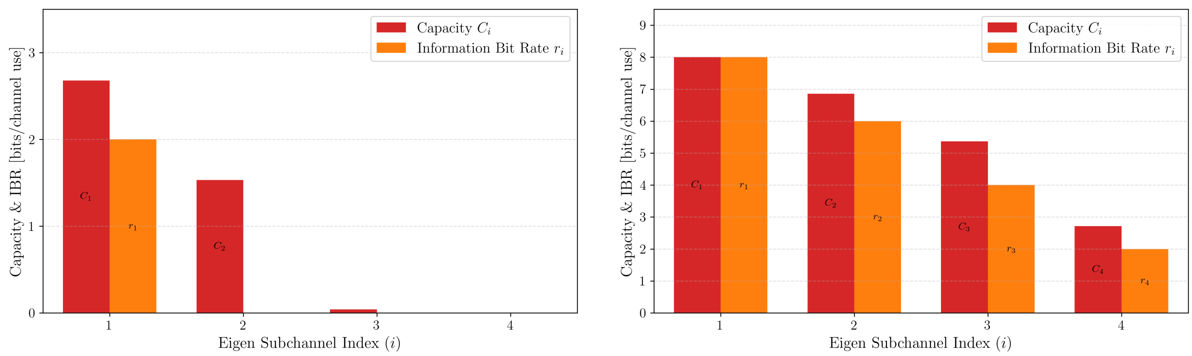


Figure 2.4: BER as a function of SNR for three SU-MIMO systems with 2 transmit and 2 receive antennas, each using a different modulation constellations on each stream: 16–PAM, 16–PSK, and 16–QAM.

is set to the largest even integer not exceeding its capacity. Figure 2.5 illustrates the resulting bit allocations for a 4×4 system at two different SNR values. At low SNR, the waterfilling solution concentrates almost all power on the strongest eigen subchannel, which is the only one capable of supporting a non-zero bit allocation. Because waterfilling maximizes the mutual information without accounting for the discrete nature of the bit allocation, a small amount of power is still directed to the weaker subchannels. Since their capacities fall below the 2-bit threshold, however, their bit allocations remain zero and the corresponding power is wasted. Jointly optimizing the power and bit allocation to avoid this inefficiency has been studied in [13, 14], but is not considered in this dissertation. At high SNR, more subchannels become active and support higher-order constellations, increasing the total IBR.

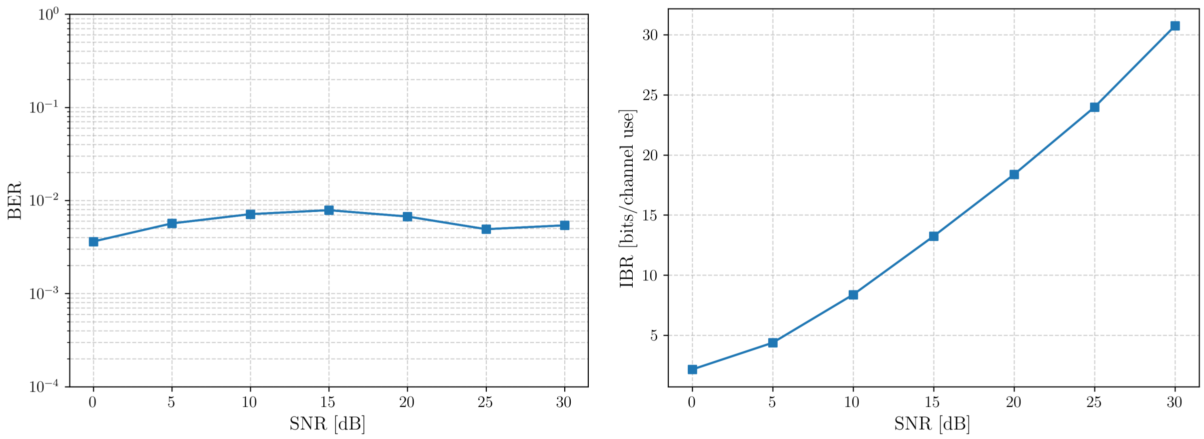


(a) 0 dB

(b) 20 dB

Figure 2.5: Adaptive bit allocation in different SNR regimes for a system with 4 transmit and 4 receive antennas. The number of bits per symbol on each stream is the largest even integer not exceeding the capacity of the corresponding eigen subchannel.

Figure 2.6 shows the BER and IBR as a function of SNR for this adaptive bit allocation strategy. In contrast to the fixed bit allocation case, the BER remains approximately constant across the entire SNR range, while the IBR increases steadily with SNR, as expected.

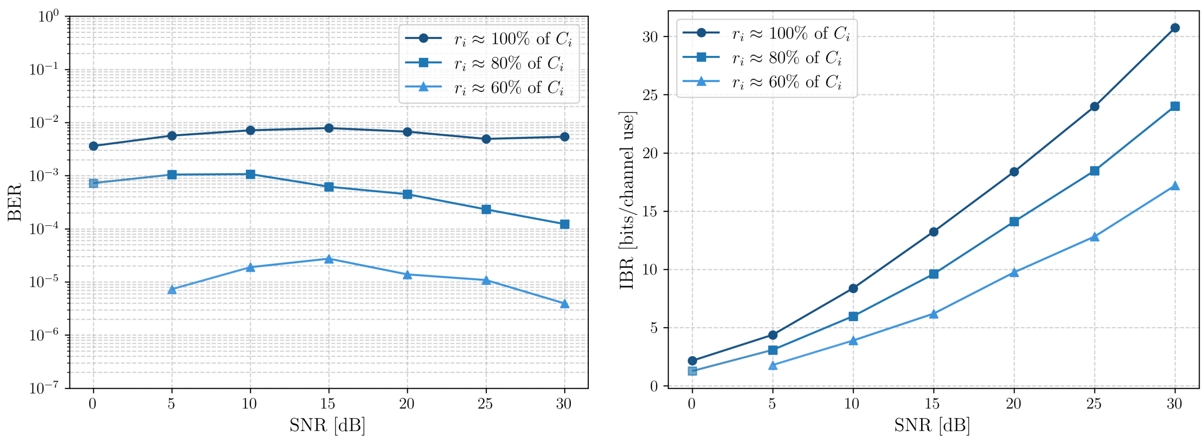


(a) **(b)**

Figure 2.6: System performance with adaptive bit allocation for a system with 4 transmit and 4 receive antennas. The BER and IBR are shown as a function of SNR.

2.3.2 Impact of the Data Rate

The previous subsection assigned to each eigen subchannel the largest even number of bits not exceeding the subchannel capacity, so that the system operated at approximately the maximum achievable rate. In practice, however, it may be preferable to transmit at a lower rate in exchange for a reduced BER. To quantify this tradeoff, we introduce a rate factor $r_f \in (0, 1]$ that specifies which fraction of the subchannel capacity is used for transmission. The bit allocation on subchannel i is then generalized to the largest even integer not exceeding $r_f\, C_i$, where C_i denotes the instantaneous capacity of that subchannel. The case $r_f = 1$ recovers the maximum-rate allocation of the previous subsection.



(a) **(b)**

Figure 2.7: Illustration of the impact of the data rate on the system performance for a system with 4 transmit and 4 receive antennas using adaptive bit allocation. The (a) BER and (b) IBR are shown as a function of SNR, for three different rate factors.

Figure 2.7 shows the BER and IBR as a function of SNR for several values of r_f on a 4×4 system. As expected, reducing r_f lowers the IBR, because fewer bits are assigned to each subchannel. At the same time, the BER decreases, since the chosen modulation order moves further below the subchannel capacity and the link operates with a larger margin. This illustrates an important tradeoff between throughput and reliability: operating below capacity sacrifices data rate in exchange for improved error performance.

For completeness, Figure 2.8 shows the analogous experiment under fixed bit allocation on a 2×2 system, where the same modulation order is applied on all eigen subchannels across the entire SNR range. The behavior is consistent: a smaller constellation size yields a lower BER at any given SNR. The slope of the BER curve is essentially unchanged by the constellation order.

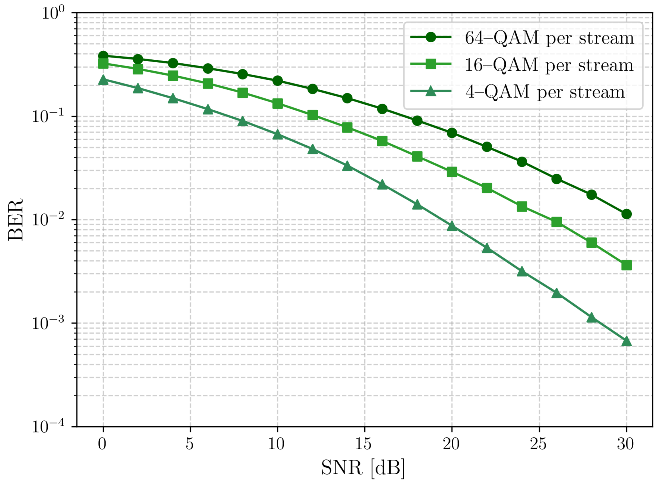


Figure 2.8: Illustration of the impact of the data rate on the system performance for a system with 2 transmit and 2 receive antennas using fixed bit allocation. The BER is shown as a function of SNR for three different modulation orders.

2.3.3 Impact of the Antenna Count

This subsection examines how the number of transmit and receive antennas affects system performance, keeping the rate factor fixed at $r_f = 1$. Figure 2.9 shows the BER and IBR as a function of SNR for several antenna configurations. The BER remains approximately constant across all configurations, consistent with the behavior observed in Section 2.3.1. By contrast, the maximum achievable rate, and thus the IBR, markedly with antenna count. Three distinct effects can be observed.

The first effect is a change in the slope of the IBR curve, which is determined by $\min(N_T, N_R)$. This follows directly from the channel rank: for an i.i.d. Rayleigh fading channel, $R_H \leq \min(N_T, N_R)$, so the number of parallel eigen subchannels cannot exceed the smaller antenna count. Consequently, adding antennas on both sides creates additional parallel streams and therefore increases the slope of the maximum achievable rate in function of the SNR

The second effect is a vertical upward shift of the IBR curve when antennas are added only on the side with a larger antenna count while $\min(N_T, N_R)$ remains unchanged. This shift occurs because, for a fixed smaller dimension, the expected squared singular values of an $N_R \times N_T$ i.i.d. Gaussian matrix increase with the larger dimension [15]. The eigen subchannels thus experience better average channel conditions, yielding a higher IBR at the same SNR while leaving the slope unchanged.

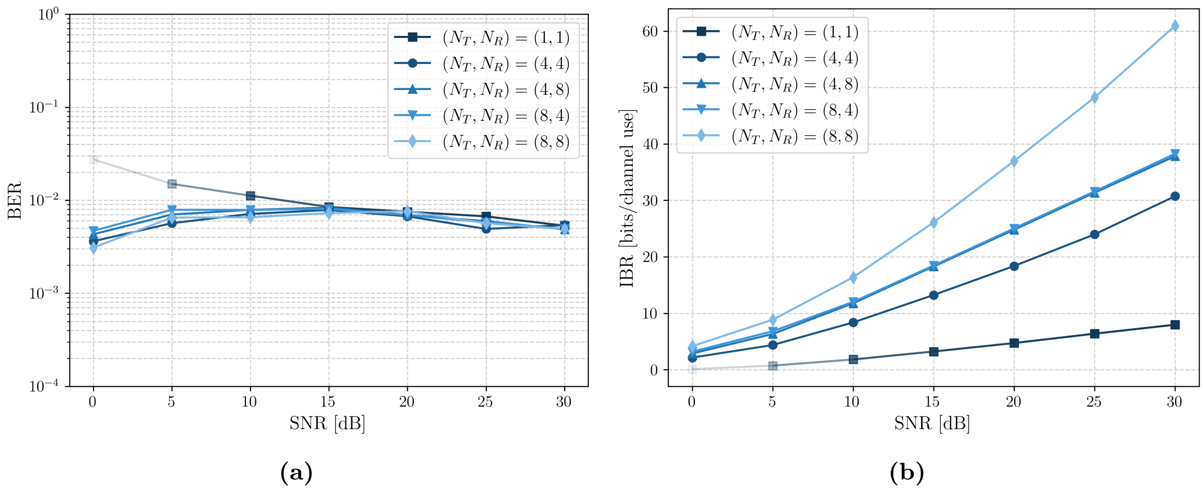


Figure 2.9: Illustration of the impact of the antenna count on the system performance for a system adaptive bit allocation. The (a) BER and (b) IBR are shown as a function of SNR, for five different antenna configurations.

Finally, configurations with the same $\min(N_T, N_R)$ and the same total number of antennas show no performance difference when the excess antennas are placed at the TX or at the RX. This symmetry follows since $\mathbf{H}\mathbf{H}^{\mathsf{H}}$ and $\mathbf{H}^{\mathsf{H}}\mathbf{H}$ have the same non-zero eigenvalues, so the singular-value distribution of an $N_R \times N_T$ i.i.d. Rayleigh channel is identical to that of its transpose.

As in the previous subsection, the rate–reliability tradeoff also applies here. Figure 2.10 illustrates this by adjusting the rate factor of each configuration so that all systems achieve approximately the same IBR at each SNR. Under this equal-rate comparison, systems with more antennas consistently achieve lower BER, because the additional spatial degrees of freedom support the same throughput with larger margins on the individual eigen subchannels.

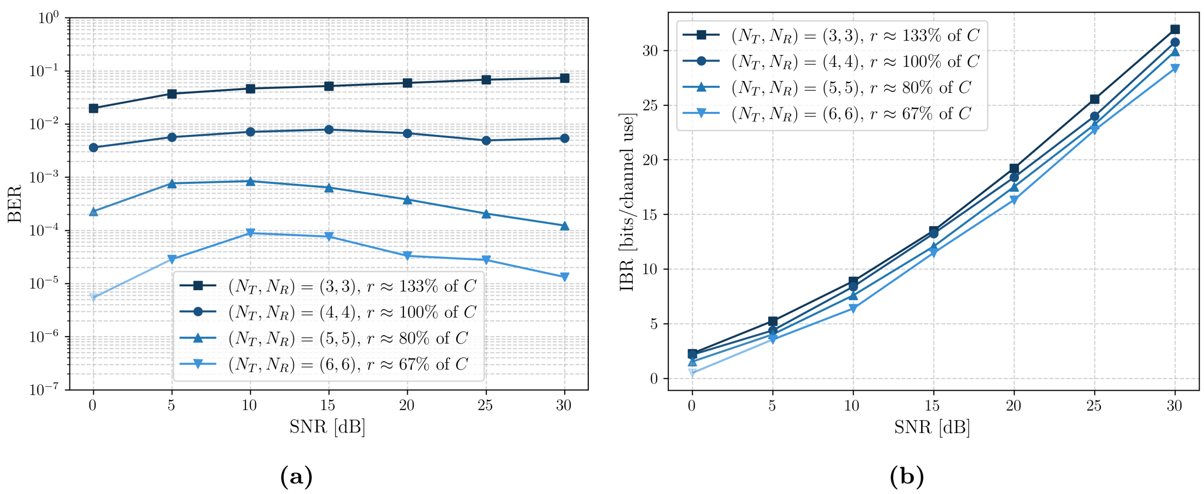


Figure 2.10: Illustration of the impact of the antenna count on the system performance for a system adaptive bit allocation. The (a) BER and (b) IBR are shown as a function of SNR, for five different antenna configurations. The rate factor of each configuration is adjusted so that all systems achieve approximately the same IBR at each SNR.

In summary, equipping both the TX and RX with more antennas enables either higher throughput or higher reliability, at the cost of increased hardware complexity and computational load.

2.3.4 Analytical Validation

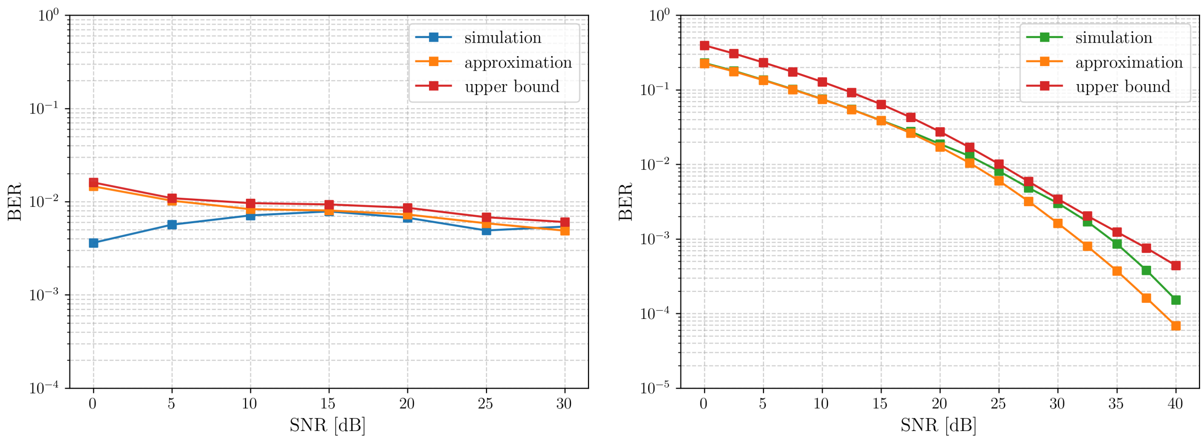
As established in Section 2.1.2, the SVD-based precoder and combiner decouple the SU-MIMO channel into R_H parallel eigen subchannels, each of which can be treated as an independent SISO link. This decomposition enables an analytical evaluation of the total system BER.

Because the number of transmitted bits can differ across eigen subchannels, the total BER is computed as a weighted average of the subchannel BERs, with weights proportional to the corresponding IBRs:

$$\mathrm{BER} = \frac{1}{\mathrm{IBR}} \sum_{i=1}^{R_H} \mathrm{IBR}_i \, \mathrm{BER}_i. \tag{2.30}$$

The BER on each eigen subchannel equals the BER of a scalar additive white Gaussian noise (AWGN) channel. For subchannel i, the effective signal power is determined by the allocated transmit power scaled by the corresponding subchannel gain, while the noise power remains unchanged. Although the subchannel gains are random variables [15], they are treated as fixed within each analytical evaluation, and the resulting BER is then averaged over multiple channel realizations. The BER of a general scalar AWGN channel is derived in Appendix B under the union upper bound and the nearest-neighbour approximation.

Figure 2.11 compares the analytical expressions with the simulated BER curves for a 4×4 system under adaptive and fixed bit allocation. In both cases, the nearest-neighbour approximation and the union upper bound closely track the simulation results, which validates the simulation implementation.



(a) adaptive bit allocation

(b) fixed bit allocation

Figure 2.11: Analytical validation of the system performance simulation results for a system with 4 transmit and 4 receive antennas. The BER is shown as a function of SNR for two different bit allocation strategies: (a) adaptive bit allocation with a unit rate factor; and (b) fixed bit allocation, where all subchannels use 4–QAM across the entire SNR range. In both cases, the simulation results are compared with the union upper bound and nearest neighbour approximation derived in Appendix B.

Chapter 3

Precoding for MU-MIMO systems

Consider the downlink MU-MIMO system introduced in Section 1.2.2. The key difference from the SU-MIMO system of the previous chapter is the presence of inter-user interference: the signal intended for one user leaks into the receive antennas of the other users.

The preceding chapter restricted attention to linear precoding, in which the transmitted signal is obtained by applying a linear transformation, represented by a matrix, to the data symbol vector. Nonlinear precoding techniques, on the other hand, apply nonlinear processing to the data symbols before transmission. While linear precoding can achieve the capacity of the SU-MIMO channel, it is no longer capacity-achieving in the MU-MIMO setting.

Two well-known nonlinear precoding techniques are dirty paper coding (DPC) [16] and Tomlinson–Harashima precoding (THP) [17, 18]. DPC is an information-theoretic technique that achieves the capacity region of the Gaussian broadcast channel by encoding each data stream such that interference from all previously encoded streams is pre-subtracted at the transmitter. THP approximates DPC at reduced complexity by successively canceling inter-user interference at the transmitter using modulo arithmetic. Despite their superior capacity, nonlinear techniques incur a significantly higher computational cost. Linear precoding, by contrast, offers near-optimal performance at a fraction of the complexity, and therefore remains the preferred approach in practice. This chapter accordingly focuses on linear precoding techniques.

The following sections presents four commonly used linear precoding strategies for the MU-MIMO downlink, arranged in order of increasing complexity. The presentation begins with three precoding techniques related to ZF, and concludes with the state-of-the-art WMMSE precoder. At the end of this chapter, their performance is compared through numerical simulations for different system configurations. In line with the previous chapter, the design is focused on maximizing the total sum rate (SR) of the system.

3.1 Zero-Forcing Precoding

The objective of zero–forcing (ZF) precoding is to cancel all interference, both inter-user and inter-stream. Moreover, no receiver-side combining is applied. Instead, all processing is done by the precoder and each UT forwards its received samples directly for detection.

Considering these restrictions, we obtain that

$$\mathbf{T} = (\mathbf{GHF}) = \sqrt{\mathbf{Q}} \quad \text{and} \quad \mathbf{G} = \mathbf{I}_{\text{rect}}, \tag{3.1}$$

where $\mathbf{T}$ is the compound MU-MIMO transfer matrix, $\sqrt{\mathbf{Q}} \in \mathbb{R}_{>0}^{K\,N_S \times K\,N_S}$ is a diagonal scaling matrix and $\mathbf{I}_{\text{rect}}$ is a rectangular diagonal matrix with unit entries on its main diagonal. The compound MU-MIMO input–output relation (1.10) then simplifies to

$$z_{k,\,i} = q_{k,i}\, a_{k,i} + n_{k,i} \qquad \text{for} \quad k = 1, \ldots, K \quad \text{and} \quad i = 1, \ldots, N_S. \tag{3.2}$$

This shows that the interference has been completely eliminated and each data stream is received over an independent scalar AWGN channel, which allows performing symbol-by-symbol detection at each UT.

It is shown in [19] that the constraints in (3.1) are satisfied when the precoder is chosen as the right pseudo-inverse of the channel matrix, scaled by $\sqrt{\mathbf{Q}}$:

$$\mathbf{F} = \mathbf{H}^{\mathsf{H}}(\mathbf{H}\mathbf{H}^{\mathsf{H}})^{-1}\, \sqrt{\mathbf{Q}} \tag{3.3}$$

This expression requires the channel matrix to have full row rank ($R_H = K\,N_S$), so that the inverse $(\mathbf{H}\mathbf{H}^{\mathsf{H}})^{-1}$ exists. For an i.i.d. Rayleigh fading channel, this condition holds almost surely provided that $N_T \geq KN_S$. In what follows, this condition is assumed to be satisfied. If it is not, the inverse $(\mathbf{H}\mathbf{H}^{\mathsf{H}})^{-1}$ is replaced by the Moore-Penrose pseudo-inverse, which is computed from the SVD of the channel matrix [20].

It remains to determine the optimal diagonal entries of the scaling matrix $\mathbf{Q}$ so as to maximize the weighted sum rate (WSR) under the total transmit power constraint. Considering (3.2), the WSR of the system can be written as

$$\text{WSR} = \sum_{k=1}^{K} \sum_{i=1}^{N_S} \mu_{k,i}\, \log_2\big(1 + \frac{q_{k,i}}{N_0}\big), \tag{3.4}$$

where $\mu_{k,i}$ is the weight assigned to the ith data stream intended for UT k. Substituting the ZF precoder into the transmit power constraint (1.8) yields

$$\sum_{k=1}^{K} \sum_{i=1}^{N_S} q_{k,i}\, \big((\mathbf{H}\mathbf{H}^{\mathsf{H}})^{-1}\big)_{n,\,n} \leq P_t \qquad \text{with} \quad n = (k-1)\,N_S + i. \tag{3.5}$$

Let us introduce the substitution

$$p_n = \gamma_n^{-1}\, q_{k,i} \quad \text{with} \quad \gamma_n^{-1} = \big((\mathbf{H}\mathbf{H}^{\mathsf{H}})^{-1}\big)_{n,\,n}, \tag{3.6}$$

such that we can write the WSR maximization problem as follows:

$$\begin{aligned} \max_{\{p_n\}} \quad & \sum_{n=1}^{K\,N_S} \alpha_n \, \log_2 \left(1 + \frac{\gamma_n}{N_0}\, p_n\right) \\ \text{s.t.} \quad & \sum_{n=1}^{K\,N_S} p_n \leq P_t \end{aligned} \tag{3.7}$$

This is exactly the optimization problem handled in Appendix A. The optimal values for p_i are therefore obtained using the waterfilling solution as described in Algorithm 5.

Note that the above power allocation strategy may allocate almost no power to the data streams of a UT with poor channel conditions. The achievable rate for that specific user will therefore be very small, compared to the other users with better channel conditions. An alternative, more fair approach is to divide the total transmit power equally across the different users, and use the waterfilling solution to allocate the power to which every user is entitled across the different data streams intended for that user.

To summarize, the steps required for computing the ZF compound precoder matrix for total SR maximization across all UTs are described in Algorithm 1.

Algorithm 1 ZF Precoding

Require:

- The compound channel matrix $\mathbf{H}$;
- The total transmit power P_t and the received noise power N_0.

Ensure: The compound precoder matrix $\mathbf{F}$.

1: Compute p_i for $i = 1, \ldots, K\,N_S$ using the waterfilling algorithm (Alg. 5) with inputs
$P_t = P_t, \quad \alpha_n = 1, \quad \text{and} \quad \gamma_n = \left(N_0 \left((\mathbf{H}\mathbf{H}^{\mathsf{H}})^{-1}\right)_{n,n}\right)^{-1}$.

2: Compute the entries of the scaling matrix as
$q_{k,i} = p_n \left((\mathbf{H}\mathbf{H}^{\mathsf{H}})^{-1}\right)_{n,n} \quad \text{with} \quad n = (k-1)\,N_S + i, \quad \text{for } k = 1, \ldots, K \text{ and } i = 1, \ldots, N_S.$

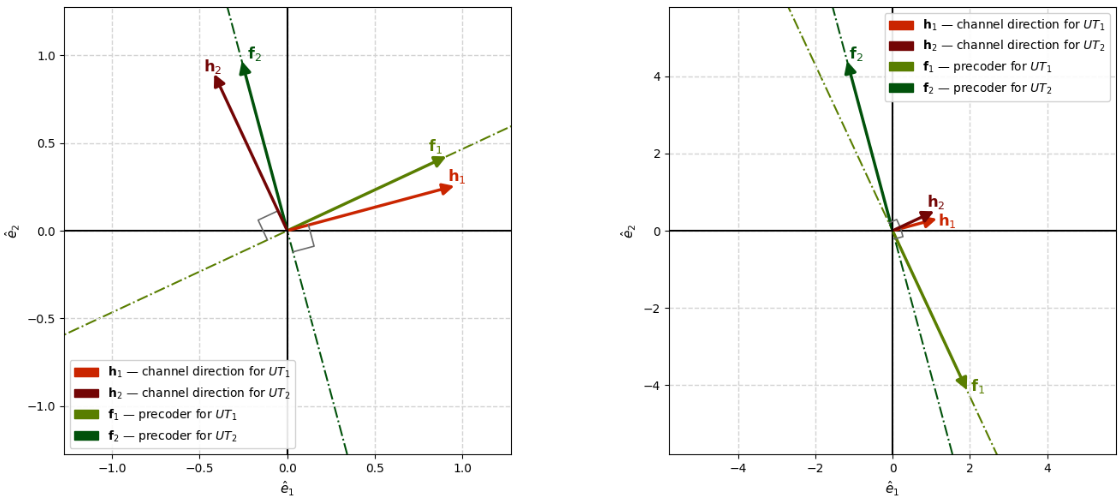
3: Compute the compound precoder matrix as
$\mathbf{F} = \mathbf{H}^{\mathsf{H}}(\mathbf{H}\mathbf{H}^{\mathsf{H}})^{-1}\sqrt{\mathbf{Q}}$

Return $\mathbf{F}$

To build an intuitive understanding of ZF precoding, consider a downlink MU-MIMO system with two single-antenna UTs and a BS with two transmit antennas. In this case, the channel matrix is formed by two row vectors, each describing the channel direction toward one user, while the precoder consists of two column vectors, each defining the transmit direction of the data stream intended for a specific UT. The ZF design chooses each precoding vector to be orthogonal to the channel vector of the non-intended user. This ensures that the signal intended for one UT does not leak into the receive antenna of the other UT, thereby eliminating inter-user interference. This geometry is illustrated in Figure 3.1, where Figure 3.1a corresponds to favorable channel conditions and Figure 3.1b to unfavorable channel conditions.

In the favorable case, the channel directions for the two UTs are nearly orthogonal. As a result, the ZF precoding vectors can remain close to the intended channel directions, which preserves a high effective channel gain. In the unfavorable case, the channel directions are strongly correlated (nearly parallel), so the ZF constraints force the precoding vectors away from the desired channel directions. This substantially reduces the effective gain.

For a fair geometric comparison, the channel vectors in both figures are normalized, and the precoding vectors are scaled such that each stream attains unit effective channel gain. Under this normalization, the much larger precoder norms in the unfavorable case indicate a significantly higher transmit-power requirement to achieve the same effective gain. With a fixed total transmit power in practice, this translates directly into lower received SNR and therefore a lower achievable SR.



(a) favorable channel conditions

(b) unfavorable channel conditions

Figure 3.1: Visualization of ZF precoder for a MU-MIMO system with 2 single-antenna UTs and a 2-antenna BS under (a) favorable and (b) unfavorable channel conditions. For both scenarios, the channel vectors are normalized and the precoding vectors are scaled such that the effective channel gain for each data stream is equal to 1. Note the much larger norm of the precoding vectors in the unfavorable scenario compared to the favorable scenario, which illustrates that a lot more power is required to achieve the same effective channel gain when the channel vectors of the UTs are strongly correlated.

Figure 3.2 illustrates what happens when the BS is equipped with an additional transmit antenna while the number of single-antenna UTs remains equal to two. The channel matrix is still formed by two row vectors, one for each user, and the precoder by two column vectors. However, these vectors now lie in a three-dimensional space. As a result, the null-space orthogonal to each user's channel direction is no longer a line but a two-dimensional plane.

In order to eliminate all interference, the precoding vector for each user can be chosen anywhere in the null-space of the other users' channel vectors, that is, in the plane orthogonal to the channel directions of the other users. The best choice is to select the precoding vectors as close as possible to the intended channel directions, which corresponds to projecting the channel vectors onto the respective orthogonal planes. This maximizes the effective channel gain for each data stream, since it allows as much power as possible to be transmitted in the direction of the intended UT.

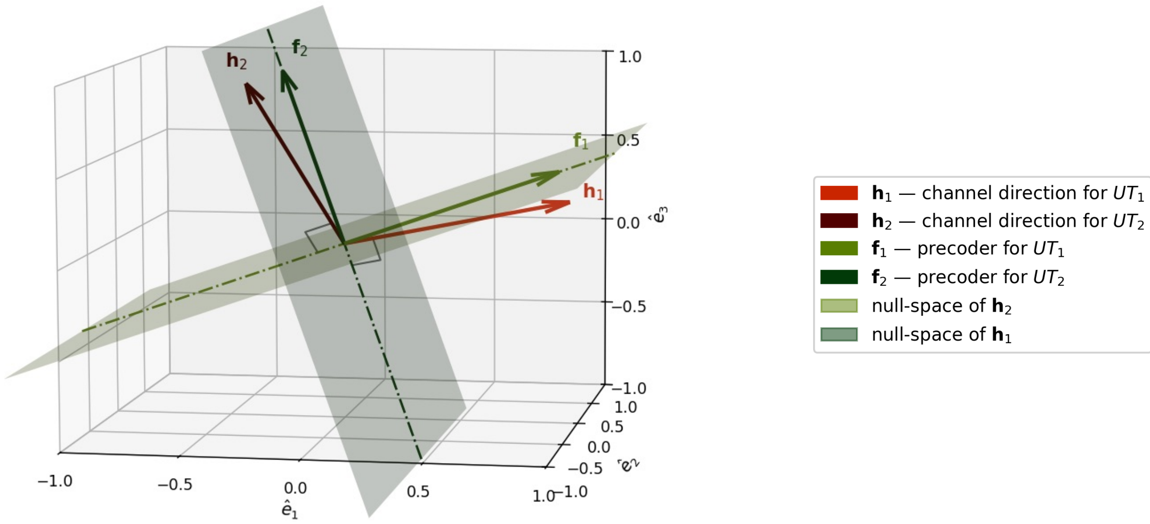


Figure 3.2: Visualization of ZF precoder for a MU-MIMO system with 2 single-antenna UTs and a 3-antenna BS. The channel vectors are normalized and the precoding vectors are scaled such that the effective channel gain for each data stream is equal to 1. With three transmit antennas, the null-space orthogonal to each user's channel direction is two-dimensional. The ZF precoding vector is chosen as the projection of the intended user's channel direction onto this plane, which minimizes the deviation from the desired channel direction while preserving the zero-forcing constraint.

3.2 Zero-Forcing Precoding with an Heuristic LSV Combiner

Pure ZF precoding is clearly suboptimal because it does not exploit receiver-side combining at the UTs. A practical extension is therefore to equip each UT with a heuristic left singular vector (LSV) combiner, while keeping the BS precoder design based on ZF. The key idea is to improve the effective channel gains seen without increasing the signaling overhead. Unlike coordinated designs, where the BS jointly computes the compound precoder and the combiners and then broadcasts the combiners to the UTs, this approach is non-coordinated: each UT computes its own combiner locally.

The procedure is as follows. First, the BS transmits pilot symbols, from which each UT estimates its downlink channel matrix $\mathbf{H}_k$. Next, each UT constructs its heuristic combiner matrix $\mathbf{G}_k$ from the N_S dominant left singular vectors of $\mathbf{H}_k$, analogous to combiner for capacity-achieving precoding in the SU-MIMO scenario. Each UT then forms its virtual channel $(\mathbf{G}_k\mathbf{H}_k)$ and feeds this matrix back to the BS. Finally, the BS stacks these virtual channels and applies the standard ZF precoding procedure from the previous section on the virtual channel.

This method was proposed by Sander Cornelis in [8] and is particularly attractive in feedforward-limited scenarios. The complete procedure is summarized in Algorithm 2.

3.3 Block Diagonalization Precoding

The last precoding technique in the class of ZF precoders is a coordinated technique called block diagonalization (BD) precoding [21]. In contrast to pure ZF, which cancels both inter-user

Algorithm 2 ZF Precoding with Heuristic LSV combiner

Require:
- The compound channel matrix $\mathbf{H}$;
- The total transmit power P_t and the received noise power N_0.

Ensure: The compound precoder matrix $\mathbf{F}$ and the combiner matrices $\mathbf{G}_k$ for each UT.

1: Compute the combiner matrices $\mathbf{G}_k$ for each UT as
$\mathbf{G}_k = \begin{bmatrix} \mathbf{u}_{k,1} & \dots & \mathbf{u}_{k,N_S} \end{bmatrix}^{\mathsf{H}}, \quad \text{with} \quad \mathbf{H}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^{\mathsf{H}}$

2: Compute the compound precoder matrix $\mathbf{F}$ as using the ZF algorithm (Alg. 1) with inputs
$\mathbf{H} = (\mathbf{GH}), \quad P_t = P_t, \quad N_0 = N_0.$

Return $\mathbf{F}, \quad \mathbf{G}_k$ for $k = 1, \dots, K$

and inter-stream interference, BD only suppresses the inter-user interference. As a result, the compound transfer matrix is no longer required to be diagonal, but only block-diagonal, so that each UT receives a signal vector free of contributions of data streams intended for other UTs. The downlink MU-MIMO channel is thereby decomposed into K parallel SU-MIMO channels. Once this first step is completed, the SU-MIMO precoding and combining techniques developed in Chapter 2 can be applied independently to each subsystem.

The zero inter-user interference condition requires that the precoder $\mathbf{F}_k$ intended for UT k lie in the null space of the channel matrices of all other users. This leads to the constraints

$$\forall\, k' \neq k : \qquad \mathbf{H}_{k'} \mathbf{F}_k = \mathbf{0}, \qquad \text{for} \quad k = 1, \dots, K. \tag{3.8}$$

To characterize these constraints more explicitly, define the interfering channel matrix $\mathring{\mathbf{H}}_k$ of UT k as the vertically stacked channel matrices of all other users:

$$\mathring{\mathbf{H}}_k = \begin{bmatrix} \mathbf{H}_1^{\mathsf{H}} & \dots & \mathbf{H}_{k-1}^{\mathsf{H}} & \mathbf{H}_{k+1}^{\mathsf{H}} & \dots & \mathbf{H}_K^{\mathsf{H}} \end{bmatrix}^{\mathsf{H}} \in \mathbb{C}^{(K-1)N_R \times N_T}. \tag{3.9}$$

The constraint in (3.8) is then equivalent to forcing the transmit directions for UT k to lie within the null-space of its interfering channel matrix.
To obtain a basis for this null space, consider the SVD of the interfering channel matrix:

$$\mathring{\mathbf{H}}_k = \mathring{\mathbf{U}}_k \, \mathring{\mathbf{\Sigma}}_k \begin{bmatrix} \mathring{\mathbf{V}}_k^{(1)} & \mathring{\mathbf{V}}_k^{(0)} \end{bmatrix}^{\mathsf{H}}. \tag{3.10}$$

The first $R_{\mathring{H}_k}$ right singular vectors of $\mathring{\mathbf{H}}_k$, collected in $\mathring{\mathbf{V}}_k^{(1)}$, span the row space of $\mathring{\mathbf{H}}_k$, whereas the remaining $(N_T - R_{\mathring{H}_k})$ right singular vectors, collected in $\mathring{\mathbf{V}}_k^{(0)}$, span its null space. By $R_{\mathring{H}_k}$ we denote the rank of the interfering channel matrix $\mathring{\mathbf{H}}_k$. Therefore, any precoder of the form

$$\mathbf{F}_k^{\mathrm{MU}} = \mathring{\mathbf{V}}_k^{(0)} \tag{3.11}$$

automatically satisfies the zero inter-user interference condition.

Data transmission to UT k is only possible if the null space of $\mathring{\mathbf{H}}_k$ is non-empty, i.e. it has a dimension greater than zero. This condition is satisfied if $R_{\mathring{H}_k} < N_T$, which is guaranteed under the standing assumption that the number of transmit antennas is greater than or equal to the total number of receive antennas across all users, i.e. $N_T \geq K N_R$. In addition, the channel as seen after the projection of $\mathbf{F}_k^{\mathrm{MU}}$, i.e. $(\mathbf{H}_k \mathring{\mathbf{V}}_k^{(0)})$, must still retain useful spatial dimensions. In other words, the channel directions of UT k should not be entirely contained in the row space of the interfering UTs. A sufficient condition is that at least one row of $\mathbf{H}_k$ is linearly independent of the rows of $\mathring{\mathbf{H}}_k$ [22]. This shows that BD, like ZF, benefits from users with sufficiently distinct spatial signatures, i.e. UTs without highly correlated channel directions. Note that pure ZF imposes the stronger requirement that all rows of $\mathbf{H}_k$ are linearly independent of $\mathring{\mathbf{H}}_k$. While this is not necessary for BD, it would result in a higher rank of $(\mathbf{H}_k \mathring{\mathbf{V}}_k^{(0)})$, and therefore greater degrees of freedom in the subsequent SU-MIMO design step.

If, in a first step, the receiver combiners are ignored and the multi-user precoder is chosen as $\mathbf{F}_k^{\mathrm{MU}} = \mathring{\mathbf{V}}_k^{(0)}$, then the compound MU-MIMO input-output relation simplifies to

$$\mathbf{y}_k = (\mathbf{H}_k \mathring{\mathbf{V}}_k^{(0)})\, \mathbf{a}_k + \mathbf{n}_k, \qquad \text{for} \quad k = 1, \ldots, K. \tag{3.12}$$

Hence, the original downlink MU-MIMO system is transformed into K parallel SU-MIMO subsystems, each with $(N_T - R_{\mathring{H}_k})$ effective transmit antennas, N_R receive antennas, and an effective channel matrix $\mathbf{H}_k \mathring{\mathbf{V}}_k^{(0)}$. The second step is then to apply a SU-MIMO precoder and combiner design to each of these effective channels. In this dissertation, the focus is on maximum-information-rate design. Therefore, we adapt the capacity-achieving precoding technique from Section 2.1.2, so that each effective channel is diagonalized by means of its SVD, followed by a global waterfilling power allocation across all resulting eigen subchannels. The full BD procedure is summarized in Algorithm 3.

Algorithm 3 BD Precoding

Require:
- The compound channel matrix $\mathbf{H}$;
- The total transmit power P_t and the received noise power N_0.

Ensure: The compound precoder matrix $\mathbf{F}$ and the combiner matrices $\mathbf{G}_k$ for each UT.

1: Construct the interfering channel matrix for each UT k as
$\mathring{\mathbf{H}}_k = \begin{bmatrix} \mathbf{H}_1^{\mathsf{H}} & \ldots & \mathbf{H}_{k-1}^{\mathsf{H}} & \mathbf{H}_{k+1}^{\mathsf{H}} & \ldots & \mathbf{H}_K^{\mathsf{H}} \end{bmatrix}^{\mathsf{H}} \in \mathbb{C}^{(K-1)N_R \times N_T}$.

2: Compute the MU precoder matrix for each UT k as
$\mathbf{F}_k^{\mathrm{MU}} = \mathring{\mathbf{V}}_k^{(0)}, \quad \text{with} \quad \mathring{\mathbf{H}}_k = \mathring{\mathbf{U}}_k \, \mathring{\mathbf{\Sigma}}_k \begin{bmatrix} \mathring{\mathbf{V}}_k^{(1)} & \mathring{\mathbf{V}}_k^{(0)} \end{bmatrix}^{\mathsf{H}}$.

3: Compute the SU precoder and combiner matrix for each UT k as
$\mathbf{F}_k^{\mathrm{SU}} = \begin{bmatrix} \bar{\mathbf{v}}_{k,1} & \ldots & \bar{\mathbf{v}}_{k,N_S} \end{bmatrix}$ and $\mathbf{G}_k = \begin{bmatrix} \bar{\mathbf{u}}_{k,1} & \ldots & \bar{\mathbf{u}}_{k,N_S} \end{bmatrix}^{\mathsf{H}}, \quad \text{with} \quad (\mathbf{H}_k \mathring{\mathbf{V}}_k^{(0)}) = \bar{\mathbf{U}}_k \bar{\mathbf{\Sigma}}_k \bar{\mathbf{V}}_k^{\mathsf{H}}$

4: Extract the eigen subchannel gains as
$\sigma_n = \left(\bar{\mathbf{\Sigma}}_k\right)_{i,i} \quad \text{with} \quad n = (k-1)\, N_S + i, \quad \text{for } k = 1, \ldots, K \text{ and } i = 1, \ldots, N_S.$

5: Compute the optimal power allocation p_i using the waterfilling algorithm (Alg. 5) with inputs
$P_t = P_t, \quad \alpha_n = 1, \quad \text{and} \quad \gamma_n = \sigma_n^2 / N_0.$

6: Form the compound precoder matrix as
$\mathbf{F} = \begin{bmatrix} \mathbf{F}_1^{\mathrm{MU}} \mathbf{F}_1^{\mathrm{SU}} & \ldots & \mathbf{F}_K^{\mathrm{MU}} \mathbf{F}_K^{\mathrm{SU}} \end{bmatrix} \sqrt{\mathbf{P}} \quad \text{with} \quad \mathbf{P} = \mathrm{diag}(p_1, \ldots, p_{K N_S})$

Return $\mathbf{F}$, $\mathbf{G}_k$ for $k = 1, \ldots, K$

As with pure ZF, the chosen power allocation may devote very little power to users whose effective channels are weak. A more balanced alternative is to divide the total transmit power equally across the users and then apply waterfilling separately over the eigen subchannels of each SU-MIMO subsystem. More generally, the second stage of the BD design need not be based on maximum sum-rate. Other SU-MIMO design criteria, such as QoS-based or MMSE-based precoding, can equally be applied to the parallel channels obtained after the block-diagonalization step.

To conclude this section, Figure 3.3 compares the achievable-rate performance of BD precoding in a downlink MU-MIMO setting with the capacity-achieving SVD precoding benchmark in a corresponding SU-MIMO setting. The reference system is a 6×2 SU-MIMO link, whereas the multi-user system is an $8 \times \{2, 2\}$ MU-MIMO link with two UTs, each equipped with two receive antennas. The BD algorithm decouples the MU-MIMO system into 2 parallel SU-MIMO subsystems with, on average, each 6 effective transmit antennas and 2 receive antennas. The rates are averaged over 1000 i.i.d. Rayleigh channel realizations.

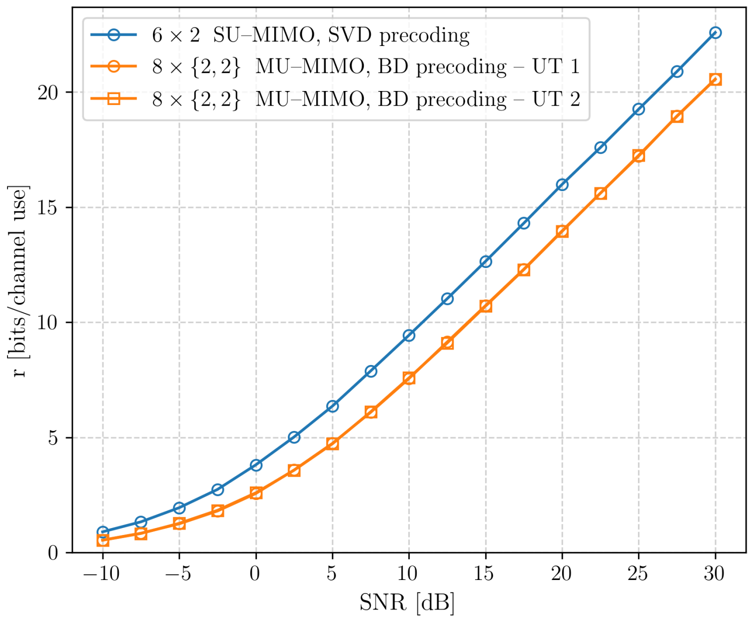


Figure 3.3: Comparison between the achievable rate of a 6×2 SU-MIMO system using SVD precoding and the achievable rates of both UTs in an $8 \times \{2, 2\}$ MU-MIMO system using BD precoding. Results are averaged over 1000 i.i.d. Rayleigh channel realizations. The SNR is defined as the ratio between the total transmit power P_t at the BS and the noise power per receive antenna N_0.

The curves show that the achievable-rate slope as a function of SNR is essentially the same for all users. In addition, the SU benchmark exhibits only a small SNR advantage over each user in the multi-user (MU) scenario. The two users in the MU scenario achieve nearly identical rates, which is expected because the MU system is statistically symmetric with respect to the users and the results are averaged over multiple channel realizations.

This comparison indicates that, on average, adding two transmit antennas at the BS and switching from SVD precoding in SU-MIMO to BD precoding in MU-MIMO enables serving an additional two-stream user with the same reliability only a minor per-user throughput penalty. From a system-design perspective, this highlights the practical spatial multiplexing capability of BD precoding. More generally, as more UTs are served simultaneously, the slope of the total SR versus SNR increases, at the cost of a modest reduction in the achievable rate of each individual UT.

3.4 Weighted Minimum Mean Square Error Precoding

The three precoding techniques discussed so far all enforce hard structural constraints on the precoders to handle the interference. Pure ZF and ZF with an LSV combiner cancel it completely through a channel inversion, while BD confines each user's precoder to the null space of the other users' channels. In both cases, satisfying these structural constraints prevents the precoder from operating in directions that could simultaneously offer high signal gain and tolerable interference, and thus limits the achievable performance. None of these designs directly maximizes the weighted sum rate (WSR) of the system. In this section, we present a precoding technique with the explicit goal of maximizing the WSR under a total transmit power constraint.

Consider the WSR maximization problem:

$$\max_{\{\mathbf{F}_k\}} \sum_{k=1}^{K} w_k R_k \qquad \text{s.t.} \quad \sum_{k=1}^{K} \operatorname{tr}\Big(\mathbf{F}_k \mathbf{F}_k^{\mathsf{H}}\Big) \leq P_t. \tag{3.13}$$

Unlike its single-user counterpart in Chapter 2, problem (3.13) is non-convex in $\{\mathbf{F}_k\}$ and admits no closed-form solution. The state-of-the-art approach, proposed by the authors of [23], is to recast it as an equivalent WMMSE problem and to solve the latter by alternating optimization.

Consider the system input–output relation for UT k from (1.9). The signals intended for the other users act, under Gaussian signaling, as additional Gaussian noise with covariance matrix

$$\mathbf{C}_{\tilde{n}_k} = N_0 \, \mathbf{I}_{N_R} + \sum_{k' \neq k} \mathbf{H}_k \mathbf{F}_{k'} \mathbf{F}_{k'}^{\mathsf{H}} \mathbf{H}_k^{\mathsf{H}}. \tag{3.14}$$

Consequently, the achievable rate of UT k can be written similarly to the SU case as

$$R_k = \log_2 \det\Big(\mathbf{I}_{N_S} + \mathbf{F}_k^{\mathsf{H}} \mathbf{H}_k^{\mathsf{H}} \mathbf{C}_{\tilde{n}_k}^{-1} \mathbf{H}_k \mathbf{F}_k\Big). \tag{3.15}$$

For a given set of precoders, the minimum mean squared error (MMSE) combiner of UT k is the Wiener filter that minimizes the mean–square estimation error in the presence of inter–user interference and noise, and is given by [24]:

$$\mathbf{G}_k^{\text{MMSE}} = \mathbf{F}_k^{\mathsf{H}} \mathbf{H}_k^{\mathsf{H}} \big(\mathbf{H}_k \mathbf{F}_k \mathbf{F}_k^{\mathsf{H}} \mathbf{H}_k^{\mathsf{H}} + \mathbf{C}_{\tilde{n}_k}\big)^{-1}. \tag{3.16}$$

The corresponding MSE matrix, defined as in Section 2.2.1 but specialized to UT k, then reduces to [25]

$$\mathbf{E}_k = \mathbb{E}\Big[(\mathbf{a}_k - \mathbf{z}_k)(\mathbf{a}_k - \mathbf{z}_k)^{\mathsf{H}}\Big] = \big(\mathbf{I}_{N_S} + \mathbf{F}_k^{\mathsf{H}} \mathbf{H}_k^{\mathsf{H}} \mathbf{C}_{\tilde{n}_k}^{-1} \mathbf{H}_k \mathbf{F}_k\big)^{-1}. \tag{3.17}$$

Comparing (3.15) and (3.17), the achievable rate of UT k can be expressed directly in terms of its MMSE matrix as [23]

$$R_k = \log_2 \det\big(\mathbf{E}_k^{-1}\big). \tag{3.18}$$

This identity is the key bridge between the WSR maximization problem and the WMMSE minimization problem. It is the multi-user analogue of the rate–MSE link already established in the generalized WMMSE design of Section 2.2.1.

Define the WMMSE optimization problem, in analogy to its SU counterpart in (2.15), as

$$\min_{\{\mathbf{F}_k\}} \sum_{k=1}^{K} \mathrm{tr}(\mathbf{W}_{\mathrm{e},\,k}\,\mathbf{E}_k) \qquad \mathrm{s.\,t.} \quad \sum_{k=1}^{K} \mathrm{tr}(\mathbf{F}_k\mathbf{F}_k^{\mathsf{H}}) \leq P_t, \tag{3.19}$$

where each $\mathbf{W}_{\mathrm{e},k} \in \mathbb{C}^{N_S \times N_S}$ is a positive-definite weight matrix associated with UT k and $\mathbf{E}_k$ is the MMSE matrix in (3.17). Whereas in the SU case the weight matrix could be chosen a priori so as to recover the maximum-information-rate design (cf. Section 2.2.2), the MU setting is more involved. Comparing the gradients of the Lagrangians of optimization problems (3.13) and (3.19) with respect to $\mathbf{F}_k$ shows that they coincide whenever [23]

$$\mathbf{W}_{\mathrm{e},\,k} = w_k\,\mathbf{E}_k^{-1} \qquad \text{for} \quad k = 1, \ldots, K. \tag{3.20}$$

At any stationary point of the WSR problem, computing $\{\mathbf{E}_k\}$ from (3.17) and selecting the weights as in (3.20) therefore yields a stationary point of the WMMSE problem with the same precoders, and vice versa. This equivalence is the MU counterpart of the special case $\mathbf{W}_{\mathrm{e}} = \boldsymbol{\Lambda}$ identified in Section 2.2.2 for the SU channel. However, in the MU case, the weights depend on the precoders themselves through $\mathbf{E}_k$, because the precoder of every user influences the interference experienced by all other users. The WSR problem can therefore not be solved by a single WMMSE design with fixed weights, but must instead be tackled iteratively by alternating between updating the weights and solving the WMMSE problem.

The WMMSE algorithm exploits the equivalence above by cycling, at each iteration, through three closed-form updates. First, the MMSE combiners are updated according to (3.16). Second, the weights are updated according to (3.20). Third, the compound precoder update solves the WMMSE problem (3.19) for fixed combiners and weights, which admits the closed-form solution [26]

$$\bar{\mathbf{F}} = \Big(\mathbf{H}^{\mathsf{H}}\mathbf{G}^{\mathsf{H}}\mathbf{W}\mathbf{G}\mathbf{H} + \mathrm{tr}(\mathbf{W}\mathbf{G}\mathbf{G}^{\mathsf{H}})\,\frac{N_0}{P_t}\,\mathbf{I}_{N_T}\Big)^{-1}\mathbf{H}^{\mathsf{H}}\mathbf{G}^{\mathsf{H}}\mathbf{W}, \tag{3.21}$$

followed by the rescaling $\mathbf{F} = \beta\,\bar{\mathbf{F}}$ with $\beta^2 = P_t/\mathrm{tr}(\bar{\mathbf{F}}\bar{\mathbf{F}}^{\mathsf{H}})$ to enforce the total power constraint with equality. The full procedure is summarized in Algorithm 4.

It is shown in [23] that the WMMSE precoder converges to a local optimum of the WSR problem. Because the problem is non-convex, the quality of the obtained solution depends on the initialization $\{\mathbf{F}_k^{(0)}\}$. Furthermore, the authors pointed out that for any set of precoders, a rate-equivalent set can be constructed such that the MSE matrices are diagonal for all users. This justifies the restriction of the weight update to the diagonal elements, and implies that each UT

Algorithm 4 WMMSE Precoding

Require:
- The compound channel matrix $\mathbf{H}$;
- The total transmit power P_t and the received noise power N_0.

Ensure: The compound precoder matrix $\mathbf{F}$ and the combiner matrices $\mathbf{G}_k$ for each UT.

1: Initialize the precoders as
$\mathbf{F}_k = \mathbf{F}_k^{(0)} \quad \text{with} \quad \mathbf{F}_k^{(0)} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}) \quad \text{for} \quad k = 1, \ldots, K.$

2: **repeat**

3: Update the combiners as
$\mathbf{G}_k = \mathbf{F}_k^{\mathsf{H}} \mathbf{H}_k^{\mathsf{H}} \left(\mathbf{H}_k \mathbf{F}_k \mathbf{F}_k^{\mathsf{H}} \mathbf{H}_k^{\mathsf{H}} + \mathbf{C}_{\tilde{n}_k}\right)^{-1} \quad \text{for} \quad k = 1, \ldots, K.$

4: Update the weights as
$\mathbf{W}_k = \mathbf{E}_k^{-1} \quad \text{for} \quad k = 1, \ldots, K.$

5: Update the precoder as
$\bar{\mathbf{F}} = \left(\mathbf{H}^{\mathsf{H}} \mathbf{G}^{\mathsf{H}} \mathbf{W} \mathbf{G} \mathbf{H} + \mathrm{tr}(\mathbf{W} \mathbf{G} \mathbf{G}^{\mathsf{H}}) \, \tfrac{N_0}{P_t} \, \mathbf{I}_{N_T}\right)^{-1} \mathbf{H}^{\mathsf{H}} \mathbf{G}^{\mathsf{H}} \mathbf{W}$
$\mathbf{F} = \beta \, \bar{\mathbf{F}} \quad \text{with} \quad \beta^2 = P_t / \mathrm{tr}(\bar{\mathbf{F}} \bar{\mathbf{F}}^{\mathsf{H}})$

6: **until** convergence

Return $\mathbf{F}$, $\mathbf{G}_k$ for $k = 1, \ldots, K$.

can detect its N_S data streams independently. The implementation used throughout this work is this diagonal-restricted version. It runs 10 iterations for 2 different random initializations and retains the precoder that yields the highest WSR.

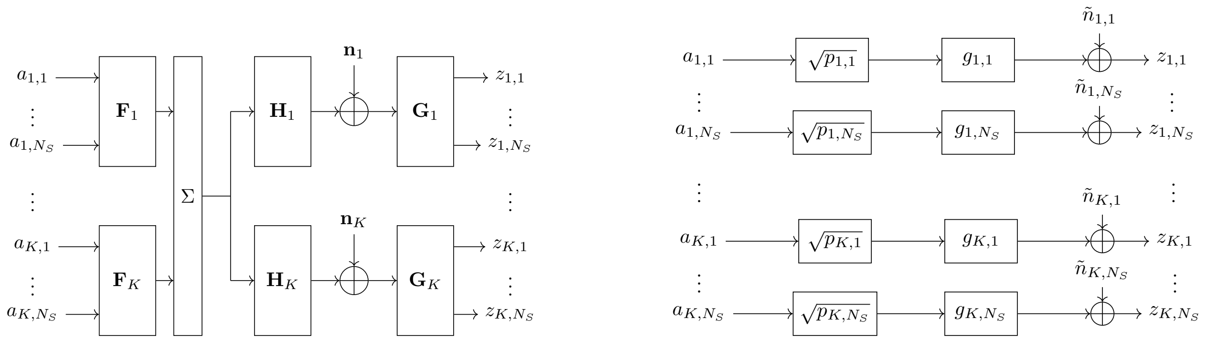
3.5 Performance Evaluation

The previous sections derived linear transceiver designs for downlink MU-MIMO sum-rate maximization under different structural constraints. This section evaluates their behavior through Monte Carlo simulations. First, the three ZF–type methods are compared through the statistics of their effective subchannel gains. Second, the achievable SR of all four linear precoding techniques is compared for two different system configurations.

The simulations are conducted under the same assumptions as in Chapter 2: the channel is modeled as an i.i.d. Rayleigh block fading channel (see Appendix C), the reported performance metrics are averaged over at least 100 independent channel realizations, and perfect CSI is assumed at both the BS and the UTs.

3.5.1 Comparison of the ZF Precoders

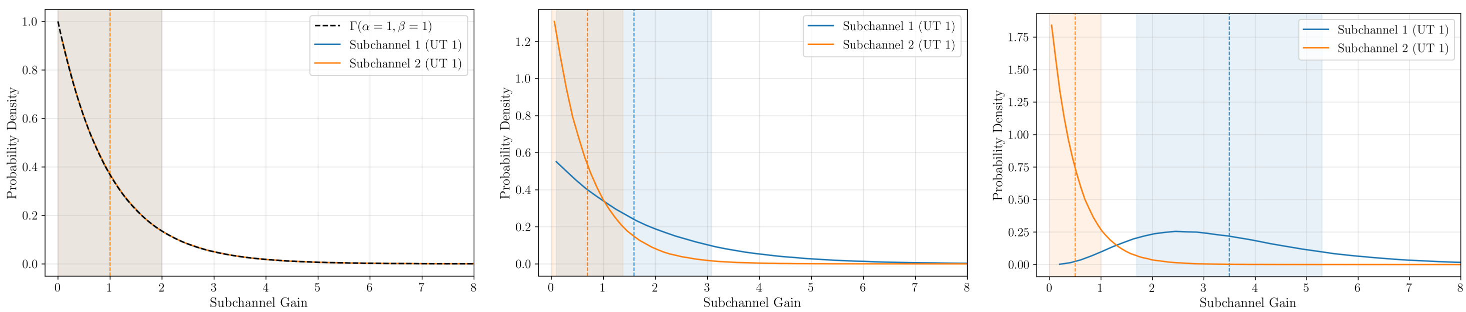
This subsection compares the three ZF–type designs under the maximum-information-rate criterion: pure ZF, ZF with heuristic LSV combining, and BD. In all three cases, the end-to-end processing transforms the original MU-MIMO channel into parallel scalar Gaussian subchannels, as illustrated in Figure 3.4. Hence, for a fixed power budget and waterfilling policy, performance is completely determined by the statistics of the effective subchannel gains g_n. The stronger these effective channel gains, the better the overall system performance.



(a) **(b)**

Figure 3.4: A block diagram of the original MU-MIMO channel (a) and its decomposition into parallel Gaussian SISO subchannels (b), as achieved by the ZF–type precoding techniques.

Figure 3.5 reports the marginal pdfs for a single user in an system with 8 transmit antennas at the BS and 4 two-antenna UTs, obtained through Monte Carlo simulations. Because users with the same number of receive antennas are statistically symmetric, this user is representative for the full set of users. For pure ZF, the effective subchannel gains are given by $g_n = \left((\mathbf{H}\mathbf{H}^{\mathsf{H}})^{-1}\right)^{-1}_{n,n}$ and, under an i.i.d. Rayleigh fading model, their distribution follows from the inverse-Wishart model [27, 28]. In particular, each g_n is Gamma-distributed with shape parameter $\alpha = N_T - KN_R + 1$ and rate parameter $\beta = 1$. For ZF with an heuristic LSV combiner and for BD precoding, no analytical distribution is derived..



(a) ZF **(b)** ZF + LSV Combiner **(c)** BD

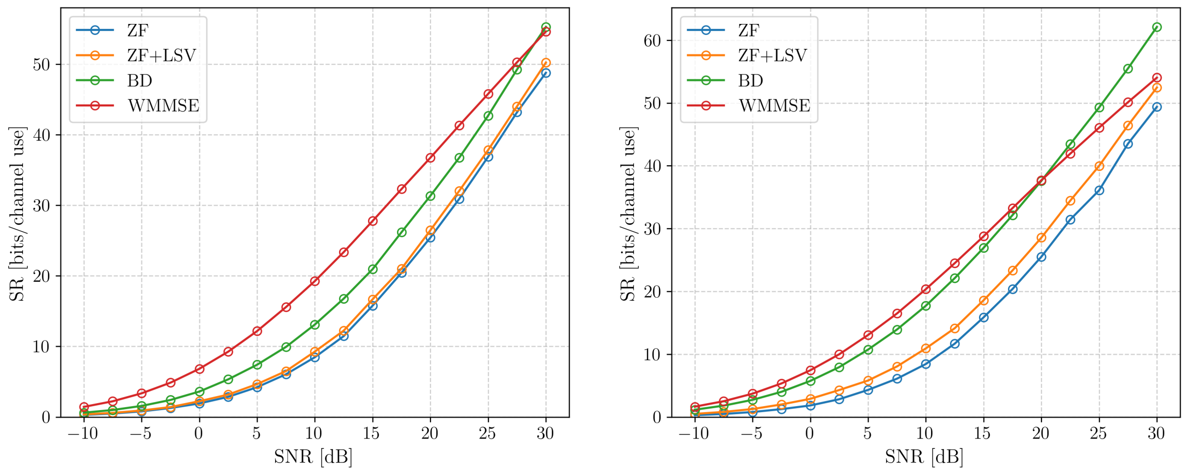
Figure 3.5: The marginal pdfs of the effective subchannel gains for a single UT in a $8 \times \{2, 2, 2, 2\}$ MU-MIMO system, compared between the different ZF–type precoding techniques. The expected value is indicated by a vertical dashed line, and the shaded area around the expected value corresponds to the standard deviation.

For pure ZF (Fig. 3.5a), the two subchannels of a single UT have identical marginal distributions, consistent with the analytical result. Adding the heuristic LSV combiner (Fig. 3.5b) breaks this symmetry between the two subchannels. The first combiner vector is aligned with the dominant left singular vector of $\mathbf{H}_k$, which concentrates the available receive-side spatial gain onto the dominant subchannel at the expense of the weaker one. This creates a slightly more favorable result compared to pure ZF. BD precoding (Fig. 3.5c) achieves substantially larger gains for the dominant subchannel than those observed under the ZF with LSV combiner technique. The sum of the expected subchannel gains is clearly the largest, due to the better utilization of the spatial degrees of freedom and the coordinated design. Similar observations hold for other antenna configurations.

These differences in subchannel gain statistics translate directly into differences in achievable sum rate, as will be shown in the next section.

3.5.2 Achievable Sum Rate Comparison

Figure 3.6 compares the achievable SR of all four precoding techniques for two fully loaded MU-MIMO configurations, where "fully loaded" means that the total number of receive antennas equals the number of transmit antennas ($KN_R = N_T$). The two configurations differ only in how the receive antennas are distributed across UTs: four UTs with two receive antennas each in Fig. 3.6a and two UTs with four receive antennas each in Fig. 3.6b.



(a) $8 \times \{2, 2, 2, 2\}$ MU-MIMO

(b) $8 \times \{4, 4\}$ MU-MIMO

Figure 3.6: Comparison of the achievable sum rate for the considered MU linear precoding techniques in two fully loaded MU-MIMO systems. The results are averaged over at least 1000 i.i.d. Rayleigh channel realizations.

In both configurations, the ranking

$$SR_{\text{WMMSE}} > SR_{\text{BD}} > SR_{\text{ZF+LSV}} > SR_{\text{ZF}}$$

holds at moderate SNR, consistent with the increasing level of design flexibility across the four methods. Pure ZF imposes the strictest constraints, requiring the precoder to invert the full compound channel matrix with no receiver-side processing. ZF with LSV combining retains the same precoder structure but introduces receive-side spatial processing. BD further relaxes the structural constraints from per-stream to per-user orthogonality and exploits receiver combining through a coordinated design, yielding a substantial gain in achievable SR. WMMSE places no structural constraint on the precoder and directly optimizes the WSR, achieving the highest SR at moderate SNR. The performance differences manifest as a horizontal SNR offset rather than a change in growth rate, because all four techniques achieve the same multiplexing gain.

A notable exception to the above ranking occurs at very high SNR in the $8 \times \{4, 4\}$ configuration, where BD eventually overtakes WMMSE. This crossover arises because BD approaches the globally optimal solution for the maximum-information-rate criterion as SNR grows [21], while WMMSE may converge to a suboptimal local optimum in this regime.

Comparing the two configurations also reveals how the distribution of receive antennas affects performance. For pure ZF, the two curves are virtually identical: since no receiver combining is applied and the total number of receive antennas is the same, the achievable SR depends only on the compound channel and is unaffected by how the antennas are grouped across UTs. For ZF with LSV combining, BD and WMMSE, concentrating the receive antennas on fewer UTs yields a consistently higher SR. The benefit of concentrating antennas is more pronounced for BD than for ZF+LSV, which reflects the fact that the coordinated BD design makes fuller use of the per-user spatial degrees of freedom than the non-coordinated LSV heuristic.

The simulation results confirm the position of WMMSE as the state-of-the-art linear MU-MIMO precoding technique under the maximum-SR criterion. BD provides a close-to-optimal coordinated alternative that does not require iterative optimization, while ZF+LSV offers a low-overhead non-coordinated baseline. The choice between these techniques in practice therefore involves a tradeoff between achievable SR, computational complexity, and signaling overhead. In the next chapters, performance evaluations are carried out using the WMMSE precoder.

Part II

Precoding for Mobile Satellite Systems

Chapter 4

High-Throughput Satellite Communication Systems

The preceding chapters developed the theory of linear precoding for generic SU- and MU-MIMO systems. Before turning to the specific problem addressed by this work, the present chapter situates these results within the context of satellite communications. We first motivate the use of precoding in modern high-throughput satellite (HTS) systems and describe the system architecture considered in the remainder of this work. We then discuss the channel features and modeling assumptions that distinguish the satellite setting from the terrestrial MU-MIMO scenario of Chapter 3. Finally, we review the main families of precoding techniques proposed in the literature and identify the specific category that is relevant for our work.

4.1 Motivation

Satellite communications have recently regained considerable attention as a key enabler of ubiquitous broadband connectivity. They provide cost-efficient coverage of large geographical areas, including regions where deploying a terrestrial infrastructure is impractical, and are envisioned as an integral component of future 5G and beyond networks [29, 30].

To meet the rapidly growing demand for data rates, modern satellite systems have evolved from single-beam architectures toward high-throughput satellite (HTS) systems, in which the coverage area is partitioned into many narrow spot beams. This multi-beam architecture enables spatial multiplexing by concurrently delivering independent data streams to geographically separated users. In conventional designs, inter-beam interference is managed through partial frequency reuse and polarization isolation, whereby adjacent beams are assigned non-overlapping frequency sub-bands or orthogonal polarizations. While effective at reducing interference, this approach significantly constrains spectral efficiency, since the full available bandwidth cannot be reused across all beams simultaneously. On the other hand, full frequency reuse maximizes the spectrum available per beam. The price to pay is a substantial increase in inter-beam interference: signals intended for one beam leak into adjacent beams through the sidelobes of the antenna radiation pattern, and the system transitions from a noise-limited regime to an interference-limited one.

Precoding has emerged as the natural solution: by exploiting CSI at the transmitter, interference can be suppressed while the available spectrum is reused across all beams [29].

4.2 Modeling Aspects

System Architecture

An illustration of the system architecture of a multi-beam satellite system is shown in Figure 4.1. It consists of three main entities: the gateway (GW), the satellite or BS, and the UTs. The GW is connected to the core network and usually performs all baseband processing. The link between the GW and the satellite is referred to as the feeder link, while the link between the satellite and the UTs is referred to as the user link. The direction from the GW throught the BS to the UTs is called the forward link, while the opposite direction is called the return link. The forward link is by far the most rate-demanding direction, and centralizing the precoder design at the GW (or BS) aligns naturally with the MU-MIMO downlink framework developed in Chapter 3.

Throughout this work, we focus on the forward link of a multi-beam HTS system serving mobile UTs.

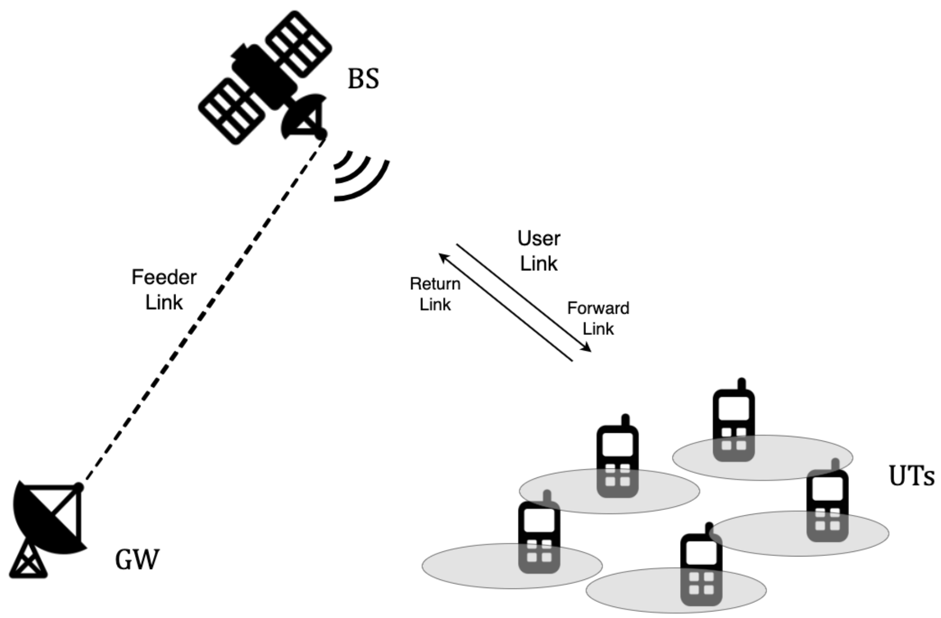


Figure 4.1: Schematic view of the multibeam HTS communication system architecture.

Features of Satellite Communications

Compared to terrestrial MU-MIMO systems, satellite communications exhibit several distinctive features. The propagation conditions are dominated by a line-of-sight (LoS) component between the satellite and the UTs, accompanied by a weaker scattered contribution arising from the local environment of the UT [31]. The free-space path loss is several orders of magnitude larger than in terrestrial links, and the received power can be further reduced by atmospheric impairments such as rain attenuation. For mobile satellite services, however, communication often takes place in the L- or S-band, where propagation losses are smaller than in the Ka-band commonly used for fixed satellite services [29]. This comes at the expense of a more limited available bandwidth, which further motivates aggressive frequency reuse. For the present work, the two most relevant features are therefore the mobility of the UTs and the large propagation delay.

The mobility of the UTs and/or the satellite makes the channel time-varying through Doppler effects. For geostationary earth orbit (GEO) satellites, which appear essentially stationary relative to the Earth's surface, the satellite-induced Doppler shift is negligible, and the dominant source of channel variation is UT motion alone. For low earth orbit (LEO) satellites, the high orbital velocity produces a substantial satellite-induced Doppler shift. However, since the satellite's trajectory is known with high accuracy, this contribution is deterministic and can be pre-compensated (see e.g. [32]). The residual, more difficult component to track is the Doppler shift induced by the motion of the UT, since it depends on the instantaneous user velocity and propagation geometry. So, in both cases, UT mobility is the primary driver of channel time-variation, and it causes the channel to fluctuate at a rate governed by the UT speed and the carrier frequency.

The large geographical distances between the GW, the satellite, and the UTs introduce a propagation delay that is several orders of magnitude larger than in terrestrial systems. For a GEO satellite, which orbits at an altitude of approximately 35,786 km, the one-way propagation delay is on the order of 250 ms. Even for a LEO constellation, the round-trip time (RTT) remains in the tens of milliseconds. Since the precoder is computed based on CSI provided by the UTs, this delay implies that the channel used to design the precoder is inevitably outdated relative to the channel experienced when the precoded signal reaches the UTs. In a static scenario, this mismatch is of limited importance. In combination with a time-varying channel, however, it forms a major challenge for precoding design.

Channel Model

No single probabilistic model captures the full range of propagation conditions encountered in mobile satellite communications, and a variety of models have been proposed in the literature for different scenarios.

In this work, we begin with a deliberately simplified model in which the channel of UT_k is represented as a frequency-flat Rician fading channel [31, 33]:

$$\mathbf{H}_k(t) = \sqrt{\frac{\kappa}{\kappa+1}}\,\mathbf{H}_k^{\mathrm{LoS}}(t) + \sqrt{\frac{1}{\kappa+1}}\,\mathbf{H}_k^{\mathrm{NLoS}}(t), \tag{4.1}$$

where $\kappa \geq 0$ is the Rician factor, $\mathbf{H}_k^{\mathrm{LoS}}(t)$ is the deterministic line-of-sight (LoS) component, and $\mathbf{H}_k^{\mathrm{NLoS}}(t)$ is the time-varying stochastic non-line-of-sight (NLoS) component.

The Rician factor quantifies the power ratio between the LoS and the NLoS contribution. It typically takes values around 5 dB in urban environments and up to 15 dB or more in rural environments. In this work, it is set to 5 dB.

The LoS component is treated as time-invariant, and the small spatial variations across the antenna arrays are neglected. So, all entries of $\mathbf{H}_k^{\mathrm{LoS}}$ share a common, constant phase offset θ_k

associated with UT k:

$$\mathbf{H}_k^{\mathrm{LoS}} = e^{j\,\theta_k}\,\mathbf{1}_{N_R \times N_T} \tag{4.2}$$

As a consequence, the LoS component does not contain any temporal or spatial variation.

The NLoS component $\mathbf{H}_k^{\mathrm{NLoS}}(t)$ captures the scattered multipath contribution. Each entry is considered i.i.d., and modeled as a zero-mean unit-variance complex Gaussian random variable consistent with the Rayleigh fading model as described in Appendix C. The temporal correlation of the NLoS entries is governed by the Doppler spread induced by UT mobility. Under the classical isotropic scattering assumption, the power spectral density of each channel entry follows the Jakes model:

$$S_{\mathbf{H}_k^{\mathrm{NLoS}}}(f) = \frac{1}{\pi f_D \sqrt{1-(f/f_D)^2}}, \quad |f| \leq f_D, \tag{4.3}$$

where $f_D = v f_c / c$ is the maximum Doppler frequency, determined by the UT speed v, the carrier frequency f_c, and the speed of light c. Further details on the Doppler power spectrum are provided in Appendix C.3, and the generation of a time-correlated random process with a prescribed autocorrelation function or power spectral density is addressed in Appendix D.

The model as stated rests on two simplifying assumptions: spatial correlation between channel entries is neglected, and the LoS Doppler shift is assumed to be perfectly pre-compensated. While these assumptions yield a tractable and well-understood baseline, they do not always hold in practice. At the end of the following chapter, we therefore describe how this model can be extended to a more realistic setting that accounts for spatial correlation and a residual LoS Doppler component, and we briefly evaluate the impact of these features on the performance of the proposed precoding techniques.

4.3 Classification

A large number of precoding techniques have been proposed for satellite communications, differing in the assumptions they make about the system and in the design objectives they pursue. A comprehensive survey can be found in [34]. The objective of this section is not to reproduce this overview, but rather to position the present work within the broader landscape and to identify the techniques that are most closely related.

A Taxonomy of Precoding Techniques

Besides the distinction between linear and nonlinear techniques, existing satellite precoding methods can be classified along several other non-exclusive axes. For each axis we briefly recall the main alternatives and indicate the choice made in this work.

A first distinction concerns the traffic model. In practical multibeam satellite systems, a single beam may simultaneously serve multiple UTs within the same frame, which leads naturally to a multi-group multicast formulation. To avoid additional complications introduced by multicast scheduling, we restrict the analysis to the unicast case, in which only a single UT is served per beam. A second axis concerns the optimization criterion. Depending on the system objective, the precoder may be designed to maximize the sum rate under a transmit power constraint, or to minimize the transmit power provided that QoS metric is met. Closely related to this is the choice of power constraint. In practical satellite payloads, each feed is typically driven by its own power amplifier, which makes per-feed power constraints the most natural formulation from an implementation perspective. However, owing to the additional complexity associated with per-feed power constraints, most existing precoding techniques are formulated under a total transmit-power constraint. In this dissertation, the emphasis is placed on sum-rate maximization subject to a total transmit power constraint, in line with the terrestrial MU-MIMO framework developed in the previous chapters. A further classification axis is the system architecture. In large multibeam systems, precoding may be distributed across multiple GWs, and part of the processing may even be transferred on board the satellite in order to relax feeder-link limitations. Here, the simplest architecture is considered: a single GW performs all precoding on the ground and serves the multibeam payload through a common feeder link.

Finally, and most importantly for the present work, precoding techniques differ in the type of CSI assumed at the transmitter. Three regimes are commonly distinguished: perfect instantaneous CSI, statistical CSI, and imperfect or outdated CSI. The vast majority of the satellite precoding literature assumes perfect instantaneous CSI at the GW. This assumption becomes unrealistic in the scenario considered here, where the combination of large propagation delay and a time-varying channel due to a UT-induced Doppler renders the CSI available at the GW systematically outdated. Precoding under outdated CSI is therefore the regime adopted in this work.

Related Work and Positioning

To the best of our knowledge, only two works have addressed the design of linear precoding for mobile multi-beam satellite systems with outdated CSI under assumptions comparable to ours.

The first is the work of Vázquez et al. [35], which considers an L-band mobile interactive multi-beam system over a GEO satellite. The authors model the feedback loop as a single delay τ on the order of 700 milliseconds and adopt a simple closed-form MMSE precoder. Their key analytical observation is that, for a fixed precoder, the random phase fluctuations of the fading process affect the desired and the interfering terms symmetrically and therefore cancel in the resulting signal-to-interference-plus-noise ratio (SINR), only the amplitude fluctuations of the fading process remain visible. As a consequence, the ergodic sum rate is essentially unchanged by outdated CSI, and the dominant practical impairment becomes the mismatch between the SINR predicted by the outdated CSI and the SINR effectively experienced by the UTs, which is handled by fixed back-off margins in the link-adaptation stage.

The second is the work of Joroughi et al. [36], which considers a maritime multi-beam scenario at L-band. The channel is decomposed into a time-invariant beam-gain matrix and a time-varying diagonal attenuation matrix, and the authors propose two CSI feedback strategies. The first is a memoryless strategy that feeds the GW with the latest delayed snapshot of the attenuation. The corresponding precoder is a conventional ZF or MMSE precoder computed from this estimate. The second is a memory-based strategy that averages the attenuation over several past snapshots, and it is paired with a novel two-stage precoder: a first stage suppresses inter-beam interference, while a second diagonal stage compensates for the residual time-varying attenuation.

While the works above tackle the same regime of outdated CSI in mobile multi-beam satellite systems, our approach differs in several aspects. First, rather than focusing on a single precoder, we evaluate the impact of outdated CSI on three different, commonly used linear precoders. Second, instead of addressing the mismatch mainly through link-adaptation margins or temporal averaging and a dedicated robust precoder structure, we seek to compensate it in a more direct way.

Chapter 5

Precoding under Outdated CSI

In Chapter 3, linear precoding was developed under the implicit assumption that the CSI used to design the precoder coincides with the CSI experienced by the UTs at the moment of transmission. As discussed in Chapter 4, this assumption breaks down in the mobile satellite scenario considered in this work, owing to the combination of a large propagation delay and a UT-induced Doppler spread. This chapter is devoted to characterizing the impact of this mismatch on precoder performance and to developing a strategy that compensates for it.

Section 5.1 formulates the problem precisely and introduces the CSI feedback delay as the key operating parameter. From there, the chapter proceeds in three steps. First, Section 5.2 quantifies the impact of outdated CSI on the achievable rate of the linear precoders developed in Chapter 3. Section 5.3 then proposes a compensation method that exploits the temporal correlation of the channel in order to predict the instantaneous channel at the time of transmission. Finally, Section 5.4 re-evaluates both the impact of outdated CSI and the proposed compensation strategy under a more realistic channel model.

To end this work, the results are applied to a set of representative example scenarios, after which the main findings are summarized in a final conclusion.

5.1 Problem Formulation

We consider the forward link of the multi-beam HTS system as described in Chapter 4. The GW serves K UTs simultaneously through the satellite, equipped with N_T feeds, by applying a linear precoder to the data symbols. Each UT is equipped with N_R receive antennas. The simulations performed in this chapter focus on a system in which the BS has 8 transmit antennas and serves 4 UTs with 2 receive antennas each. The channel of UT k at time t is denoted by $\mathbf{H}_k(t) \in \mathbb{C}^{N_R \times N_T}$ and follows the Rician model from (4.1). The received signal at UT k reads

$$\mathbf{y}_k(t) = \mathbf{H}_k(t)\ \mathbf{F}(t)\,\mathbf{a}(t) + \mathbf{n}_k(t), \tag{5.1}$$

where $\mathbf{a}(t)$ collects the data symbols of each stream intended for UT k into a column vector, and $\mathbf{n}_k(t) \sim \mathcal{CN}(\mathbf{0}, N_0\mathbf{I}_{N_R})$ is additive noise. The precoder satisfies the total transmit-power constraint $\mathrm{tr}\big(\mathbf{F}(t)\mathbf{F}^{\mathsf{H}}(t)\big) \leq P_T$ at any moment in time.

In practice, the channel must first be estimated at the UTs and fed back to the GW, before it can be used to design the precoder. Denoting the aggregate elapsed time by the CSI feedback delay τ_{CSI}, the precoder applied at time t is necessarily based on the outdated channel state $\mathbf{H}(t-\tau_{\mathrm{CSI}})$. How severely this mismatch degrades performance depends on how much the channel has evolved during τ_{CSI}.

The temporal variability of the channel is characterised through the normalised autocorrelation function of its NLoS component, denoted by $R_h(\tau)$. It is shown in Appendix C.3 that, under Jakes' model, the autocorrelation between two channel realizations separated by a lag τ equals the zeroth-order Bessel function of the first kind, evaluated in $2\pi f_D \tau$. The coherence time of the NLoS component is defined as the lag at which the autocorrelation first falls to one half of its peak value:

$$T_c \triangleq \max\{\tau > 0 \, : \, R_h(\tau) \geq \frac{1}{2}\}. \tag{5.2}$$

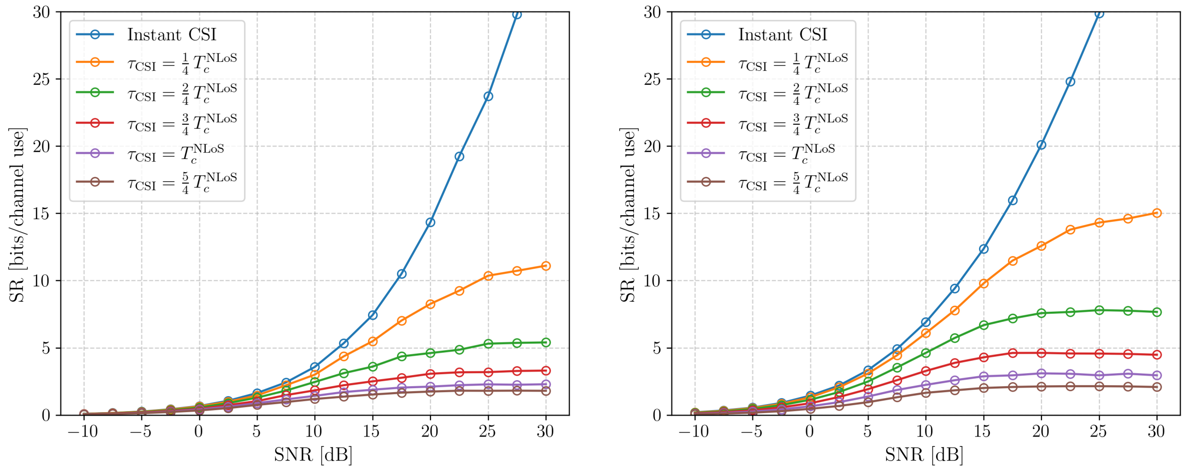
Rather than studying the impact of τ_{CSI} in absolute terms, the CSI feedback delay is expressed as a fraction of the coherence time throughout the analysis. Two operating points sharing the same τ_{CSI}/T_c exhibit, by construction, the same correlation between the outdated and the instantaneous channel, irrespective of the absolute values of τ_{CSI} and T_c. Reporting results as a function of τ_{CSI}/T_c therefore yields conclusions that apply to any combination of carrier frequency, UT velocity and orbital altitude consistent with the underlying Rician model.

5.2 Impact of Outdated CSI

In order to quantify the impact of outdated CSI on the system performance, the achievable rate obtained by linear precoders developed in Chapter 3 is simulated as a function of the SNR. Each precoder is designed based on the outdated channel state $\mathbf{H}(t - \tau_{\mathrm{CSI}})$ and applied to the instantaneous channel $\mathbf{H}(t)$. The resulting sum rate is averaged over more than 100 channel realizations, and over multiple coherence periods of the fading process for each realization. Figures 5.1 and 5.2 report the results for a CSI feedback delay τ_{CSI} that varies from $0\,T_c$ (instantaneous CSI, used as a reference) up to $5/4\,T_c$, slightly beyond the coherence time.

All three precoders are very sensitive to outdated CSI. Even a delay of only a quarter of the coherence time causes a substantial loss on the achievable rate at moderate and high SNR values. The loss with respect to instantaneous CSI grows monotonically with τ_{CSI}/T_c, confirming the intuition that the more outdated the CSI, the worse the performance. Once the CSI feedback delay surpasses the coherence time of the NLoS component, the system becomes essentially unusable. This aligns with the definition of T_c. Beyond the point $\tau_{\mathrm{CSI}} \geq T_c$, the channel available at the GW and the channel actually experienced by the UTs are only weakly correlated such that the precoder is in effect designed for a channel that bears little resemblance to the true one. As a consequence, the precoder fails to exploit the spatial dimension of the channel and the sum-rate collapses.

A closer look at Figure 5.1 shows that both ZF and BD precoding saturate at high SNR once the CSI is even slightly outdated. Both ZF and BD precoders completely eliminate the interference between the users, no matter the ratio of the transmit power to the noise power. With outdated CSI, the precoder nulls interference along the wrong directions, leaving a residual inter-user interference that is proportional to the transmit power. As the SNR increases, the system transitions from a noise-limited regime to an interference-limited one and the desired signal and the residual interference grow at the same rate. Therefore, the SINR saturates and the achievable rate converges to a finite ceiling, that is determined solely by the magnitude of the channel evolution over the feedback delay.



(a) ZF Precoding

(b) BD Precoding

Figure 5.1: The achievable sum-rate of a $8 \times \{2, 2, 2, 2\}$ MU-MIMO system as a function of the SNR, for ZF and BD precoding. The different curves correspond to different values of the CSI feedback delay τ_{CSI}, expressed as a fraction of the coherence time T_c.

The WMMSE precoder exhibits a qualitatively different behaviour. The achievable rate reaches a maximum and decreases slightly beyond it when the precoder design relies on outdated CSI, as is clearly visible in Figure 5.2. Unlike ZF and BD, WMMSE precoding adapts its solution to the operating SNR. At low SNR it trades interference suppression against desired-signal power, while at high SNR the regularization vanishes and it converges to the interference-free BD solution. The algorithm then increasingly concentrates power in directions that it expects to be beneficial, but which, due to the channel mismatch, actively generate inter-user interference in the actual channel. The ZF-type precoders do not exhibit this behaviour because they steer in the wrong direction just as aggressive for any SNR value.

A final observation concerns the reference curve obtained with instantaneous CSI. Even in this idealised case, the achievable rate lies noticeably below the values obtained in Chapter 3 under a purely Rayleigh fading channel. This accounts for all three precoding techniques, and is a direct consequence of the LoS contribution in a Rician channel. Indeed, the LoS component is rank-one, such that the largest fraction of the channel energy is concentrated along a single spatial

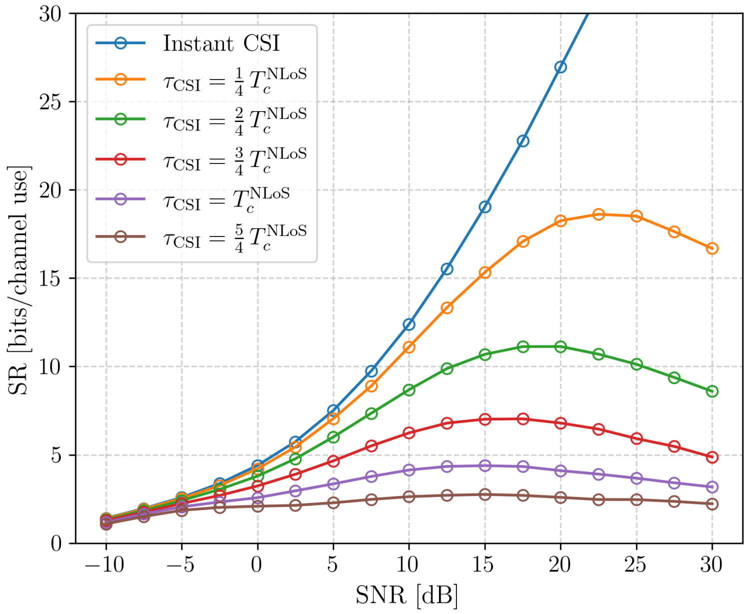


Figure 5.2: Similar figure as Fig. 5.1, but for WMMSE precoding.

direction. A purely Rayleigh fading channel, by contrast, provides rich spatial diversity and a well-conditioned channel matrix, granting the precoder significantly more spatial multiplexing gain. Therefore, the LoS component imposes an inherent limit on the spatial degrees of freedom available, regardless of the quality of the CSI.

Taken together, these results show that even a small feedback delay poses a threat to the achievable rate at moderate and high SNR, and that no choice of precoder can restore the linear growth observed with instantaneous CSI by itself. The next section therefore turns to the design of a compensation strategy that targets the source of the problem, which is the temporal mismatch between $\mathbf{H}(t - \tau_{\text{CSI}})$ and $\mathbf{H}(t)$, rather than its consequences.

5.3 Compensation for Outdated CSI

As established in the previous section, the degradation observed in Section 5.2 stems from the mismatch between the channel estimate available at the GW and the channel experienced at the time of transmission. However, the channel is a temporally correlated random process, and the history of past channel estimates therefore carries useful information about the current channel state. In what follows, we propose a simple compensation strategy that exploits the temporal correlation to predict, at each UT, the channel that will be experienced after the CSI feedback delay, and feed this predicted channel back to the GW instead of the most recent channel estimate. The performance gain introduced by this strategy will again be evaluated by means of a simulation.

5.3.1 Compensation Strategy

Among the many possible prediction methods, a linear autoregressive (AR) model is adopted here as a simple and tractable baseline. It leads to a closed-form predictor, requires only knowledge of the autocorrelation function, and entails a very limited computational complexity. The prediction is performed locally at each UT, while the GW continues to compute the precoder exactly as in Chapter 3. From the perspective of the GW, the precoding procedure is therefore unchanged. Only the channel information on which it relies, is replaced by a prediction of the future channel state. As a result, the additional complexity introduced by the compensation strategy is borne entirely by the UTs.

Under the simplified channel model of Chapter 4, the temporal variation is entirely contained in the NLoS component. Moreover, spatial correlation within the NLoS component is neglected, so that the entries of $\mathbf{H}_k^{\mathrm{NLoS}}(t)$ evolve independently and share the same temporal correlation function. Therefore, we choose to carry out the prediction element-wise, using the same scalar predictor for each channel coefficient. Let $\hat{h}_{k,\,(i,j)}[n]$ denote the estimate of the (i,j)-th entry of the channel matrix of UT k at the n-th pilot instant. A p-th order AR predictor of the channel Δ pilot intervals ahead is then written as

$$\hat{h}_{k,\,(i,j)}[n+\Delta] = \sum_{m=0}^{p-1} \alpha_m \, \hat{h}_{k,\,(i,j)}[n-m]. \tag{5.3}$$

The prediction horizon Δ is chosen such that the predicted sample corresponds to the actual transmission time, that is, $\Delta\, T_p = \tau_{\mathrm{CSI}}$, where T_p denotes the pilot period.

The predictor coefficients $\{\alpha_m\}_{m=1}^{p}$ are obtained from the Yule–Walker equations associated with the autocorrelation function $R_h(\tau)$ of the channel process:

$$\begin{bmatrix} R_h(0) & R_h(T_p) & \cdots & R_h((p-1)T_p) \\ R_h(T_p) & R_h(0) & \cdots & R_h((p-2)T_p) \\ \vdots & \vdots & \ddots & \vdots \\ R_h((p-1)T_p) & R_h((p-2)T_p) & \cdots & R_h(0) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_p \end{bmatrix} = \begin{bmatrix} R_h(\Delta T_p) \\ R_h((\Delta+1)T_p) \\ \vdots \\ R_h((\Delta+p-1)T_p) \end{bmatrix}. \tag{5.4}$$

Since the autocorrelation function is fully specified by the assumed channel model, these coefficients can be computed offline once the predictor order p, the pilot period T_p, and the delay τ_{CSI} have been fixed.

Three assumptions underlie the proposed compensation strategy. First, channel-estimation errors are neglected, so that the channel estimates available at the UTs are assumed to be perfect. Second, each UT is assumed to know the second-order statistics of its channel. In the present model, this amounts to knowing the autocorrelation function $R_h(\tau)$, which under Jakes' model is fully determined by the maximum Doppler frequency f_D. This is a reasonable assumption, since f_D can be estimated locally from the carrier frequency and the UT velocity. Third, the pilot period T_p is treated as a free design parameter and is not constrained by a pilot-overhead budget. This idealization makes it possible to characterize the intrinsic potential of the compensation strategy independently of the resource limitations of a specific system implementation.

5.3.2 Parameter Optimization

The predictor of (5.3) is controlled by two free parameters. Firstly, the pilot period T_p, or equivalently the pilot rate $1/T_p$, which determines the temporal resolution at which the channel is observed. Secondly, the time span or context window length $L \triangleq p\,T_p$ covered by the p past samples, which controls how far back in time the predictor reaches. For a fixed window length, a higher pilot rate raises the model order p by providing finer temporal resolution, while for a fixed pilot rate, a longer window increases the model order by incorporating older observations. The order p of the AR model is fixed once T_p and L are chosen through the relation $p = L/T_p$, and thus not considered as a third design parameter. Both model parameters are expressed relative to the coherence time of the channel, in line with the normalization of the CSI feedback delay adopted throughout this chapter.

Rather than performing a joint grid search over T_p and L, the two parameters are tuned sequentially. The pilot rate is first optimised with the context window length held fixed at a conservative value, after which the context window length is tuned with the pilot rate frozen at the value found in the first step. Both optimisations are evaluated for the $8 \times \{2, 2, 2, 2\}$ system using WMMSE precoding at a single operating point with an SNR equal to 20 dB, for different values of CSI feedback delay.

For the first sweep, the context window length fixed at $2\,T_c$. This is a generous choice, made to ensure that no potentially useful information would be excluded. Figure 5.3 shows the achievable sum-rate as a function of the pilot rate for four values of the CSI feedback delay. The achievable sum-rate increases steadily as the pilot rate grows, reflecting the improvement in temporal resolution available to the predictor. When too few samples are available within the temporal correlation window to capture the channel dynamics accurately, the achievable sum-rate drops rapidly, especailly for the larger feedback delays. At 6 messages per T_c, all four curves reach their respective maximum and converge to nearly the same value, indicating that the compensation is

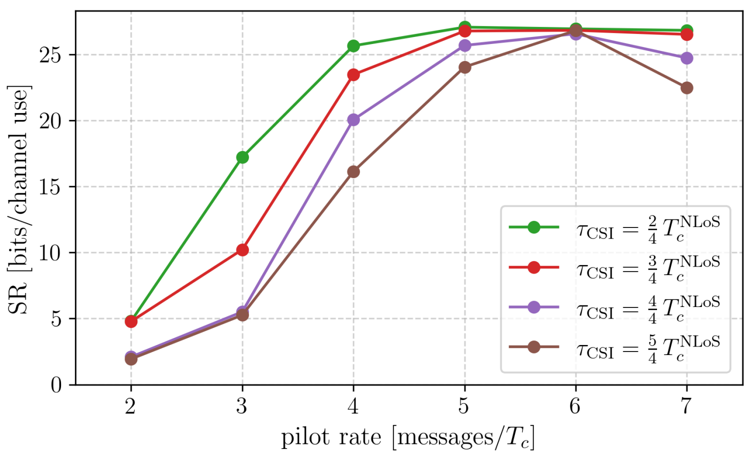


Figure 5.3: The achievable sum-rate plotted as a function of the pilot rate. The $8 \times \{2, 2, 2, 2\}$ system uses a WMMSE precoder and AR-based channel prediction model at the UT, and operates at a SNR equal to 20 dB. The context window length is fixed to $2\,T_c$. Each curve corresponds to a different CSI feedback delay.

equally effective regardless of the feedback delay. Increasing the pilot rate further than 6 messages per coherence time does not improve performance and even causes a slight degradation for the larger delays, most likely because the resulting high model order renders the Yule-Walker system ill-conditioned. The autocorrelation matrix then becomes increasingly close to singular. A pilot rate of 6 messages per coherence time is therefore selected.

With the pilot rate fixed at 6 messages per T_c, the context window length is varied. Figure 5.4 shows that a too short window prevents the predictor from accessing enough of the temporal correlation of the channel and, as expected, severely degrades the achievable sum-rate. Also, the curves saturate beyond a certain prediction horizon. This aligns with the intuition that, after a certain point, the channel history is no longer significantly correlated with the value being predicted, such that the additional information contained in older samples is negligible. This saturation point depends strongly on the CSI feedback delay: for shorter feedback delays, a shorter window length sufficies. Note that a longer context window length results in a larger memory footprint at the UT. A context window length of one coherence period is therefore adopted for all subsequent simulations.

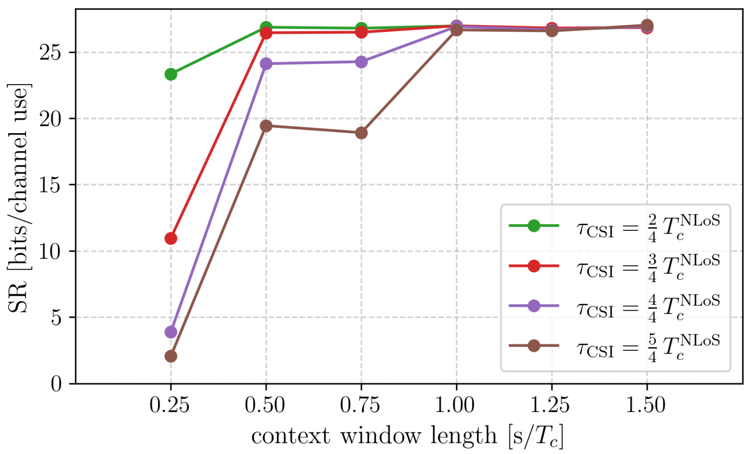
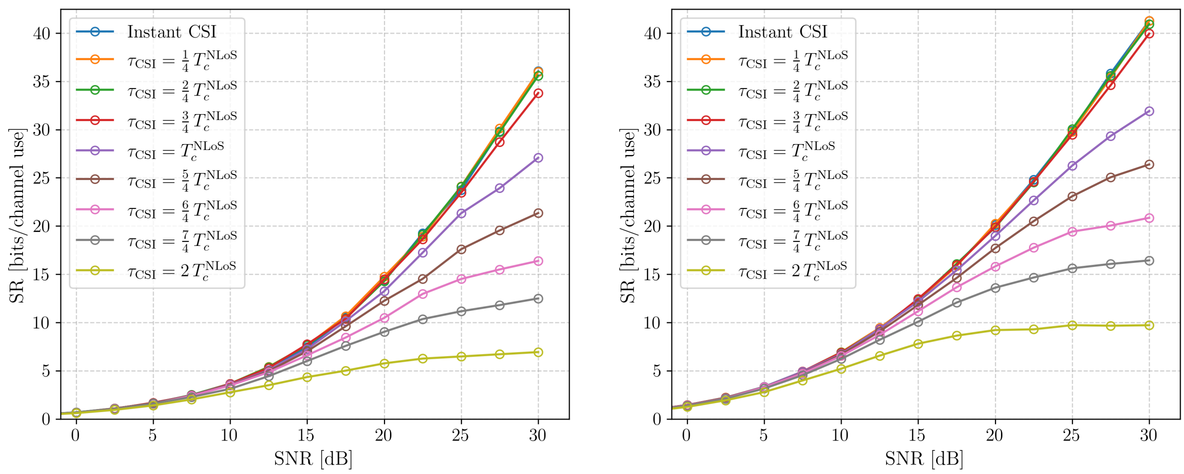


Figure 5.4: The achievable sum-rate plotted as a function of the context window length. The pilot rate is set to 6 messages per coherence period. Other than that, the system operates under the exact same configurations in Fig. 5.3.

5.3.3 Performance Evaluation

This subsection quantifies the gain achieved by the proposed AR-based compensation strategy. The predictor settings are those selected in Section 5.3.2: a pilot rate of 6 messages per coherence period and a context window length equal to one coherence period. All other system configurations are kept identical to Section 5.2, so that the effect of compensation can be isolated directly. Figure 5.5 shows the achievable sum-rate for ZF and BD precoding, and Figure 5.6 provides an explicit three-way comparison for WMMSE precoding. Each panel shows the achievable sum-rate under perfect CSI, outdated CSI, and predicted CSI for a single feedback delay.

Figures 5.5 and 5.6 show a consistent trend across all three precoding techniques. For feedback delays up to approximately one coherence time, the compensation mechanism restores the



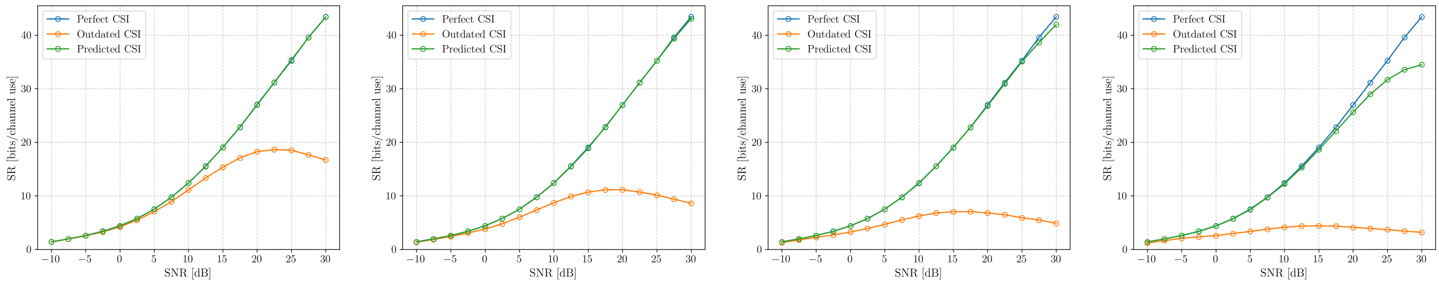
(a) ZF Precoding **(b)** BD Precoding

Figure 5.5: The achievable sum-rate as a function of the SNR for ZF and BD precoding with AR-based channel prediction at the UTs. The predictor uses the parameter pair selected in Section 5.3.2. The different curves correspond to different CSI feedback delays, from no delay up to a delay equal to 2 coherence periods of the NLoS component.

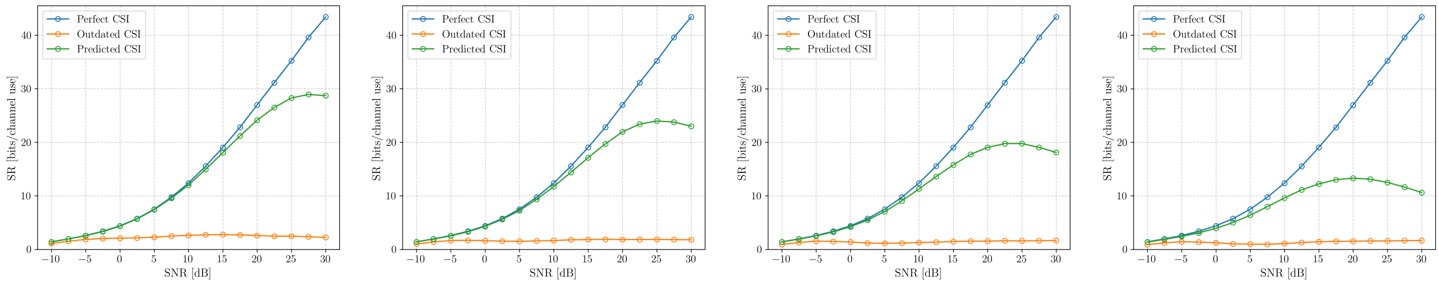
performance obtained with instantaneous CSI almost completely. This confirms that the dominant part of the mismatch can be mitigated as long as the predictor operates within a time interval where the NLoS component remains sufficiently correlated. When the feedback delay exceeds the coherence time, the gain gradually decreases. This behaviour is expected: for $\tau_{\mathrm{CSI}} > T_c$, the channel samples available to the predictor become weakly correlated with the target value, so the predicted channel cannot accurately track the channel realization experienced at the time of transmission. The compensation strategy remains beneficial in this regime, but the residual mismatch grows with delay and the achievable sum-rate drops accordingly.

An remarkeble difference between Figure 5.1 and Figures 5.5 is the disappearance of the rate ceiling. The compensation strategy restores the growth of the achievable sum-rate with increasing SNR for almost all considered CSI feedback delays. Only when τ_{CSI} approaches $2\,T_c$, the residual prediction error is large enough to reintroduce an interference floor, and the familiar saturation behaviour of §5.2 reappears. A similar observation can be made for WMMSE precoding. Only for feedback delays well above T_c, the sum-rate exhibits the peak-and-decrease behaviour identified in Section 5.2.

Altogether, these results show that channel prediction at the UTs is an effective and low-complexity mitigation mechanism for CSI aging under the Rician fading channel model. The largest performance gains are obtained when the feedback delay is on the order of, or smaller than, the coherence time of the NLoS component.



(a) $\tau_{\mathrm{CSI}} = \frac{1}{4}T_c$ **(b)** $\tau_{\mathrm{CSI}} = \frac{2}{4}T_c$ **(c)** $\tau_{\mathrm{CSI}} = \frac{3}{4}T_c$ **(d)** $\tau_{\mathrm{CSI}} = \frac{4}{4}T_c$



(e) $\tau_{\mathrm{CSI}} = \frac{5}{4}T_c$ **(f)** $\tau_{\mathrm{CSI}} = \frac{6}{4}T_c$ **(g)** $\tau_{\mathrm{CSI}} = \frac{7}{4}T_c$ **(h)** $\tau_{\mathrm{CSI}} = \frac{8}{4}T_c$

Figure 5.6: A comparison of the WMMSE precoding performance under perfect CSI, outdated CSI, and compensated CSI obtained through AR-based prediction at the UTs. The compensation gain is strongest for $\tau_{\mathrm{CSI}} \leq T_c$ and decreases progressively as the feedback delay exceeds the channel coherence time.

5.4 Evaluation on a More Realistic Channel

5.4.1 Channel Model

The Rician model used in Sections 5.2 and 5.3 provides a useful baseline, but it neglects two physically relevant effects in practical mobile satellite links. First, the antenna array geometry at both the satellite and the UTs and the propagation directions introduce spatial correlation between channel coefficients. Second, the motion of a mobile UT induces a deterministic Doppler shift on the LoS component that cannot always be pre-compensated. The ray-tracing based channel model introduced in this section extends the Rician model and incorporates both effects. It is based on the channel model from [37, 38, 39].

Both the satellite and each UT are equipped with a uniform planar array (UPA), as illustrated in Figure 5.7 The satellite array has $N_T = N_T^x \times N_T^y$ elements, with inter-element spacings d_T^x and d_T^y along the x- and y-axis, respectively. Fully analog, the UT array has $N_R = N_R^x \times N_R^y$ elements, with spacings d_R^x and d_R^y. For a wavefront arriving from elevation angle θ and azimuth angle ϕ, the array steering vector of a UPA with $N = N_x \times N_y$ elements reads

$$\mathbf{a}(\theta, \phi)_{(n_x N_y + n_y)} = \frac{1}{\sqrt{N}} \exp\left(j \frac{2\pi}{\lambda} \left(n_x d_x \sin\theta \cos\phi + n_y d_y \sin\theta \sin\phi \right) \right), \tag{5.5}$$

for $n_x = 0, \ldots, N_x - 1$ and $n_y = 0, \ldots, N_y - 1$, where λ is the carrier wavelength.

Because the scattering on ground takes place only within a few kilometers around each UT, the angle of departure (AoD) for different propagation paths to the same UT are nearly identical

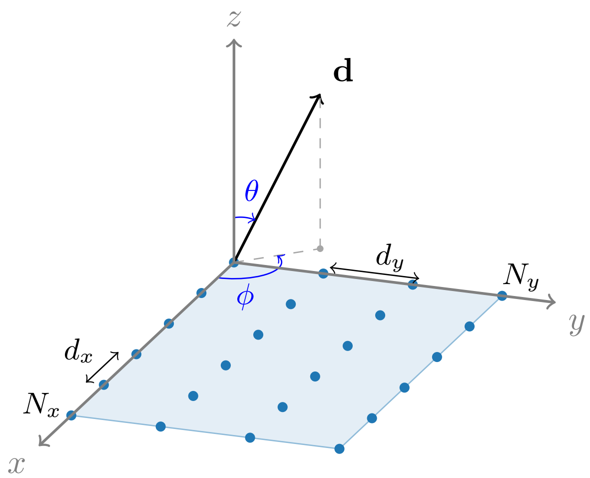


Figure 5.7: The uniform planar array (UPA) antenna configuration considered at both the satellite and the UTs. The vector $\mathbf{d}$ points in the satellite-to-user direction.

due to the exceptional propagation distance in satellite communications [38]. Therefore, a single satellite-side steering vector $\mathbf{a}_k^{\text{sat}} = \mathbf{a}(\theta_k^{\text{sat}}, \phi_k^{\text{sat}}) \in \mathbb{C}^{N_T}$ applies to every component of the channel, where $(\theta_k^{\text{sat}}, \phi_k^{\text{sat}})$ is the elevation-azimuth angle pair to UT k from a satellite point-of-view. By contrast, from a UT perspective scattering is local, so each ray within a cluster arrives from a distinct direction described by an individual steering vector $\mathbf{a}_{k,c,\ell}^{\text{ut}} = \mathbf{a}(\theta_{k,c,\ell}^{\text{ut}}, \phi_{k,c,\ell}^{\text{ut}}) \in \mathbb{C}^{N_R}$.

The channel matrix for UT k is modelled as

$$\begin{aligned} &\mathbf{H}_k(t) = \sqrt{\frac{\kappa}{\kappa+1}}\,\mathbf{H}_k^{\text{LoS}}(t) + \sqrt{\frac{1}{\kappa+1}}\,\mathbf{H}_k^{\text{NLoS}}(t) \\ &\text{with} \quad \mathbf{H}_k^{\text{LoS}}(t) = \alpha_k^{\text{LoS}}(t)\;\mathbf{a}_k^{\text{ut}}\left(\mathbf{a}_k^{\text{sat}}\right)^{\mathsf{H}} \\ &\text{and} \quad \mathbf{H}_k^{\text{NLoS}}(t) = \sum_{c=1}^{C}\sqrt{w_c}\cdot\frac{1}{\sqrt{L_c}}\sum_{\ell=1}^{L_c}\alpha_{k,c,\ell}^{\text{NLoS}}(t)\;\mathbf{a}_{k,c,\ell}^{\text{ut}}\left(\mathbf{a}_k^{\text{sat}}\right)^{\mathsf{H}}, \end{aligned} \tag{5.6}$$

where κ is the Rician factor, c indexes the NLoS scattering clusters with relative powers w_c, and L_c is the number of rays in cluster c.

The LoS gain $\alpha_k^{\text{LoS}}(t)$ incorporates the Doppler shift induced by the UT's motion:

$$\alpha_k^{\text{LoS}}(t) = \exp\!\big(j\left(\psi_k + 2\pi\nu_k^{\text{LoS}}\,t\right)\big), \tag{5.7}$$

where $\psi_k \sim \mathcal{U}(0, 2\pi)$ is a random initial phase and $\nu_k^{\text{LoS}} = f_D\,\cos(\pi/2 - \theta_k^{\text{sat}})$ is the Doppler frequency of the LoS path. Here, $f_D = v_{\text{UT}}\,f_c/c$ is the maximum Doppler frequency, determined by the carrier frequency f_c, the UT velocity v_{UT}, and the speed of light c. The factor $\sin\theta_k^{\text{sat}}$ accounts for the projection of the UT's horizontal velocity onto the satellite-to-user direction. Unlike the simplified model of (4.1), the LoS component is no longer time-invariant. It rotates in phase at a rate that depends on the UT's velocity and elevation angle.

Each cluster consists of L_c rays whose temporal evolution follows the same Jakes isotropic-scattering model as in the simplified channel of Section 5.1.

$$\alpha_k^{\mathrm{NLoS}}(t) \sim \mathcal{CN}(0,1) \quad \text{with} \quad R_{\alpha_k^{\mathrm{NLoS}}}(\Delta t) = J_0(2\pi f_D \Delta t). \tag{5.8}$$

The angle of arrival (AoA) at UT k of a ray ℓ within cluster c is

$$\theta_{k,c,\ell}^{\mathrm{ut}} = \theta_k^{\mathrm{ut}} + \Delta\theta_{k,c} + \sigma_\theta \, u_{k,c,\ell}, \tag{5.9}$$

$$\phi_{k,c,\ell}^{\mathrm{ut}} = \phi_k^{\mathrm{ut}} + \Delta\phi_{k,c} + \sigma_\phi \, v_{k,c,\ell}, \tag{5.10}$$

where $(\theta_k^{\mathrm{ut}}, \phi_k^{\mathrm{ut}})$ is the LoS direction as seen at the UT, $(\Delta\theta_{k,c}, \Delta\phi_{k,c})$ is the mean angular offset from the LoS direction for cluster c, σ_θ and σ_ϕ are the intra-cluster angular spreads, and $u_{k,c,\ell}$, $v_{k,c,\ell}$ are independent Laplace-distributed random variables with zero mean and unit variance. We restrict ourselves to 2 clusters. The first cluster is centred on the LoS direction ($\Delta\theta_{k,1} = \Delta\phi_{k,1} = 0$), while the second cluster is offset by a user-specific, randomly drawn displacement $(\Delta\theta_{k,2}, \Delta\phi_{k,2})$. The elevation angle of the LoS path is the same at both the satellite and the UT, i.e. $\theta_k^{\mathrm{ut}} = \theta_k^{\mathrm{sat}}$. The orientation of the UT's array, i.e. ϕ_k^{ut}, relative to the satellite's array is determined by a rotation ρ around the vertical axis, which is uniformly distributed in $[0, 2\pi)$ and independent of the satellite-side azimuth.

Since every term in (5.6) shares the same satellite-side steering vector $\mathbf{a}_k^{\mathrm{sat}}$, the channel matrix of a single UT is of rank one at any moment in time. Consequently, regardless of the number of receive antennas, the channel of each UT supports at most one spatial stream. This significantly alters the behavior of the precoding schemes considered in this work, since the receive-side spatial multiplexing gain is no longer available. Because the purpose of this section is not to study this rank limitation itself, the subsequent evaluation is restricted to system configurations in which each UT is equipped with a single receive antenna.

A second distinction from the simplified model concerns the spatial structure of the channel. Under the ray-based model of (5.6), the entries of the channal matrix for UT k are no longer spatially independent. In principle, a predictor that exploits this spatial correlation jointly with the temporal one could yield more accurate predictions than the element-wise approach of Section 5.3.1. However, developing such a predictor would significantly increase the complexity at each UT. To maintain a fair and direct comparison with the results of Section 5.3, the same element-wise AR predictor is applied here without modification, with the same parameter values established in Section 5.3.2.

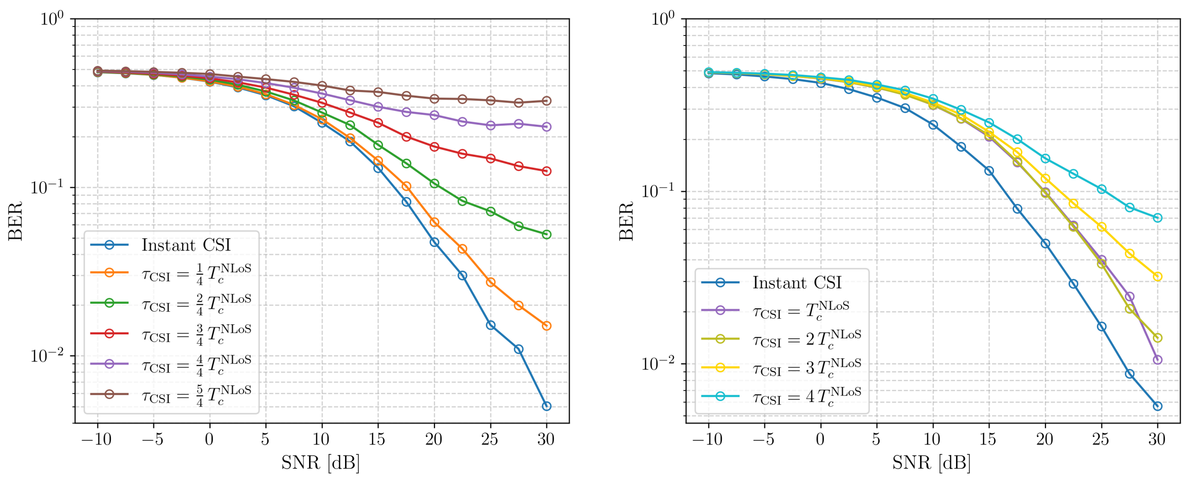
5.4.2 Performance Evaluation

In this section, we perform a first evaluation of the BER of an $8 \times \{1,1,1,1\}$ system using a ZF precoder. Each UT is equipped with a single receive antenna and is served with one data stream carrying 16-QAM symbols. In analogy with Section 5.2, the CSI feedback delay is expressed as a fraction of the coherence time T_c of the NLoS component, as defined in (5.2), so that the results remain independent of any particular choice of UT velocity or carrier frequency.

The left panel of Figure 5.8 shows the BER under outdated CSI as a function of the SNR for several values of the feedback delay. The qualitative behavior closely mirrors the observations made in Section 5.2 for the simplified Rician model. As the feedback delay grows, the BER at any given SNR degrades monotonically. For delays approaching or exceeding T_c^{NLoS}, the residual inter-user interference caused by the channel mismatch begins to dominate the noise at high SNR, and the BER curves flatten. The system has an extremely poor reliability, no matter the avalaible transmit power.

The right panel of Figure 5.8 shows the same system with AR-based channel prediction. The compensation restores a substantial fraction of the performance lost due to the feedback delay, in line with the findings of Section 5.3.3. A notable difference with respect to the simplified model, however, is that the predictor remains effective over a wider range of CSI feedback delays. This extended compensation range can be attributed to the deterministic LoS component. Because the LoS phase rotates at a fixed rate ν_k^{LoS} that is fully determined by the satellite geometry and the UT velocity, the AR model can track this contribution accurately even when the coherence time of the NLoS component has been exceeded.

These first results are promising: the AR-based compensation strategy, developed and tuned under the simplified Rician model, is capable of significantly reducing the BER for feedback delays up to 4 times the NLoS coherence time. At the same time, these simulations only scratch the surface of what a dedicated analysis of this channel model could reveal. A thorough parameter study, an investigation of the spatial and temporal correlation structure on precoding performance, and a comparison across multiple precoding strategies are needed to draw sound conclusions, and are left as natural directions for future work.



(a) Outdated CSI **(b)** Predicted CSI

Figure 5.8: title

5.5 Conclusion

This chapter addressed a major limitation of linear precoding in mobile satellite systems: the CSI used at the GW is inevitably outdated at transmission time. After restating the problem in terms of the normalized delay τ_{CSI}/T_c, we quantified the resulting mismatch, proposed a compensation strategy based on channel prediction at the UTs, and evaluated its robustness under a more realistic ray-based satellite channel model.

The analysis of Section 5.2 showed that all considered precoders are highly sensitive to CSI aging. Even moderate delay values cause a substantial loss at medium and high SNR, and once τ_{CSI} approaches the coherence time of the NLoS component, the system becomes strongly interference-limited.

To mitigate this effect, Section 5.3 introduced an element-wise AR predictor at the UTs, with unchanged precoding processing at the GW. After tuning the pilot rate and context window, the proposed strategy was shown to recover most of the performance loss for delays up to about one coherence time, and to remain beneficial beyond that range. Hence, channel prediction offers a practical and low-complexity way to extend the operating region of linear precoding under delayed CSI feedback.

Section 5.4 then replaced the simplified Rician model by a ray-based model including spatial correlation and residual LoS Doppler. The same compensation mechanism still provided clear performance gains. The additional limitation caused by the user channel being rank-deficient was avoided by restricting the evaluation to single-antenna UTs.

Finally, let us map the normalized-delay results to representative mobile satellite system (MSS) operating points to provide a more practical insight. The total CSI feedback delay is modelled as four times the one-way propagation delay between the UT and the satellite. This factor of four accounts for the downlink pilot transmission to the UTs, the transmission of the estimated or predicted channel over the return link from the UT to the GW through the satellite, and finally the transmission of the precoder matrix over the forward feeder link back to the satellite. MSS typically operate in the L-band [29], with carrier frequencies between 1 and 2 GHz, and corresponding wavelengths ranging from 15 to 30 cm. Table 5.1 summarizes the maximum Doppler frequency and the NLoS coherence time for different UT velocities.

UT velocity	f_D	T_c^{NLoS}
5 km/h	6.95 Hz	34.84 ms
15 km/h	20.85 Hz	11.61 ms
20 km/h	27.80 Hz	8.71 ms
30 km/h	41.70 Hz	5.81 ms
40 km/h	55.59 Hz	4.36 ms

Table 5.1: The maximum Doppler frequency f_D and NLoS coherence time T_c^{NLoS} as a function of UT velocity, for a carrier frequency in the L-band ($f_c = 1.5$ GHz).

For a LEO satellite at an altitude of 550 km, the one-way propagation delay is $\tau_{\mathrm{prop}} = 1.84$ ms, which results in a total CSI feedback delay of $\tau_{\mathrm{CSI}} = 7.34$ ms. Comparing this value against Table 5.1, the condition $\tau_{\mathrm{CSI}} \leq T_c$ is satisfied for all UT moving slower than approximately 20 km/h. In this regime, the AR-based compensation operates at its most effective and the achievable rate is restored to near-perfect-CSI performance. For higher velocities, such as 30 km/h or 40 km/h, the feedback delay exceeds the coherence time and perfect restoration is no longer possible. However, the compensation still provides a substantial gain: whereas uncompensated outdated CSI would render the system essentially unusable at these speeds, the predicted CSI preserves a meaningful link quality. The AR-based strategy is therefore well-suited to LEO satellite scenarios across a wide range of UT velocities, under the assumption of a Rician fading channel as in 4.1.

For a GEO satellite at an altitude of 35 786 km, the one-way propagation delay is $\tau_{\text{prop}} = 119.4$ ms, yielding $\tau_{\text{CSI}} = 477.5$ ms. Even for the slowest UT velocity considered (5 km/h, $T_c = 34.84$ ms), this corresponds to $\tau_{\text{CSI}}/T_c \approx 13.7$, far beyond the range where temporal prediction is effective. The AR-based compensation strategy is therefore not applicable to GEO links. In this regime, the mismatch between the available and the actual channel is too large to be bridged the predictor. More robust precoding formulations that explicitly account for severe CSI uncertainty are required.

Part III

Appendices

Appendix A

Waterfilling for Power Allocation

In this appendix, we solve the problem of optimal power allocation across independent scalar Gaussian channels using the method of Lagrangian multipliers. The key principles of this method are explained alongside the derivation, because it is also employed to solve other optimization problems encountered in this dissertation. The solution to the optimal power allocation problem gives rise to the waterfilling algorithm, which is discussed at the end of this appendix.

A.1 Optimal Power Allocation

Consider N independent SISO Gaussian transmission channels. Optimal power allocation across these channels can be formulated as a constrained maximization problem. The objective is to distribute the total available transmit power among the different channels such that the combined achievable rate is maximized.

Consider the following general optimization problem:

$$\begin{aligned} \max_{\{p_n\}} \quad & \sum_{n=1}^{N} \alpha_n \log_2 \left(1 + \gamma_n \, p_n\right) \\ \text{s. t.} \quad & \sum_{n=1}^{N} p_n = P_t \quad \text{and} \quad p_n \geq 0 \quad \forall n = 1, \ldots, N \end{aligned} \tag{A.1}$$

where p_n denotes the power allocated to channel n, and P_t denotes the total available transmit power. The coefficients α_n are non-negative weighting factors that allow different channels to be prioritized unequally, whereas $\gamma_n > 0$ denotes the ratio between the channel gain and the noise power of channel n.

To solve the constrained optimization problem in (A.1), the method of Lagrangian multipliers is employed. The core idea of this method is to convert a constrained optimization problem into an unconstrained one by absorbing the constraints into a modified objective function, known as the Lagrangian. This is achieved by introducing auxiliary variables, called Lagrange multipliers, one for each constraint, which penalize violations of the constraints.

Since the optimization problem in (A.1) involves both an equality constraint (total power budget) and inequality constraints (non-negativity of the individual power allocations), both types must be incorporated into the Lagrangian. Equality constraints enter with an unconstrained multiplier $\mu \in \mathbb{R}$, whereas inequality constraints of the form $p_n \geq 0$ are rewritten as $-p_n \leq 0$ and enter with a non-negative multiplier $\lambda_n \geq 0$, ensuring that the penalty is only active when the constraint is binding. The resulting Lagrangian is

$$\mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = \sum_{n=1}^{N} \alpha_n \log_2 \left(1 + \gamma_n \, p_n\right) - \mu \left(\sum_{n=1}^{N} p_n - P_t \right) + \sum_{n=1}^{N} \lambda_n p_n \tag{A.2}$$

where μ is the Lagrange multiplier associated with the total power constraint, and λ_n are the multipliers associated with the non-negativity constraints on the individual power allocations.

Since the objective function is concave and all constraints are affine, the optimization problem is convex. For convex problems, the Karush-Kuhn-Tucker (KKT) conditions are both necessary and sufficient for a solution to be globally optimal [40, 41]. These conditions are given by

1. Stationarity: $$\nabla_{\mathbf{p}} \, \mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = 0 \tag{A.3a}$$
2. Primal feasibility: $$p_n \geq 0 \quad \text{and} \quad \sum_{n=1}^{N} p_n = P_t \tag{A.3b}$$
3. Dual feasibility: $$\lambda_n \geq 0 \tag{A.3c}$$
4. Complementary slackness: $$\begin{cases} p_n > 0 \implies \lambda_n = 0 \\ p_n = 0 \implies \lambda_n \geq 0 \end{cases} \tag{A.3d}$$

From the KKT conditions, we can derive the optimal power allocation for each SISO channel. The derivation proceeds by successively applying the stationarity, complementary slackness, and primal feasibility conditions.

The stationarity condition requires that the gradient of the Lagrangian with respect to each optimization variable vanishes at the optimum. This expresses the balance between the marginal gain in the achievable rate and the penalties introduced by the power constraints. Applying this condition to the Lagrangian yields:

$$\nabla_{\mathbf{p}} \, \mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = 0 \quad \iff \quad \frac{\partial \mathcal{L}}{\partial p_n} = \frac{\alpha_n \gamma_n}{\ln(2) \left(1 + \gamma_n p_n\right)} - \mu + \lambda_n = 0, \quad \forall n = 1, \dots, N \tag{A.4}$$

The complementary slackness condition links the non-negativity constraint on the allocated power to its associated Lagrange multiplier. When a positive amount of power is allocated to channel n, the corresponding multiplier λ_n is zero, and the stationarity condition reduces to:

$$\frac{\alpha_n \gamma_n}{\ln(2)(1 + \gamma_n p_n)} - \mu = 0 \tag{A.5a}$$

$$\iff \quad p_n = \frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \tag{A.5b}$$

The primal feasibility condition enforces the non-negativity of the power allocations. As a result, the above expression for p_n must be clipped to zero whenever it becomes negative:

$$p_n = \left(\frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \right)_+ \tag{A.6}$$

Finally, the primal feasibility condition also requires the total allocated power to equal the available transmit power P_t. By substituting the expression for p_n from (A.6) into the total power constraint, we obtain an implicit equation for the Lagrangian multiplier μ:

$$\sum_{n=1}^{N} \left(\frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \right)_+ = P_t \tag{A.7}$$

Since μ cannot be obtained in closed form, it is determined iteratively such that the total allocated power satisfies (A.7). Owing to the monotonic relationship between μ and the total allocated power, a bisection method is typically employed to compute the optimal value of μ numerically.

A.2 Waterfilling Algorithm

A more efficient alternative to the bisection method exploits the geometric structure of (A.7). Associate with each channel n a container of width α_n and floor height $1/(\alpha_n \gamma_n)$. The total available transmit power P_t is represented as water that is poured simultaneously into all containers. Because all containers share a common water level, channels with a lower floor and a wider container receive more water. A container remains empty if its floor height lies above the common water level, meaning no power is allocated to that channel. The water volume in each channel container is exactly equal to the power allocated to that channel under the optimal solution [42]. This interpretation is illustrated in Fig. A.1 for a system with $N = 4$ channels.

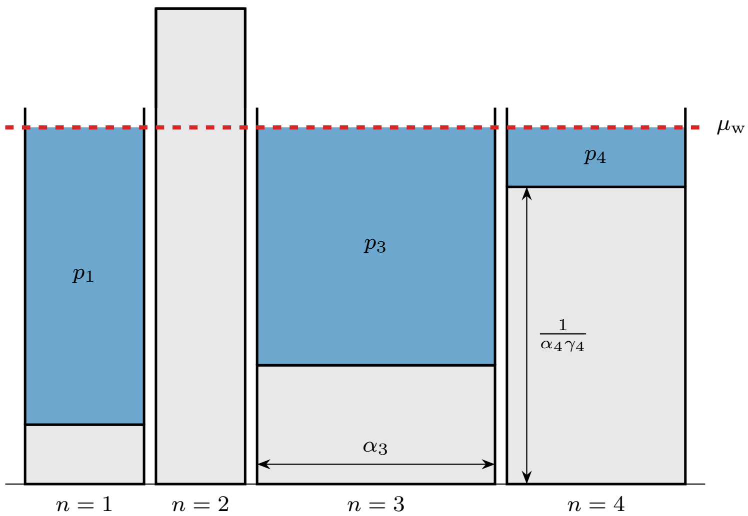


Figure A.1: A visualization of the waterfilling algorithm for optimal power allocation strategy in a system with $N = 4$ channels.

The complete procedure for determining the water level and the corresponding power allocation is summarized in Algorithm 5.

Algorithm 5 Waterfilling Algorithm for power allocation across Gaussian SISO subchannels

Require:
- The total transmit power: P_t;
- The weight factor of each channel: α_n for $n = 1, \dots, N$;
- The ratios between channel gain and noise power: $\gamma_n = g_n^2/N_0$ for $n = 1, \dots, N$.

Ensure: The optimal power allocation p_n for $n = 1, \dots, N$.

Initialization: Determine the width and height of all channel containers.
Find the permutation $\mathcal{P} : n \to i$ between the original index n and the sorted index i that sorts the weighted ratios between channel gain and noise power $\alpha_n \gamma_n$ in decreasing order ($\mathrm{O}(N \log N)$).

$w_i \leftarrow \alpha_{\mathcal{P}(n)}, \quad h_i \leftarrow 1/(\alpha_{\mathcal{P}(n)} \gamma_{\mathcal{P}(n)})$

Iteration: Determine the optimal number of active channels ($\mathrm{O}(\log N)$).

$N^{\text{LB}}, N^{\text{UB}} \leftarrow 0, N$
while $N^{\text{UB}} - N^{\text{LB}} > 1$ **do**
  Update the number of active channels:
  $N^{\text{iter}} \leftarrow \lfloor (N^{\text{UB}} + N^{\text{LB}})/2 \rfloor$
  Compute the minimum power required to activate N^{iter} channels:
  $P_{\min} \leftarrow \sum_{i=1}^{N^{\text{iter}}} w_i \left(h_{N^{\text{iter}}} - h_i\right)$
  Update the bound on the number of active channels:
  if $P_{\min} < P_t$ **then** $N^{\text{LB}} \leftarrow N^{\text{iter}}$ **else** $N^{\text{UB}} \leftarrow N^{\text{iter}}$
end while

Termination: Compute the power allocated to each channel.
Fill the N^{LB} active stream containers:
$p_i \leftarrow w_i \left(h_{N^{\text{LB}}} - h_i\right) \quad \text{for } i = 1, \dots, N^{\text{LB}}$
Distribute the remaining power among the active channels according to their weight:
$p_i \leftarrow p_i + (w_i / \sum_{i'=1}^{N^{\text{LB}}} w_{i'}) \cdot (P_t - \sum_{i'=1}^{N^{\text{LB}}} p_{i'}) \quad \text{for } i = 1, \dots, N^{\text{LB}}$
Set the power of the inactive channels to zero:
$p_i \leftarrow 0 \quad \text{for } i = N^{\text{LB}} + 1, \dots, N$
Use the inverse permutation $\mathcal{P}^{-1}$ to map the allocated power back to the original index n.
$p_n \leftarrow p_{\mathcal{P}^{-1}(i)}$

Return p_n for $n = 1, \dots, N$

A.3 Examples

Piecewise Frequency-Selective Channel

Consider a channel with a piecewise constant frequency response, partitioned into N non-overlapping subbands. Subband n spans the frequency interval $[B_{n-1}, B_n]$ and has a bandwidth $\beta_n = |B_n - B_{n-1}|$, a constant channel gain g_n, and experiences additive white Gaussian noise with power spectral density N_0.

The capacity of this channel can then be written as

$$C = \max_{|S(f)|^2} \int_f \log_2\left(1 + \frac{|H(f)|^2}{N_0}|S(f)|^2\right) df \qquad \text{s.t.} \quad \int_f |S(f)|^2 df \leq P_t \tag{A.8a}$$

$$= \max_{|S(f)|^2} \sum_{n=1}^{N} \int_{B_{n-1}}^{B_n} \log_2\left(1 + \frac{g_n}{N_0}|S(f)|^2\right) df \qquad \text{s.t.} \quad \int_f |S(f)|^2 df \leq P_t \tag{A.8b}$$

$$= \max_{p_n} \sum_{n=1}^{N} \beta_n \log_2\left(1 + \frac{g_n}{N_0}\frac{p_n}{\beta_n}\right) \qquad \text{s.t.} \quad \sum_{n=1}^{N} p_n \leq P_t \tag{A.8c}$$

where $S(f)$ denotes the power spectral density of the transmitted signal and $H(f)$ the frequency response of the channel. In the final step, Jensen's inequality and the concavity of the logarithm imply that the capacity-achieving power spectral density is constant within each subband. Accordingly, if p_n denotes the total power allocated to subband n, then $|S(f)|^2 = p_n/\beta_n$, for $f \in [B_{n-1}, B_n]$.

The optimization problem in (A.8c) is a special case of (A.1) with $\alpha_n = \beta_n$ and $\gamma_n = g_n/(N_0\beta_n)$. Hence, the capacity-achieving power allocation can be obtained by applying the waterfilling algorithm described in the previous section.

Single-User MIMO Channel

As shown in Section 2.1.1, a SU-MIMO channel can be decomposed into R_H independent parallel SISO eigen subchannels by means of the SVD of the channel matrix.

The capacity optimization problem takes the form

$$C = \max_{\{p_n\}} \sum_{n=1}^{N_s} \log_2\left(1 + \frac{\sigma_n}{N_0} p_n\right) \qquad \text{s.t.} \quad \sum_{n=1}^{N_s} p_n \leq P_t,$$

where p_n is the power allocated to the n-th eigen subchannel with channel gain σ_n and N_0 is the noise power on each receive antenna.

This is a special case of (A.1) with $\alpha_n = 1$ and $\gamma_n = \sigma_n/N_0$. Hence, the optimal power allocation, in the sense that it achieves the capacity, can be obtained by applying the waterfilling algorithm described in the previous section.

Appendix B

Analytical BER of an AWGN Channel

In this appendix, an analytical expression for the bit error rate (BER) of an uncoded transmission over an additive white Gaussian noise (AWGN) channel is derived.

The TX maps m_c independent and identically distributed (i.i.d.) bits onto a data symbol a drawn from the constellation $\mathcal{A}$, which contains $M = 2^{m_c}$ distinct symbols. The constellation is assumed to be normalized, i.e. $\mathbb{E}[|a|^2] = 1$. Let p denote the transmit power per symbol. The actual channel input is then given by $\sqrt{p}\,a$. Transmission takes place over an AWGN channel, so that the received signal can be written as

$$y = \sqrt{p}\,a + n, \tag{B.1}$$

where n is a zero-mean CS complex Gaussian noise sample with variance N_0. For convenience, the receiver works with the normalized decision variable $z = y/\sqrt{p}$, which allows the detection problem to be formulated directly in terms of the normalized constellation $\mathcal{A}$ and an equivalent noise sample w with variance N_0/p.

Based on the decision variable z and the constellation $\mathcal{A}$, the detector at the RX decides in favor of a symbol $\hat{a}$. The optimal detector is the one that minimizes the probability of decision error. For an AWGN channel and under the assumption of i.i.d. input bits, the optimal decision rule reduces to the minimum-distance decision rule, which selects the constellation symbol closest to z in Euclidean distance. Finally, the demapper maps the detected symbol $\hat{a}$ back to m_c bits, denoted by $\hat{\mathbf{b}}$. A block diagram of this transmission chain is shown in Figure B.1.

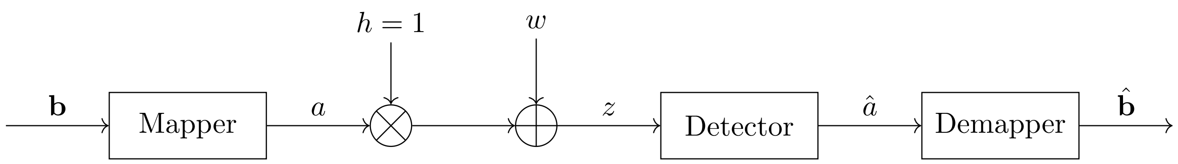


Figure B.1: Block diagram of an AWGN channel

The mapper, detector, and demapper are deterministic functions, whereas the transmitted symbol a and the noise sample n are random variables. Consequently, the decision variable z and the detected symbol $\hat{a}$ are also random variables, both of which depend on the transmitted symbol and the noise realization. Conditioned on the transmitted symbol a, the decision variable is CS complex Gaussian with mean a and variance N_0/p, i.e.

$$z \mid a \sim \mathcal{CN}\left(a, \frac{N_0}{p}\right). \tag{B.2}$$

The BER is defined as the probability that a transmitted bit is detected incorrectly. Equivalently, it can be expressed as the average number of bit errors per detected symbol, normalized by the number of bits per symbol:

$$\mathrm{BER} = \frac{1}{\log_2(M)} \sum_{(\hat{\alpha},\alpha)\in\mathcal{A}^2} \mathrm{N}(\hat{\alpha}, \alpha)\, \mathrm{Prob}[\hat{a} = \hat{\alpha} \wedge a = \alpha]. \tag{B.3}$$

Here, $\mathcal{A}^2$ denotes the Cartesian product of the constellation with itself, and thus contains all possible transmitted-detected symbol pairs. The quantity $\mathrm{N}(\hat{\alpha}, \alpha)$ denotes the number of bit errors incurred when the symbol α is transmitted and $\hat{\alpha}$ is detected. In other words, it equals the Hamming distance between the binary labels of α and $\hat{\alpha}$. Finally, $\mathrm{Prob}[\hat{a} = \hat{\alpha} \wedge a = \alpha]$ is the joint probability that α is transmitted and $\hat{\alpha}$ is detected.

Since the transmission is uncoded and the input bits are i.i.d., all constellation symbols are equiprobable. The BER expression can therefore be simplified using Bayes' rule as

$$\begin{aligned}
\mathrm{BER} &= \frac{1}{\log_2(M)} \sum_{(\hat{\alpha},\alpha)\in\mathcal{A}^2} \mathrm{N}(\hat{\alpha}, \alpha)\, \mathrm{Prob}[\hat{a} = \hat{\alpha} \mid a = \alpha]\, \mathrm{Prob}[a = \alpha] && \text{(B.4a)}\\
&= \frac{1}{M\log_2(M)} \sum_{(\hat{\alpha},\alpha)\in\mathcal{A}^2} \mathrm{N}(\hat{\alpha}, \alpha)\, \mathrm{Prob}[\hat{a} = \hat{\alpha} \mid a = \alpha]. && \text{(B.4b)}
\end{aligned}$$

B.1 Union Upper Bound

Consider the pairwise error event in which the transmitted symbol α is detected as a different symbol $\hat{\alpha}$, with $\hat{\alpha} \neq \alpha$. Under the minimum-distance decision rule, the detector decides in favor of $\hat{\alpha}$ if and only if the decision variable z falls inside the decision region associated with $\hat{\alpha}$. Hence,

$$\mathrm{Prob}[\hat{a} = \hat{\alpha} \mid a = \alpha] = \mathrm{Prob}[z \in \mathcal{D}(\hat{\alpha}) \mid a = \alpha], \tag{B.5}$$

where $\mathcal{D}(\hat{\alpha})$ denotes the decision region of $\hat{\alpha}$, i.e. the set of all points in the complex plane that are closer to $\hat{\alpha}$ than to any other constellation symbol.

An exact evaluation of this conditional decision probability is generally cumbersome because the decision region may be bounded by several neighboring constellation points. A more tractable expression is obtained by replacing the full decision event by a necessary condition. Indeed, if $\hat{\alpha}$ is detected when α was transmitted, then the decision variable must at least be closer to $\hat{\alpha}$ than to α. This condition is necessary, but not sufficient, since yet another constellation point may still be closer to z than $\hat{\alpha}$. Therefore,

$$\mathrm{Prob}[\hat{a} = \hat{\alpha} \mid a = \alpha] \leq \mathrm{Prob}[|z - \hat{\alpha}| < |z - \alpha| \mid a = \alpha]. \tag{B.6}$$

Geometrically, this upper bound corresponds to the probability that the noise pushes the decision variable across the perpendicular bisector between α and $\hat{\alpha}$, rather than all the way into the decision region of $\hat{\alpha}$.

The right-hand side of (B.6) can be simplified as

$$\begin{aligned}
&\mathrm{Prob}[|z - \hat{\alpha}| < |z - \alpha| \mid a = \alpha] \\
&\quad = \mathrm{Prob}\left[|(\alpha + w) - \hat{\alpha}|^2 < |(\alpha + w) - \alpha|^2\right] && \text{(B.7a)} \\
&\quad = \mathrm{Prob}\left[|w - (\hat{\alpha} - \alpha)|^2 - |w|^2 < 0\right] && \text{(B.7b)} \\
&\quad = \mathrm{Prob}\left[|\hat{\alpha} - \alpha|^2 - 2\Re\{w\,(\hat{\alpha} - \alpha)^*\} < 0\right] && \text{(B.7c)} \\
&\quad = \mathrm{Prob}\left[\Re\{w\,\frac{(\hat{\alpha} - \alpha)^*}{|\hat{\alpha} - \alpha|}\} > \frac{|\hat{\alpha} - \alpha|}{2}\right] && \text{(B.7d)} \\
&\quad = \mathrm{Q}\left(\sqrt{\frac{p}{2N_0}}\,|\hat{\alpha} - \alpha|\right) && \text{(B.7e)}
\end{aligned}$$

In the last step, the conditional inequality is expressed as the tail probability of a real Gaussian random variable. Indeed, the quantity inside the probability operator is the projection of the CS complex Gaussian noise w onto a unit-magnitude complex direction. Such a multiplication only rotates w in the complex plane and therefore leaves its distribution unchanged. Consequently, the projected quantity is a zero-mean real Gaussian random variable with variance equal to half the variance of w, i.e. $N_0/(2p)$.

Substituting (B.7e) into (B.4) yields the following upper bound on the BER:

$$\mathrm{BER} \leq \frac{1}{M \log_2(M)} \sum_{(\hat{\alpha},\alpha) \in \mathcal{A}^2} \mathrm{N}(\hat{\alpha}, \alpha)\, \mathrm{Q}\left(\sqrt{\frac{p}{2N_0}}\,|\hat{\alpha} - \alpha|\right). \tag{B.8}$$

This expression is known as the union upper bound.

B.2 Nearest Neighbour Approximation

The union upper bound in (B.8) sums over all possible transmitted–detected symbol pairs. For moderate to high SNR, however, the Q-function decays rapidly with increasing argument, so the dominant contributions arise from pairs $(\hat{\alpha}, \alpha)$ with the smallest inter-symbol distance. It is therefore common to retain only the nearest-neighbour terms in the sum, i.e. those located at the minimum Euclidean distance $d_{\min}$ between the constellation points in $\mathcal{A}$. Letting $\mathcal{S}(\alpha)$ denote the set of nearest neighbours of symbol α in $\mathcal{A}$, the approximation reads

$$\mathrm{BER} \approx \frac{1}{M \log_2(M)}\, \mathrm{Q}\left(\sqrt{\frac{p}{2N_0}}\, d_{\min}\right) \sum_{\alpha \in \mathcal{A}} \sum_{\hat{\alpha} \in \mathcal{S}(\alpha)} \mathrm{N}(\hat{\alpha}, \alpha). \tag{B.9}$$

For uncoded transmission, the optimal mapping from bits to symbol is given by Gray coding, which ensures that the binary labels of any two nearest-neighbour symbols differ in only one bit position, i.e. their Hamming distance is equal to one. Hence,

$$\mathrm{BER} \approx \frac{1}{M \log_2(M)} \, \mathrm{Q}\!\left(\sqrt{\frac{p}{2N_0}}\, d_{\min}\right) \sum_{\alpha \in \mathcal{A}} |\mathcal{S}(\alpha)|. \tag{B.10}$$

where $|\mathcal{S}(\alpha)|$ denotes the number of nearest neighbours of symbol α. The remaining sum counts the total number of nearest-neighbour pairs in the constellation. Dividing by M gives the average number of nearest neighbours per symbol,

$$K = \frac{1}{M} \sum_{\alpha \in \mathcal{A}} |\mathcal{S}(\alpha)|, \tag{B.11}$$

which is a constant that depends only on the type and size of the constellation. The values of K and $d_{\min}$ for all common modulation constellations, normalized to unit average symbol energy, are listed in Table B.1.

The final nearest-neighbour BER approximation reads

$$\mathrm{BER} \approx \frac{K}{\log_2(M)} \, \mathrm{Q}\!\left(\sqrt{\frac{p}{2N_0}}\, d_{\min}\right). \tag{B.12}$$

Constellation	K	$d_{\min}$
M-PAM	$\frac{2(M-1)}{M}$	$\sqrt{\frac{12}{M^2-1}}$
M-PSK	$\begin{cases} 1, & M=2 \\ 2, & M>2 \end{cases}$	$2\sin\left(\frac{\pi}{M}\right)$
M-QAM	$\frac{4(\sqrt{M}-1)}{\sqrt{M}}$	$\sqrt{\frac{6}{M-1}}$

Table B.1: Average number of nearest neighbours K and minimum inter-symbol distance $d_{\min}$ for common modulation constellations, normalized to unit average symbol energy.

Appendix C

Rayleigh Fading Channel Model

This appendix derives the Rayleigh channel model, which is commonly used to characterize wireless propagation in environments with many scatterers. Understanding the physical origin of this model provides useful intuition, since it serves as a fundamental building block for the more sophisticated channel models encountered throughout the dissertation. The derivation proceeds in three steps. First, the Doppler shift experienced by a moving RX receiving a signal reflected by a single scatterer is derived in Section C.1. Second, this result is extended to a large number of independent scatterers in Section C.2, and it is shown that the resulting channel gain converges to a circularly symmetric complex Gaussian random variable as the number of scatterers grows large. Finally, we explore the time-varying nature of the channel in Section C.3, and derive the autocorrelation function and power spectral density of the fading process.

C.1 Doppler Shift

Consider a TX transmitting a baseband signal $x(t)$, modulated onto a carrier of frequency f_c. The signal reaches the RX after being reflected by a single scatterer, introducing an attenuation a_α. For now, the RX is assumed to be stationary, so the path length d_0 and hence the propagation delay $\tau_0 = d_0/c$ are constant. The received passband signal is given by

$$r_{\mathrm{PB}}(t) = \Re\Big\{a_\alpha \, x(t-\tau_0)\, e^{j2\pi f_c (t-\tau_0)}\Big\}. \tag{C.1}$$

Now suppose the RX moves with velocity v, and let α denote the angle between the direction of motion and the direction of the incoming wave. The component of velocity along the propagation path is $v\cos(\alpha)$, so the path length, and hence the propagation delay, become time-varying. This scenario is illustrated in Figure C.1.

The time-varying propagation delay is given by

$$\tau(t) = \frac{d_0 - v\cos(\alpha)\, t}{c} = \tau_0 - \frac{v\cos(\alpha)}{c}\, t. \tag{C.2}$$

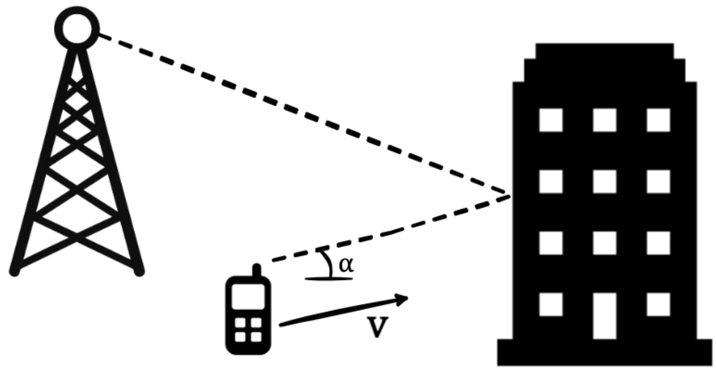


Figure C.1: An illustration of the scenario where the receiver experiences a Doppler shift due to its motion relative to the incoming signal reflected by a scatterer.

Substituting (C.2) into the expression for the received passband signal and expanding the phase term gives

$$r_{\mathrm{PB}}(t) = \Re\Big\{a_\alpha\, x\Big((1+\tfrac{v\cos(\alpha)}{c})t - \tau_0\Big)\; e^{j2\pi(f_c + f_D\cos(\alpha))\,t}\, e^{-j2\pi f_c\tau_0}\Big\}, \tag{C.3}$$

where $f_D = v/\lambda$ is the maximum Doppler frequency and $\lambda = c/f_c$ is the carrier wavelength.

Two simplifying assumptions are now introduced to obtain a tractable baseband model. The first is the narrowband assumption: the bandwidth B of $x(t)$ is assumed to be much smaller than the carrier frequency, $B \ll f_c$. Since $x(t)$ varies slowly compared to the carrier, the small time dilation $v\cos(\alpha)/c \ll 1$ causes negligible distortion of the baseband envelope. The second assumption is perfect synchronization at the RX, so that the initial delay τ_0 can be neglected as well. Together, these two approximations give

$$x\Big(\Big(1+\tfrac{v\cos(\alpha)}{c}\Big)\,t - \tau_0\Big) \approx x(t). \tag{C.4}$$

Applying both approximations and downconverting to baseband yields the received signal

$$y(t) = h(t)\, x(t), \qquad \text{with} \quad h(t) = a_\alpha\, e^{\,j2\pi f_D\cos(\alpha)\,t\,+\,j\phi_\alpha}, \tag{C.5}$$

where $\phi_\alpha = -2\pi f_c\tau_0$ is the initial carrier phase accumulated over the path length d_0.

The channel $h(t)$ is a complex exponential with constant amplitude $|h(t)| = a_\alpha$ and a phase that advances linearly at the Doppler frequency $f_D\cos(\alpha)$. In practice, even sub-wavelength changes in d_0 cause variations of order 2π in ϕ_α, so it is customary to model ϕ_α as uniformly distributed over $[0, 2\pi)$ and independent across different scatterers.

C.2 CS Complex Gaussian Channel Gain

Consider the general scenario in which the signal from the TX reaches the RX via N independent scatterers. The scatterers are assumed to be uniformly distributed on a sphere around the RX, so that the angles of arrival α_n are independent and identically distributed. Each scatterer n is further characterized by an initial phase ϕ_n, which is uniformly distributed over $[0, 2\pi)$ and independent across scatterers, as established in Section C.1. The individual attenuations are set to $a_n = 1/\sqrt{N}$, so that the total expected received power remains normalized to unity for any value of N. By superposition of the single-scatterer result in (C.5), the aggregate channel gain reads

$$h(t) = \sum_{n=1}^{N} a_n \, e^{j(2\pi f_D \cos(\alpha_n)\, t \,+\, \phi_n)}. \tag{C.6}$$

To characterize the distribution of $h(t)$, the real and imaginary parts are analyzed separately. Let $\theta_n \triangleq 2\pi f_D \cos(\alpha_n)\, t$ denote the deterministic phase contribution of scatterer n at time t, and define

$$X(t) = \sum_{n=1}^{N} a_n \cos(\theta_n + \phi_n), \qquad Y(t) = \sum_{n=1}^{N} a_n \sin(\theta_n + \phi_n), \tag{C.7}$$

so that $h(t) = X(t) + j\,Y(t)$. The mean, variance, and covariance of these components are computed below for a fixed time instant t.

Mean:
Since ϕ_n is uniformly distributed over $[0, 2\pi)$ and independent of α_n, the expectation of each summand can be evaluated conditionally on α_n using the law of total expectation:

$$\mathbb{E}[X(t)] = \sum_{n=1}^{N} \mathbb{E}[a_n \cos(\theta_n + \phi_n)] \tag{C.8a}$$

$$= \sum_{n=1}^{N} \mathbb{E}[\mathbb{E}[a_n \cos(\theta_n + \phi_n) \mid \alpha_n]] \tag{C.8b}$$

$$= \sum_{n=1}^{N} \mathbb{E}\left[a_n \int_0^{2\pi} \cos(\theta_n + \phi_n)\, \frac{1}{2\pi}\, \mathrm{d}\phi_n\right] \tag{C.8c}$$

$$= \sum_{n=1}^{N} \mathbb{E}[a_n \; 0] \tag{C.8d}$$

Therefore, $\mathbb{E}[X(t)] = 0$. An identical argument with sine in place of cosine gives $\mathbb{E}[Y(t)] = 0$, and thus $\mathbb{E}[h(t)] = 0$.

Variance:
Since $\mathbb{E}[X(t)] = 0$, the variance reduces to the second moment, which can be expanded as

$$\mathrm{Var}[X(t)] = \sum_{n=1}^{N} \sum_{m=1}^{N} \mathbb{E}[\mathbb{E}[a_n \cos(\theta_n + \phi_n)\, a_m \cos(\theta_m + \phi_m) \mid \alpha_n, \alpha_m]]\,, \tag{C.9}$$

where the law of total expectation is applied again to evaluate the expectation of each summand conditionally on α_n and α_m. For $n \neq m$, the independence of ϕ_n and ϕ_m allows the expectation to factorize into a product of two zero-mean terms, so the cross terms vanish. For $n = m$,

$$\mathbb{E}\big[a_n^2 \cos^2(\theta_n + \phi_n) \mid \alpha_n\big] = a_n^2 \int_0^{2\pi} \cos^2(\theta_n + \phi_n)\, \frac{1}{2\pi}\, \mathrm{d}\phi_n = \frac{1}{2}\, a_n^2. \tag{C.10}$$

Summing the diagonal terms and substituting $a_n = 1/\sqrt{N}$ gives $\mathrm{Var}[X(t)] = \frac{1}{2}\sum_{n=1}^{N} a_n^2 = 1/2$. An analogous calculation gives $\mathrm{Var}[Y(t)] = 1/2$.

Covariance:
Since $\mathbb{E}[X(t)] = \mathbb{E}[Y(t)] = 0$, the covariance reduces to the cross moment, which can be expanded as

$$\mathrm{Cov}[X(t), Y(t)] = \sum_{n=1}^{N}\sum_{m=1}^{N} \mathbb{E}[\mathbb{E}[a_n \cos(\theta_n + \phi_n)\, a_m \sin(\theta_m + \phi_m) \mid \alpha_n, \alpha_m]]\,. \tag{C.11}$$

Similarly to the variance calculation, the cross terms vanish for $n \neq m$, and for $n = m$,

$$\mathbb{E}\big[a_n^2 \cos(\theta_n + \phi_n) \sin(\theta_n + \phi_n) \mid \alpha_n\big] = \frac{1}{2} a_n^2 \int_0^{2\pi} \sin(2\theta_n + 2\phi_n)\, \frac{1}{2\pi}\, \mathrm{d}\phi_n = 0. \tag{C.12}$$

Hence, $\mathrm{Cov}[X(t), Y(t)] = 0$, confirming that the real and imaginary parts of the channel gain are uncorrelated.

Each summand in $X(t)$ is a bounded, zero-mean random variable, and the summands are mutually independent. By the Central Limit Theorem [43], as the number of scatterers grows large ($N \to \infty$), the distribution of $X(t)$ converges to a Gaussian with mean 0 and variance 1/2. The same holds for $Y(t)$. Since $X(t)$ and $Y(t)$ are independent Gaussian random variables, they are jointly Gaussian, and the channel gain $h(t) = X(t) + j\,Y(t)$ is therefore a circularly symmetric complex Gaussian random variable with zero mean and unit variance:

$$h(t) \sim \mathcal{CN}(0,\, 1). \tag{C.13}$$

A standard transformation to polar coordinates shows that the amplitude $|h(t)|$ follows a Rayleigh distribution, which is the origin of the model's name. The phase $\angle h(t)$ is uniformly distributed over $[0, 2\pi)$, as a direct consequence of the circular symmetry of $\mathcal{CN}(0, 1)$. Finally, the instantaneous power $|h(t)|^2$ follows an exponential distribution with unit mean.

C.3 Doppler Spectrum

Section C.2 characterized the distribution of $h(t)$ at a fixed time instant t. To describe the temporal variation of the fading process, the autocorrelation function (acf) and the power spectral density (PSD) of $h(t)$ are derived in this section.

As established in Section C.2, the channel gain at time t can be expressed as

$$h(t) = \sum_{n=1}^{N} a_n\, e^{\,j(2\pi f_D \cos(\alpha_n)\, t \,+\, \phi_n)}, \tag{C.14}$$

where the expected power is normalized, i.e., $\sum_{n=1}^{N} \mathbb{E}[a_n^2] = 1$ and $\phi_n \sim \text{Uniform}(0, 2\pi)$ are independent across scatterers. The scatterers are further assumed to be uniformly distributed on a sphere around the RX, so that the angles of arrival are identically distributed as $\alpha_n \sim \text{Uniform}(0, 2\pi)$.

Autocorrelation Function

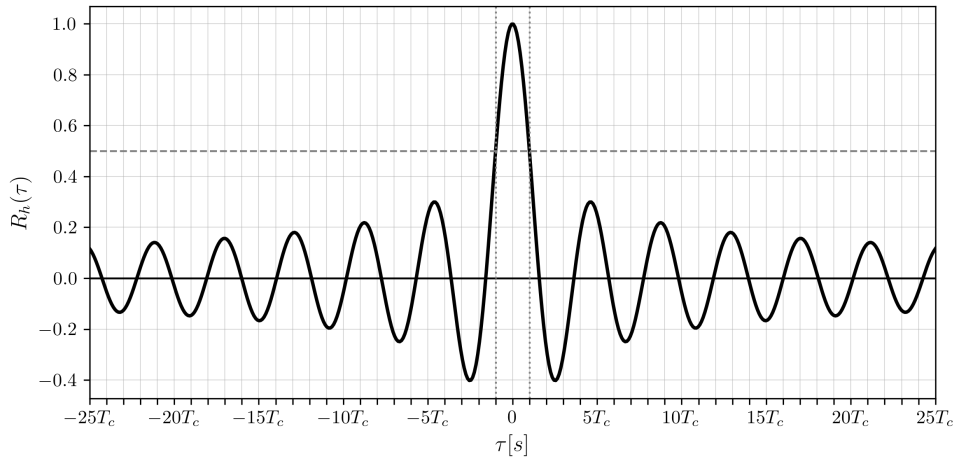
Let us calculate the acf $R_h(\tau) = \mathbb{E}[h(t)h^*(t-\tau)]$ of the fading process:

$$
\begin{aligned}
&R_h(\tau) \\
&\quad = \mathbb{E}\Big[\Big(\sum_{n=0}^{N} a_n \exp(j\,2\pi\, f_D \cos(\alpha_n)\, t + j\phi_n)\Big)\Big(\sum_{m=0}^{N} a_m \exp(-j\,2\pi\, f_D \cos(\alpha_m)\,(t-\tau) - j\phi_m)\Big)\Big] \\
&\quad = \sum_{n=0}^{N}\sum_{m=0}^{N} \mathbb{E}\big[a_n \exp(j\,2\pi\, f_D \cos(\alpha_n)\, t + j\phi_n)\; a_m \exp(-j\,2\pi\, f_D \cos(\alpha_m)\,(t-\tau) - j\phi_m)\big] \\
&\quad = \sum_{n=0}^{N}\sum_{m=0}^{N} \mathbb{E}\big[a_n a_m \exp(j\,2\pi\, f_D\,(\cos\alpha_n - \cos\alpha_m)\, t)\; \exp(j\,2\pi\, f_D \cos(\alpha_m)\,\tau)\; \exp(j\phi_n)\; \exp(-j\phi_m)\big] \\
&\quad = \sum_{n=0}^{N}\sum_{m=0}^{N} \mathbb{E}\big[a_n a_m \exp(j\,2\pi\, f_D\,(\cos\alpha_n - \cos\alpha_m)\, t)\; \exp(j\,2\pi\, f_D \cos(\alpha_m)\,\tau)\big]\; \mathbb{E}\big[\exp(j(\phi_n - \phi_m))\big]
\end{aligned}
$$

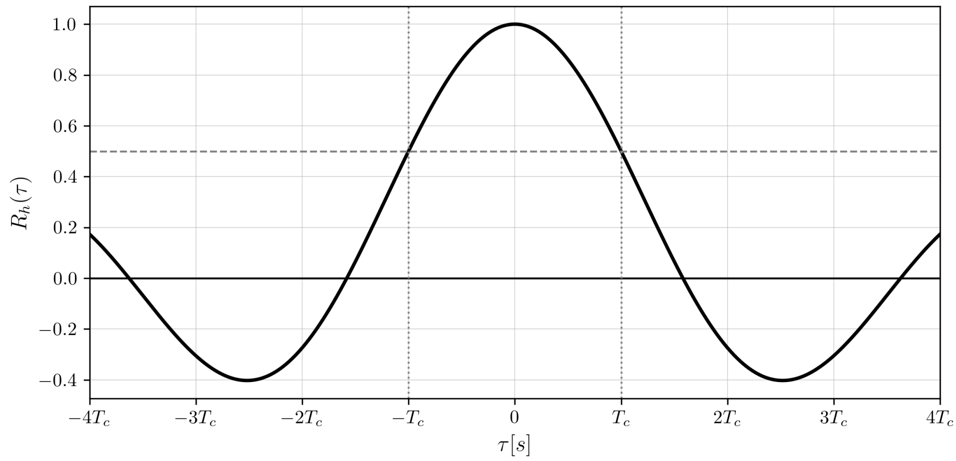
- $\mathbb{E}\big[\exp(j(\phi_n - \phi_m))\big] = \begin{cases} \mathbb{E}\left[e^{j(\phi_n - \phi_n)}\right] = 1 & \text{if } n = m \\ \mathbb{E}\left[e^{j\phi_n}\right]\,\mathbb{E}\left[e^{-j\phi_m}\right] = 0 & \text{if } n \neq m \end{cases}$

$$
\begin{aligned}
&\quad = \sum_{n=0}^{N} \mathbb{E}\big[a_n^2\, e^{j\,2\pi\, f_D \cos(\alpha_n)\,\tau}\big] \\
&\quad = \sum_{n=0}^{N} \mathbb{E}\Big[\mathbb{E}\big[a_n^2\, e^{j\,2\pi\, f_D \cos(\alpha_n)\,\tau}\big|\alpha_n\big]\Big] \\
&\quad = \sum_{n=0}^{N} \mathbb{E}\Big[a_n^2 \int_0^{2\pi} e^{j\,2\pi\, f_D \cos(\alpha_n)\,\tau}\, \frac{1}{2\pi}\, d\alpha_n\Big] \\
&\quad = \sum_{n=0}^{N} \mathbb{E}\big[a_n^2\big]\, \frac{1}{2\pi} \int_0^{2\pi} e^{j\,2\pi\, f_D \cos(\alpha_n)\,\tau}\, d\alpha \\
&\quad = \frac{1}{2\pi} \int_0^{2\pi} e^{j\,2\pi\, f_D \cos(\alpha)\,\tau}\, d\alpha \\
&\quad = J_0(2\pi f_D \tau).
\end{aligned}
$$

The acf of the channel gain is therefore given by the zeroth-order Bessel function of the first kind, $J_0(2\pi f_D \tau)$, which quantifies the degree of correlation between the channel gain at two time instants separated by τ. The acf equals one at $\tau = 0$, since the channel gain is perfectly correlated with itself at zero lag, and decays as τ increases. The channel coherence time T_c is defined as the lag at which the acf drops to a prescribed threshold, here taken as $R_h(T_c) = 1/2$. For $\tau \ll T_c$, consecutive channel samples are highly correlated and the channel remains approximately constant over the interval. For $\tau \gg T_c$, the acf approaches zero and the samples are approximately uncorrelated. Figure C.2 illustrates the acf as a function of the normalized time lag τ/T_c.



(a)



(b)

Figure C.2: The autocorrelation function of the channel gain fading process, relative to the coherence time T_c. (a) shows the autocorrelation for a maximum time lag of $25T_c$, while (b) zooms in on the region around the coherence time.

The relationship between T_c and f_D is obtained as follows:

$$R_h(T_c) = J_0(2\pi f_D T_c) = \frac{1}{2} \quad \Rightarrow \quad T_c \approx \frac{0.25}{f_D}. \tag{C.15}$$

The coherence time is therefore inversely proportional to the maximum Doppler frequency. This makes intuitive sense: the faster the RX moves, and thus the higher the Doppler frequency $f_D = v/\lambda$, the more rapidly the channel fluctuates, and hence the shorter the coherence time.

When the symbol duration is much smaller than the coherence time ($T_s \ll T_c$), the channel gain is approximately constant over many symbols. In this slow fading regime, it is common to adopt the block fading model, in which $h(t)$ is assumed to be constant over a block of L symbols and drawn independently from $\mathcal{CN}(0,1)$ for each successive block.

Power Spectral Density

By the Wiener-Khinchin theorem, the PSD of a wide-sense stationary process equals the Fourier transform of its acf. Let us compute the PSD $S_h(f) = \mathcal{F}\{R_h(\tau)\}$ of the fading process:

$$
\begin{aligned}
&S_h(f) \\
&= \int_{-\infty}^{+\infty} R_h(\tau)\, e^{-j2\pi f\tau}\, d\tau \\
&= \int_{-\infty}^{+\infty} \left(\frac{1}{2\pi} \int_0^{2\pi} e^{j\,2\pi\, f_D \cos(\alpha)\, \tau}\, d\alpha \right) e^{-j2\pi f\tau}\, d\tau \\
&= \frac{1}{2\pi} \int_0^{2\pi} \left(\int_{-\infty}^{+\infty} e^{j\,2\pi\, f_D \cos(\alpha)\, \tau}\, e^{-j2\pi f\tau}\, d\tau \right) d\alpha \\
&= \frac{1}{2\pi} \int_0^{2\pi} \left(\int_{-\infty}^{+\infty} e^{-j\,2\pi\,(f - f_D \cos(\alpha))\,\tau}\, d\tau \right) d\alpha
\end{aligned}
$$

- definition Dirac delta function:

$$\delta(\xi) = \int_{-\infty}^{+\infty} e^{-j\,2\pi\,\xi\, t}\, dt \iff \int_{-\infty}^{+\infty} e^{-j\,2\pi\,(f - f_D\cos(\alpha))\,\tau}\, d\tau = \delta(f - f_D \cos(\alpha))$$

$$= \frac{1}{2\pi} \int_0^{\pi} \delta(f - f_D \cos(\alpha))\, d\alpha \;+\; \frac{1}{2\pi} \int_{\pi}^{2\pi} \delta(f - f_D \cos(\alpha))\, d\alpha$$

- substitution:

$$
\begin{aligned}
& u = f_D \cos(\alpha) \\
& \Leftrightarrow\; du = -f_D \sin(\alpha)\, d\alpha \\
& \Leftrightarrow\; du = -f_D\ \operatorname{sgn}(\sin\alpha)\, \sqrt{1 - \cos^2(\alpha)}\, d\alpha \\
& \Leftrightarrow\; du = -f_D\ \operatorname{sgn}(\sin\alpha)\, \sqrt{1 - (\frac{u}{f_D})^2}\, d\alpha
\end{aligned}
$$

$$\alpha = 0, \alpha = 2\pi \implies u = f_D; \quad \alpha = \pi \implies u = -f_D$$

$$
\begin{aligned}
&= \frac{1}{2\pi} \int_{f_D}^{-f_D} \frac{\delta(f-u)}{-f_D\sqrt{1 - (\frac{u}{f_D})^2}}\, du \;+\; \frac{1}{2\pi} \int_{-f_D}^{f_D} \frac{\delta(f-u)}{f_D\sqrt{1 - (\frac{u}{f_D})^2}}\, du \\
&= \frac{1}{\pi} \int_{-f_D}^{f_D} \frac{\delta(f-u)}{f_D\sqrt{1 - (\frac{u}{f_D})^2}}\, du \\
&= \frac{1}{\pi} \left. \frac{1}{f_D\sqrt{1 - (\frac{u}{f_D})^2}} \right|_{u=f}, \qquad \text{for} \quad |f| < f_D \\
&= \frac{1}{\pi f_D \sqrt{1 - (\frac{f}{f_D})^2}}, \qquad \text{for} \quad |f| < f_D
\end{aligned}
$$

The resulting PSD is known as the Doppler spectrum. It is nonzero only for $|f| \leq f_D$, confirming that the fading process is band-limited to the maximum Doppler frequency: since no scattering angle can produce a Doppler shift exceeding $\pm f_D$. The total Doppler spread equals $2f_D$.

The spectrum exhibits a characteristic "bathtub" shape, which is a direct consequence of the mapping $f = f_D \cos(\alpha)$. Near $\alpha = 0$ and $\alpha = \pi$, $\cos(\alpha)$ varies slowly as a function of α, so a wide range of angles maps to a narrow frequency band near $\pm f_D$, concentrating power at the spectral edges. Near $\alpha = \pi/2$, $\cos(\alpha)$ varies rapidly, so comparatively few angles contribute per unit frequency around $f = 0$. Figure C.3 shows the resulting Doppler spectrum.

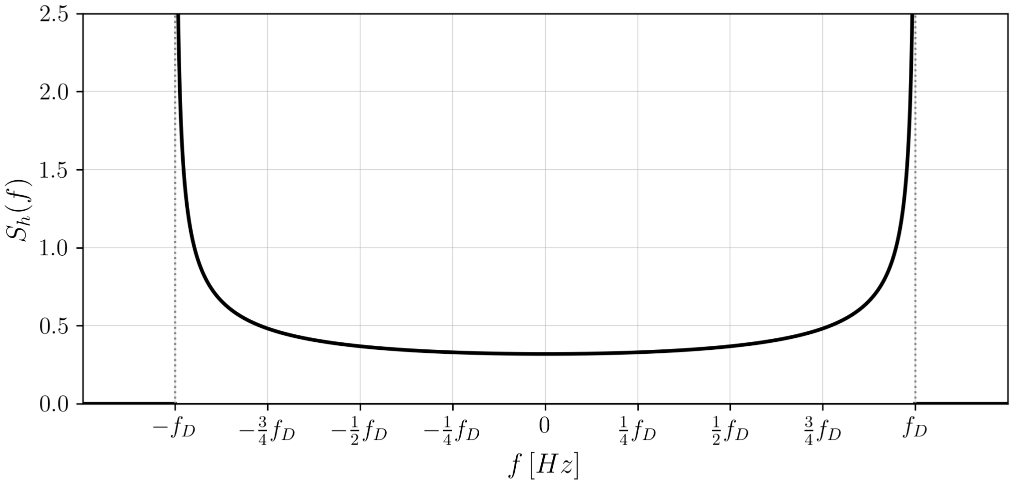


Figure C.3: The Doppler spectrum, relative to the maximum Doppler frequency f_D.

Appendix D

Computer Simulation of Time–Correlated Random Processes

When generating channel realizations for a time-varying channel, it is necessary to simulate a zero-mean complex Gaussian random process with a prescribed correlation in time. This appendix presents two practical methods for simulating time–correlated random processes on a computer. The first method uses the autocorrelation function as input and is based on the Cholesky decomposition of the covariance matrix. The second method uses the power spectral density as input and is based on filtering a white Gaussian process with a linear time-invariant system. In the simulation framework developed for this dissertation, both methods are employed to generate time–correlated channel realizations.

D.1 Cholesky Decomposition Method

The Cholesky decomposition method for generating a zero-mean unit-variance complex Gaussian process $\mathbf{h}$ of length N with a specified autocorrelation function $R_h(\tau)$ consists of the following steps:

Step 1: Build the $N \times N$ covariance matrix $\mathbf{C}$ with entries $C_{i,j} = R_h((i-j)T_s)$.

Step 2: Compute the Cholesky decomposition of the covariance matrix $\mathbf{C} = \mathbf{L}\mathbf{L}^H$, where $\mathbf{L}$ is a lower triangular matrix.

Step 3: Generate a column vector $\mathbf{w}$ of N i.i.d. white complex Gaussian random variables with zero mean and unit variance.

Step 4: Obtain the desired Gaussian process as $\mathbf{h} = \mathbf{L}\mathbf{w}$.

We show that the Cholesky decomposition method effectively generates a zero-mean unit-variance complex Gaussian process $\mathbf{h}$ of length N with a specified autocorrelation function $R_h(\tau)$.

First, let us verify that the generated process $\mathbf{h}$ is a complex Gaussian process with zero mean and the specified autocorrelation function.

Each element $h_n = \sum_{k=1}^{n} L_{k,n} w_k$ of the process $\mathbf{h}$ is a complex Gaussian random variable, since w_k are i.i.d. complex Gaussian random variables and the linear combination of complex Gaussian random variables is itself also a complex Gaussian random variable. Let us now calculate the mean and covariance matrix of the generated process $\mathbf{h}$.

The expected value of each element h_n is given by:

$$\mathbb{E}[h_n] = \mathbb{E}\left[\sum_{k=1}^{n} L_{k,n} w_k\right] = \sum_{k=1}^{n} L_{k,n}\mathbb{E}[w_k] = 0 \tag{D.1}$$

The covariance matrix of $\mathbf{h}$ is given by:

$$\begin{aligned}\mathrm{Cov}[\mathbf{h}] &= \mathbb{E}[(\mathbf{h} - \mathbb{E}[\mathbf{h}])(\mathbf{h} - \mathbb{E}[\mathbf{h}])^H] = \mathbb{E}[\mathbf{h}\mathbf{h}^H] \\ &= \mathbb{E}[(\mathbf{L}\mathbf{w})(\mathbf{L}\mathbf{w})^H] = \mathbb{E}[\mathbf{L}\mathbf{w}\mathbf{w}^H\mathbf{L}^H] = \mathbf{L}\,\mathbb{E}[\mathbf{w}\mathbf{w}^H]\,\mathbf{L}^H \\ &= \mathbf{L}\,\mathbf{I}_N\,\mathbf{L}^H = \mathbf{L}\mathbf{L}^H = \mathbf{C}\end{aligned} \tag{D.2}$$

From (D.1) and (D.2), the mean of the generated process $\mathbf{h}$ is zero and the covariance of $\mathbf{h}$ matches the specified autocorrelation function $R_h(\tau)$ by construction. This confirms that the Cholesky decomposition method effectively generates a zero-mean unit-variance complex Gaussian process with the desired correlation properties:

$$\mathbf{h} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}) \tag{D.3}$$

Second, let us demonstrate that the Cholesky decomposition of the covariance matrix $\mathbf{C}$ always exists. It is well-known that the Cholesky decomposition of a matrix $\mathbf{C}$ exists and is unique if and only if $\mathbf{C}$ is an Hermitian matrix and a positive-definite matrix, i.e.:

$$\exists!\,\mathbf{L} : \mathbf{C} = \mathbf{L}\mathbf{L}^H \iff \mathbf{C} = \mathbf{C}^H \quad \text{and} \quad \forall \mathbf{x} \in \mathbb{C}^N \setminus \{\mathbf{0}\} : \mathbf{x}^H\mathbf{C}\mathbf{x} > 0 \tag{D.4}$$

Consider a autocorrelation function $R_h(\tau)$ of a wide-sense stationary process, i.e. R_h does not depend on the absolute time t, but only on the time difference τ. Therefore, the covariance matrix $\mathbf{C}$ is a real and symmetric Toeplitz matrix by construction. Thus, $\mathbf{C}$ is a Hermitian matrix.

We prove that $\mathbf{C}$ is positive-definite by contradiction. First, let us simplify the expression $\mathbf{x}^H\mathbf{C}\mathbf{x}$:

$$\begin{aligned}\mathbf{x}^H\mathbf{C}\mathbf{x} &= \sum_{i=1}^{N}\sum_{j=1}^{N} x_i^* C_{i,j} x_j = \sum_{i=1}^{N}\sum_{j=1}^{N} x_i^* R_h((i-j)T_s) x_j \\ &= \sum_{i=1}^{N}\sum_{j=1}^{N} x_i^* \left(\int_{-\infty}^{\infty} S_h(f) e^{j2\pi f(i-j)T_s} df\right) x_j \\ &= \int_{-\infty}^{\infty} S_h(f) \left(\sum_{i=1}^{N} x_i^* e^{j2\pi f i T_s}\right)\left(\sum_{j=1}^{N} x_j e^{-j2\pi f j T_s}\right) df\end{aligned}$$

$$= \int_{-\infty}^{\infty} S_h(f) \left| \sum_{i=1}^{N} x_i e^{-j2\pi f i T_s} \right|^2 df$$
$$= \int_{-\infty}^{\infty} S_h(f) \left| X(f) \right|^2 df \tag{D.5}$$

We consider a autocorrelation function $R_h(\tau)$ that corresponds to a power spectral density that makes physical sense, i.e. $S_h(f)$ is positive for all frequencies f and strictly positive for a set of positive measure. Then, it is easy to see that $\mathbf{C}$ is positive semi-definite, since the integrand is non-negative for all frequencies f.

Assume that $\mathbf{C}$ is not positive-definite, then there exists a non-zero vector $\mathbf{x} \in \mathbb{C}^N$ such that $\mathbf{x}^H \mathbf{C} \mathbf{x} = 0$. For that to be true, the integrand must be zero, which implies that $X(f) = 0$ for all frequencies f where $S_h(f) > 0$, which is a set of positive measure. $X(f)$ can be seen as the $\mathcal{Z}$-transform of $\mathbf{x}$ evaluated on the unit circle, which corresponds to a complex polynomial of degree at most $N-1$. By the Fundamental Theorem of Algebra, such a nonzero polynomial has at most $N-1$ roots. Therefore, $\mathbf{x}$ must be the zero vector, which contradicts the assumption. Hence, $\mathbf{C}$ is positive-definite.

D.2 Linear Filtering Method

The linear filtering method for generating a zero-mean unit-variance complex Gaussian process with a specified power spectral density $S_h(f)$ consists of the following steps:

Step 1: Evaluate the target PSD $S_h(f)$ on an FFT frequency grid of N_g uniformly spaced normalized frequencies $\nu_k \in [-1/2, 1/2)$. To avoid aliasing, the sampling period T_s must satisfy the Nyquist condition $f_D T_s < 1/2$, where f_D denotes the maximum Doppler frequency.

Step 2: Construct the frequency-domain shaping response by taking the nonnegative square root of the sampled PSD: $G[\nu_k] = \sqrt{S_h[\nu_k]}$.

Step 3: Apply an inverse discrete Fourier transform to the shaping response $G[\nu_k]$ in order to obtain the impulse response of a finite-length FIR filter, $g[n] = \mathcal{F}^{-1}\{G[\nu_k]\}$, and normalize this filter to unit energy.

Step 4: Generate a sequence of white proper complex Gaussian random variables $w[n]$ with zero mean and unit variance per sample.

Step 5: Obtain the desired time-correlated output process by convolving the white-noise sequence with the shaping filter: $h[n] = \sum_m g[m]\, w[n-m]$.

The underlying idea behind this method is that passing white complex Gaussian noise through a linear time-invariant filter with frequency response $G(f)$ yields an output process whose power spectral density is equal to $|G(f)|^2$. Consequently, by choosing $G(f) = \sqrt{S_h(f)}$, the output process is shaped so as to have the desired power spectral density. The derivation below makes this argument precise.

First, note that the output process $h(t)$ is a linear transformation of the white Gaussian input process $w(t)$ and is therefore itself a complex Gaussian process. Moreover, because the input process has zero mean and the filter is linear, the mean of the output process is also zero.

Let us now determine the autocorrelation function of the output process $h(t)$:

$$
\begin{aligned}
R_h(\tau) &= \mathbb{E}[h(t)\, h^*(t-\tau)] \\
&= \mathbb{E}\left[\int_{-\infty}^{+\infty} g(u)\, w(t-u)\, du \int_{-\infty}^{+\infty} g(v)\, w^*(t-\tau-v)\, dv\right] \\
&= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} g(u)\, g^*(v)\, \mathbb{E}\big[w(t-u)\, w^*(t-\tau-v)\big]\, du\, dv
\end{aligned}
$$

- $\mathbb{E}\big[w(t-u)\, w^*(t-\tau-v)\big]$
  $= \mathbb{E}\big[w(t-u)\, w^*((t-u)-(\tau-v-u))\big] = R_w(\tau+v-u) = \delta(\tau+v-u)$

$$
\begin{aligned}
&= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} g(u)\, g^*(v)\, \delta(\tau+v-u)\, du\, dv \\
&= \int_{-\infty}^{+\infty} g(u)\, g^*(u-\tau)\, du \\
&= R_g(\tau) \qquad \text{(D.6)}
\end{aligned}
$$

Equation (D.6) shows that the autocorrelation function of the output process is equal to the autocorrelation function of the shaping filter. In particular, because the filter is normalized to unit energy, the variance of the output process is given by $R_h(0) = R_g(0) = \int_{-\infty}^{+\infty} |g(u)|^2 du = 1$. Hence, the generated process has unit variance.

Next, we compute the Fourier transform of the autocorrelation function in order to obtain the power spectral density of the output process:

$$
\begin{aligned}
S_h(f) &= \int_{-\infty}^{+\infty} R_g(\tau)\, e^{-j2\pi f\tau}\, d\tau \\
&= \int_{-\infty}^{+\infty}\left(\int_{-\infty}^{+\infty} g(u)\, g^*(u-\tau)\, du\right) e^{-j2\pi f\tau}\, d\tau \\
&= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} g(u)\, g^*(u-\tau)\, e^{-j2\pi f\tau}\, du\, d\tau \\
&= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} g(u)\, g^*(v)\, e^{-j2\pi f(u-v)}\, du\, dv \\
&= \left(\int_{-\infty}^{+\infty} g(u)\, e^{-j2\pi fu}\, du\right)\left(\int_{-\infty}^{+\infty} g^*(v)\, e^{j2\pi fv}\, dv\right) \\
&= G(f)\, G^*(f) \\
&= |G(f)|^2 \qquad \text{(D.7)}
\end{aligned}
$$

Equation (D.7) confirms that the power spectral density of the output process equals the squared magnitude of the filter's frequency response. Therefore, choosing the shaping filter such that $G(f) = \sqrt{S_h(f)}$ ensures that the generated process has the desired power spectral density.

In a practical computer implementation, the process cannot be generated in continuous time and with an infinite-length filter. Instead, the shaping filter is approximated by a finite-length discrete-time filter obtained from sampled values of the target PSD via the inverse discrete Fourier transform (DFT). It is important to choose the filter length N_g sufficiently large, so that truncation effects become negligible and the finite-length filter accurately approximates the target correlation properties of the process.

Appendix E

Simulation Framework

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